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J-20-1228



Quote Number	Project Name	Project Location
J-20-1228	Raytheon DASR	CT

From: Rich Cupillo
14 Connecticut South Drive
East Granby, CT 06026
Phone: 860.844.2220
Fax: 860.392.0222
Email: rcupillo@kinsleypower.com
Cell: 860.930.3479

Qty	Bill of Material Summary
-----	--------------------------

- | | |
|---|--|
| 1 | Kohler 100REOZJF Generator
Output- 102.0kW, 128 KVA, 277/480 Volt, 153 Amps, 60 Hz, Three Phase
Fuel Type - Diesel
Enclosure - Open
Warranty - 2 Year Basic |
| 1 | 25 Gallon Day Tank |
| 1 | 100 KW Load Bank |
| 1 | Kohler KBS-DMWA-0260S Transfer Switch
Type - Bypass\Isolation, Open Transition, 480 Volts/60Hz, 260 Amps
Poles - 4-Pole, 4-Wire, Overlapping Neutral
Enclosure - NEMA 1 Enclosure
Warranty - 2 Year Basic |
| 1 | Distributor Start-Up
Start-up Service with NFPA 110 Load Test |

Generator

Qty	Description
1	100REOZJF Generator System 100REOZJF Generator Set Includes the following: ES Description 01 Literature Languages Approvals and Listings Engine Nameplate Rating Voltage Alternator Cooling System Skid and Mounting Air Intake Controller Starting Aids, Installed Cooling System Acc.,Installed Electrical Accy.,Installed Electrical Accy.,Installed Electrical Accy.,Installed Electrical Accy.,Installed Electrical Accy.,Installed Electrical Accy.,Installed Rating, LCB 1 Amps, LCB 1 Trip Type, LCB 1 Interrupt Rating LCB 1 Aux Trip, LCB 1 Amps, LCB 2 LOAD BANK Trip Type, LCB 2 Interrupt Rating LCB 2 Aux Trip, LCB 2 LCB Accy. Installed Fuel Lines, Installed Fuel System Acc.,Installed Miscellaneous Accy,Installed Miscellaneous Accy,Installed Warranty
	24VDC Starting System English UL2200 Listing 100REOZJF, 12V, 60Hz Standby 60Hz, 277/480V, Wye, 3Ph, 4W 4R12X Unit Mounted Radiator, 50C Skid, 34" Heavy Duty APM603 1500W,120V Radiator Duct Flange Battery, 1/12V, Wet Battery Charger, 10A Run Relay Failure Relay w/Harness,1Fault Generator Heater 15 Relay I/O Board 100% Rated 200 Thermal Magnetic 18kA at 480V Shunt Trip 150 Thermal Magnetic 18kA at 480V Shunt Trip Shunt Trip Wiring Flexible Fuel Lines Fuel Pressure Gauge Air Cleaner Restriction Ind. Coolant in Genset 2 Year

The Energy Solutions Company

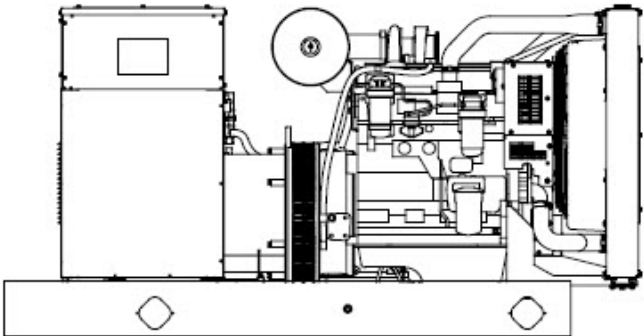
	Testing, Additional	Power Factor Test,0.8,3Ph Only
1	Silencer,Critical,4"NPT,SIEO Flexible	
1	Exhaust Conn,4"NPT Battery Charger	
1	Temp. Comp. Sensor Remote	
1	Emergency Stop Switch	
3	Lit Kit, General Maint, 100REOZJF	

Automatic Transfer Switch

Qty	Description	
	ATS 1500 Transfer Switch System	
1	KBS-DMWA-0260S	
	Includes the following:	
	Literature Languages	English
	Mechanism	Bypass Type B
	Transition	Standard
	Logic	1500
	Voltage	480V / 60 Hz
	Poles & Wires	4 Pole/4Wire, Overlapping Neut
	Enclosure	Nema 1
	Amps	260 Amps
	Connection	Standard
	Miscellaneous Acc.,Installed	Input/Output Module, Qty 1
	Miscellaneous Acc.,Installed	Current Sensing
	Warranty	2-YR BASIC
3	Lit Kit, ATS Production, KBS/KBP/KBC	
1	Module, Input/Output Mounting	



Spec Sheets



Standard Features

- Kohler Co. provides one-source responsibility for the generating system and accessories.
- The generator set and its components are prototype-tested, factory-built, and production-tested.
- The 60 Hz generator set offers a UL 2200 listing.
- The generator set accepts rated load in one step.
- The 60 Hz generator set meets NFPA 110, Level 1, when equipped with the necessary accessories and installed per NFPA standards.
- A one-year limited warranty covers all systems and components. Two-, five-, and ten-year extended warranties are also available.
- Tier 3 EPA-certified for Stationary Emergency Applications
- Alternator Protection
- Battery Rack and Cables
- Customer Connection (standard with Decision-Maker ♦ 6000 controller only)
- Local Emergency Stop Switch
- Oil Drain Extension

Alternator Features

- Operation and Installation Literature
- The unique Fast-Response X excitation system delivers excellent voltage response and short circuit capability using a rare-earth, permanent magnet (PM)-excited alternator.

Other Features

- Kohler designed controller for guaranteed system integration and remote communication.
- The low coolant level shutdown prevents overheating (standard on radiator models only).
- Integral vibration isolation eliminates the need for under-unit vibration spring isolators.
- Mount up to three circuit breakers to allow circuit protection of selected priority loads.

Generator Set Ratings

					Standby130C Ratings	
Alternator	Voltage	Ph	Hz	Peak kVA	kW/kVA	Amps
4R12X	277/480	3	60	448	102 / 128	153

RATINGS: All three-phase units are rated at 0.8 power factor. All single-phase units are rated at 1.0 power factor.
Standby Ratings: The standby rating is applicable to varying loads for the duration of a power outage. There is no overload capability for this rating.
Prime Power Ratings: At varying load, the number of generator set operating hours is unlimited. A 10% overload capacity is available for one hour in twelve.
Ratings are in accordance with ISO-8528-1 and ISO-3046-1. For limited running time and continuous ratings, consult the factory.
Obtain technical information bulletin (TIB-101) for ratings guidelines, complete ratings definitions, and site condition derates.
The generator set manufacturer reserves the right to change the design or specifications without notice and without any obligation or liability whatsoever.

Model: 100REOZJF, continued

Alternator Specifications

Specifications	Alternator
Alternator manufacturer	Kohler
Type	4-Pole, Rotating-Field
Exciter type	Brushless, Rare-Earth Permanent-Magnet
Leads, quantity	12, Reconnectable
Voltage regulator	Solid State, Volts/Hz
Insulation	NEMA MG1
Insulation: Material	Class H
Insulation: Temperature Rise	130°C, Standby
Bearing: quantity, type	1, Sealed
Coupling	Flexible disc
Amortisseur windings	Full
Voltage regulation, no-load to full-load RMS	Controller Dependent
One-Step Load Acceptance	100% of rating
Unbalanced load capability	100% of Rated Standby Current
• NEMA MG1, IEEE, and ANSI standards compliance for temperature rise and motor starting. • Sustained short-circuit current of up to 300% of the rated current for up to 10 seconds. • Sustained short-circuit current enabling down stream circuit breakers to trip without collapsing the alternator field. • Self-ventilated and dripproof construction. • Vacuum-impregnated windings with fungus-resistant epoxy varnish for dependability and long life. • Superior voltage waveform from a two-thirds pitch stator and skewed rotor.	

Engine

Engine Specification

Engine Manufacturer	John Deere
Engine Model	4045HF285I
Engine: type	4-Cycle, Turbocharged, Charge Air-Cooled
Cylinder arrangement	4 Inline
Displacement, L (cu. in.)	4.5 (276)
Bore and stroke, mm (in.)	106 x 127 (4.19 x 5.00)
Compression ratio	19:01
Piston speed, m/min. (ft./min.)	457 (1500)
Main bearings: quantity, type	5, Replaceable Insert
Rated rpm	1800
Max. power at rated rpm, kWm (BHP)	118 (158)
Cylinder head material	Cast Iron
Crankshaft material	Forged Steel
Valve (exhaust) material Intake	Chromium-Silicon Steel
Valve (exhaust) material Exhaust	Stainless Steel
Governor: type, make/model	JDEC Electronic L16 Denso HP3
Frequency regulation, no-load to-full load	Isochronous
Frequency regulation, steady state	±0.25%
Frequency	Fixed
Air cleaner type, all models	Dry

Model: 100REOZJF, continued

Exhaust

Exhaust System

Exhaust Manifold Type	Dry
Exhaust flow at rated kW, m3/min. (cfm)	22.8 (805)
Exhaust temperature at rated kW, dry exhaust, °C (°F)	580 (1076)
Maximum allowable back pressure, kPa (in. Hg)	7.5 (2.2)
Exh. outlet size at eng. hookup, mm (in.)	98 (3.86)

Engine Electrical

Engine Electrical System

Battery charging alternator	12 Volt
Battery charging alternator: Ground (negative/positive)	Negative
Battery charging alternator: Volts (DC)	12
Battery charging alternator: Ampere rating	65
Starter motor rated voltage (DC)	12
Battery, recommended cold cranking amps (CCA): Qty., CCA rating each	One, 640
Battery voltage (DC)	12

Fuel

Fuel System

Fuel type	Diesel
Fuel supply line, min. ID, mm (in.)	11.0 (0.44)
Fuel return line, min. ID, mm (in.)	6.0 (0.25)
Max. lift, fuel pump: type, m (ft.)	Engine-Driven, 1.8 (6.0)
Max. fuel flow, Lph (gph)	74.6 (19.7)
Max. return line restriction, kPa (in. Hg)	20 (5.9)
Fuel prime pump	Manual
Fuel Filter Secondary	2 Microns@ 98% Efficiency
Fuel Filter Primary	30 Microns
Fuel Filter Water Separator	Yes
Recommended fuel	#2 Diesel

Lubrication

Lubrication System

Type	Full Pressure
Oil pan capacity, L (qt.)	14.7 (15.5)
Oil pan capacity with filter, L (qt.)	15.6 (16.5)
Oil filter: quantity, type	1, Cartridge
Oil cooler	Water-Cooled

Model: 100REOZJF, continued

Cooling

Radiator System

Ambient temperature, °C (°F)	50 (122)
Engine jacket water capacity, L (gal.)	8.5 (2.25)
Radiator system capacity, including engine, L (gal.)	20.1 (5.3)
Engine jacket water flow, Lpm (gpm)	182 (48)
Heat rejected to cooling water at rated kW, dry exhaust, kW (Btu/min.)	62 (3544)
Heat rejected to air charge cooler at rated kW, dry exhaust, kW (Btu/min.)	20 (1127)
Water pump type	Centrifugal
Fan diameter, including blades, mm (in.)	600 (23.6)
Fan, kWm (HP)	6.6 (8.8)
Max. restriction of cooling air, intake and discharge side of radiator, kPA (in. H2O)	0.125 (0.5)

* Enclosure with internal silencer reduces ambient temperature capability by 5°C (9°F).

Operation Requirements

Air Requirements

Radiator-cooled cooling air, m3/min. (scfm) *	142 (5000)
Combustion air, m3/min. (cfm)	8.2 (288)
Heat rejected to ambient air: Engine, kW (Btu/min.)	25.0 (1420)
Heat rejected to ambient air: Alternator, kW (Btu/min.)	11.6 (660)

*Air density = 1.20 kg/m3 (0.075 lbm/ft3)

Fuel Consumption

Diesel, Lph (gph), at % load	Rating
Standby Fuel Consumption at 100% load	31.0 Lph (8.2 gph)
Standby Fuel Consumption at 75% load	25.0 Lph (6.6 gph)
Standby Fuel Consumption at 50% load	17.8 Lph (4.7 gph)
Standby Fuel Consumption at 25% load	9.5 Lph (2.5 gph)



The APM603 generator set controller provides advanced control, system monitoring, and system diagnostics for a single generator set or paralleling multiple generator sets. The APM603 interfaces the generator set to other power system equipment and network management systems using standard industry network communications. It uses a patented digital voltage regulator and unique software logic to manage alternator thermal overload protection as well as serves as an overcurrent protective relay, features normally requiring additional hardware. The APM603 controller meets NFPA 110, Level 1.

Display, Interface, and Accessibility

- A 7-inch color TFT touchscreen for easy local access to data.
 - Home screen can be customized to show critical data at a glance.
 - Create a custom favorites list for quick access to important data
- Measurements are selectable in metric or English units.
- Supports Modbus® protocol through serial bus and Ethernet networks, and supports SNMP and BACnet® through Ethernet networks.

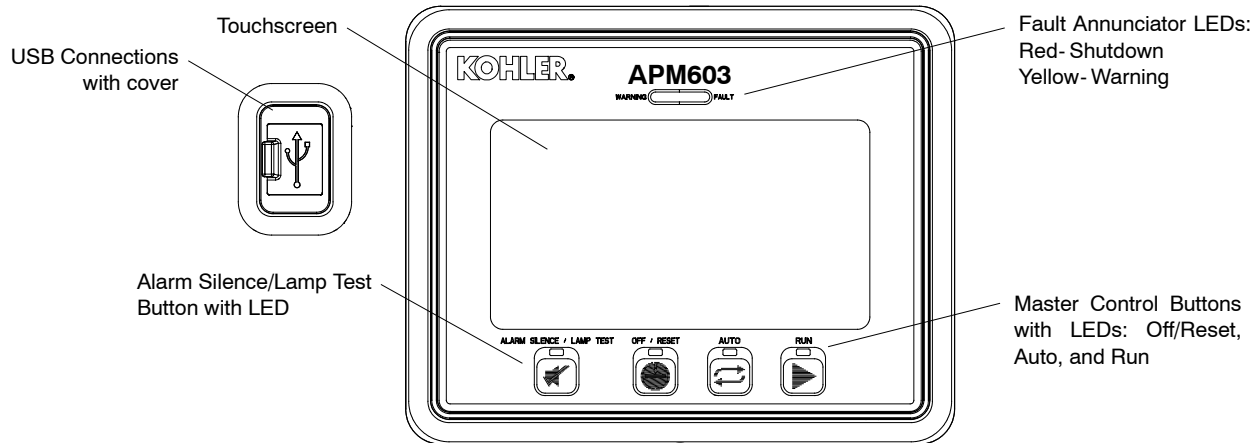
Global Support

- Sales, installation, and service support from more than 800 Kohler and SDMO service providers around the world.

On-board Diagnostics

- Immediate visibility of warnings and faults with text description and code display.
 - 15 seconds of critical data are captured around each warning and fault
 - Critical data can be viewed on the display and downloaded
- Store up to 10,000 events locally along with historical data logging of successful starts.
 - Accurate time stamp from real-time clock
 - Event log can be downloaded
- Data logging of customized parameter list for report generation and advanced troubleshooting.
 - Store to external USB drive for easy transfer to another device

Modbus® is a registered trademark of Schneider Electric.
BACnet® is a registered trademark of ASHRAE.



Controller Features

AC Output Voltage Regulator Adjustment	Maximum of $\pm 10\%$ of the system voltage
Alarm Silence	For NFPA-110 application or user convenience
Alternator Protection	Generator set overload and short circuit protection
Cyclic Cranking	Provides automatic restart after a failed start attempt with programmable on/off time and number of attempts
ECU Diagnostics	Displays engine ECU fault codes and descriptions for engine troubleshooting
Engine Start Aid	Control for an optional engine starting aid
Environmentally Sealed Membrane Keypad	Three master control buttons with LEDs: Off/Reset, Auto, and Run
Patented High-Speed RMS Digital Voltage Regulator	$\pm 0.25\%$ no-load to full-load regulation with three-phase RMS sensing
Lamp Test	Verifies functionality of the indicator LEDs
Real-time Clock	Includes battery back-up to retain date and time through controller power cycle
Remote Reset	Allows remote fault resets and restarting of the generator set
Remote Monitoring Panel	Compatible with the Kohler® Remote Serial Annunciator
Run Time Hourmeter	Displays generator set run time
Run Relay	Indicates that the generator set is running
Time Delay Engine Cooldown (TDEC)	Time delay before the generator set shuts down
Time Delay Engine Start (TDES)	Time delay before the generator set starts

Communication

USB Port	(1) Mini-USB port for PC connection (1) USB port for storage device
Serial (RS-485) Port	(1) Non-isolated for RSA III (1) Isolated for Modbus devices (1) Isolated for paralleling communication
Ethernet Port	(1) RJ45 for Modbus TCP, SNMP, and BACnet

Controller Specifications

Nominal voltage	12 or 24 VDC protected against reverse battery connection
Power	800 mAmps at 12 VDC 400 mAmps at 24 VDC
Operating Temperature	- 40°C to 70°C (- 40°F to 158°F)
Storage Temperature	- 40°C to 85°C (- 40°F to 185°F)
Humidity	5% to 95% non-condensing
Display Size, W x H	154 x 86 mm (6.0 x 3.4 inches)
Protection Index	IP65 Front

Paralleling Features

- Isochronous control with real and reactive load sharing with other APM603 controller equipped generator sets
 - Supports paralleling up to 8 generators
- Random first-on logic to prevent two or more generator sets from closing to a dead bus and provides the fastest response for a single generator online
- Automatic synchronizer with dead bus closing
- Soft loading and unloading for generator management
- Protective relay functions:
 - Synch check (25C)
 - Over current (51)
 - Over frequency (81O)
 - Over power (32O)
 - Over voltage (59)
 - Reverse power (32R)
 - Reverse reactive power (32RQ)
 - Under frequency (81U)
 - Under voltage (27)
- Generator management to allow the start and stop of generators based on load demand or state of other generators
 - Fuel level
 - Run time
 - Manual order
 - Time of day
 - Efficiency
- Simplified paralleling system view from any generator controller in the system

Overcurrent Protective Device

- Provides protection against line-to-line and line-to-neutral faults
- Uses thermal and instantaneous current limit settings for alternator protection
- Includes a maintenance mode for arc flash reduction per NEC 240.87

Load Management Features

- Programmable outputs included to command the connect and disconnect of loads based on generator or paralleling system state
 - Loads connected based on available capacity
 - Loads disconnected at system startup
 - Loads disconnected based on a maximum kW setting or underfrequency setting
- Supports up to 16 prioritized load steps per system
 - Can be used on a single generator system
 - Can be combined in a paralleling system for a total system load control capability
- Simplified load management system view from any generator controller in the system
- Requires input/output module option

Advanced Programmable I/O

- Configurable inputs and outputs can be programmed for customer specific use
- PLC-like capability for applying logic to customize generator system behavior

Troubleshooting Features

- 15 seconds of key data automatically captured around each warning and shutdown
 - Data can be exported for detailed analysis
 - Data can be viewed on controller for convenient on-site troubleshooting support
- Configurable data logger will allow you to select parameters to monitor
 - Data stored to USB device for flexibility on amount of data stored and ability to export for detailed analysis
 - Data capture controlled by user to allow capturing specific data required

NFPA 110 Requirements

In order to meet NFPA 110, Level 1 requirements, the generator set controller monitors the engine/generator functions/faults shown below.

- Engine functions:
 - Overcrank
 - Low coolant temperature warning
 - High coolant temperature warning
 - High coolant temperature shutdown
 - Low oil pressure shutdown
 - Low oil pressure warning
 - High engine speed
 - Low fuel (level or pressure) *
 - Low coolant level
 - EPS supplying load
 - High battery voltage
 - Low battery voltage
- General functions:
 - Master switch not in auto
 - Battery charger fault *
 - Lamp test
 - Contacts for local and remote common alarm
 - Audible alarm silence button
 - Remote emergency stop *

* Function requires optional input sensors or kits and is engine dependent, see Engine Data.

Standards

The generator set controller has been tested and verified for compliance with the following standards.

- NFPA 99
- NFPA 110, Level 1
- CSA 282-09
- UL 6200
- ASTM B117 (salt spray test)

Controller Functions

The controller displays warning, shutdown, and status messages. **All functions are available as relay outputs.**

Warning causes the yellow fault LED to show and sounds the alarm horn, signaling an impending problem.

Shutdown causes the red fault LED to show, sounds the alarm horn, and stops the generator set.

The controller communicates with the engine ECU and supports a large number of warning and shutdown events that are not listed here. This table highlights the items required for NFPA 110.

Event	Warning	Shutdown
Alternator Thermal Protection †		●
Battery Charger Fault *	▲	
CAN Option Board1 Comm Loss	▲	
Critically Low Fuel Level (diesel) *	▲	
ECU Diagnostic Event	▲	
ECU Mismatch Shutdown †		●
Fuel Leak Alarm (diesel) *	▲	
High Battery Voltage Warning	▲	
High Coolant Temperature Shutdown †		●
High Coolant Temperature Warning	▲	
High Fuel Level Warning (diesel) *	▲	
High Oil Temperature Shutdown †		●
High Oil Temperature Warning	▲	
Local Emergency Stop Shutdown †		●
Loss ECU Comms Shutdown †		●
Loss of Signal Low Coolant Level Voltage	▲	
Low Battery Voltage Warning	▲	
Low Coolant Level Shutdown		●
Low Coolant Temperature Warning	▲	
Low Fuel Level Shutdown (diesel) * †		●
Low Fuel Level Warning (diesel) * †	▲	
Low Fuel Pressure Warning (gas) * †	▲	
Low Oil Pressure Shutdown †		●
Low Oil Pressure Warning	▲	
Low RTC (clock) Battery Voltage	▲	
Maintenance Reminder1	▲	
Maintenance Reminder2	▲	
Maintenance Reminder3	▲	
Maximum Power Shutdown †		●
Maximum Power Warning	▲	
Not In Auto Alarm	▲	
Over Crank Shutdown †		●
Over Current Shutdown (L1, L2, L3) †		●
Over Current Warning (L1, L2, L3)	▲	
Over Frequency Shutdown †		●
Over Frequency Warning	▲	
Over Power Shutdown †		●
Over Power Warning	▲	
Over Speed Shutdown †		●
Over Voltage Shutdown (L- L, L- N, each phase)		●
Over Voltage Warning (L- L, L- N, each phase)	▲	

Event	Warning	Shutdown
Remote Emergency Stop Shutdown †		●
Reverse Power Shutdown †		●
Reverse VAR Shutdown †		●
Under Frequency Shutdown †		●
Under Frequency Warning	▲	
Under Voltage Shutdown (L- L, L- N, each phase) †		●
Under Voltage Warning (L- L, L- N, each phase)	▲	
Weak Cranking Battery	▲	
Status Messages		
Auto Button Pressed		
EPS Supplying Load		
Generator Running		
Generator Started		
Generator Stopped		
GFCI Warning *		
Load Shed Overload		
Load Shed Under Frequency		
Off Button Pressed		
RSA Event Programmable Digital Inputs, 1- 8		
Run Button Pressed		
* Function requires optional input sensors or kits		
† Items included with common fault shutdown		

John Deere Engine-Powered Models

Inputs and Outputs

Standard Dedicated User Inputs	Input Type
Auxiliary Fault (Shutdown)	Digital Input
Auxiliary Warning	
Battery Charger Fault	
Breaker Closed *	
Breaker Open *	
Excitation Over Voltage (350 kW and up)	
Fuel Leak Alarm	
Low Fuel Level Switch	
Remote Emergency Stop	
Remote Engine Start	Two-wire input
Speed Bias	Analog Voltage Input, Scalable up to +/- 10 VDC
Voltage Bias	

Standard Dedicated User Outputs	Output Type
Close Breaker *	Relay Driver Output
Common Failure	
Run	
Trip Breaker / Shunt Trip *	
* Only with remote-mounted electrically operated circuit breakers.	

Optional Configurable User Inputs and Outputs	
User Configurable Inputs	2 Analog, 0-5 VDC 4 Dry Contact Digital
User Configurable Relay Outputs	14 NO/NC Relays 1 Common Fault Relay
Note: Programmable I/O is configurable by a Kohler-authorized technician	

JD Engine Data

The following John Deere engine data is displayed on the APM603 controller.

Parameter
Engine Model Number
Engine Serial Number
ECU Serial Number
Coolant Temperature
Engine Speed
Fuel Pressure
Fuel Consumption Rate
Oil Pressure
Run Time Hours

KD Models

Inputs and Outputs

Standard Dedicated User Inputs	Input Type
Auxiliary Fault (Shutdown)	Digital Input
Auxiliary Warning	
Battery Charger Fault	
Breaker Closed *	
Breaker Tripped/ Open *	
Fuel Leak Alarm	
Fuel Level	
Idle Switch	
Key Switch Enable	
Low Fuel Level Switch	
Low Oil Level	
Remote Emergency Stop	
Remote Reset	
Remote Engine Start	Two-wire input
Speed Bias	Analog Voltage Input, Scalable up to +/- 10 VDC
Voltage Bias	

Standard Dedicated User Outputs	Output Type
Close Breaker *	Relay Driver Output
Common Failure	
Common Warning	
EPS Supplying Load	
Generator Running	
Horn	
Low Coolant Temperature	
Not in Auto	
System Ready	
Trip Breaker / Shunt Trip *	
* Only with remote-mounted electrically operated circuit breakers.	

Optional Configurable User Inputs and Outputs	
User Configurable Inputs	16 Dry Contact Digital
User Configurable Relay Outputs	8 NO/NC Relays
Note: Programmable I/O is configurable by a Kohler-authorized technician.	

KD Engine Data

The following Kohler Diesel engine data is displayed on the APM603 controller.

Parameter
Engine Model Number
Engine Serial Number
Ambient Temperature
Charge Air Pressure
Charge Air Temperature
Common Rail Fuel Pressure
Coolant Level
Coolant Temperature
Crankcase Pressure
Engine Speed
Fuel Consumption Rate
Fuel Pressure
Fuel Temperature
Intercooler Coolant Temperature (K175 engines only)
Oil Temperature
Oil Pressure
Run Time Hours

Doosan Engine-Powered Models Inputs and Outputs

Standard Dedicated User Inputs	Input Type
Auxiliary Fault (Shutdown)	Digital Input
Auxiliary Warning	
Battery Charger Fault	
Breaker Closed *	
Breaker Tripped/Open *	
Emergency Stop, Local	
Emergency Stop, Remote	
Excitation Over Voltage	
Ground Fault Relay	
Fuel Type	
Low Fuel Pressure	
Remote Engine Start	Two-wire input
Speed Bias	Analog Voltage Input, Scalable up to +/- 10 VDC
Voltage Bias	

Standard Dedicated User Outputs	Output Type
Close Breaker *	Relay Driver Output
Common Failure	
Common Warning	
Crank	
High Coolant Temperature	
Horn	
Run	
Trip Breaker / Shunt Trip *	
* Only with remote-mounted electrically operated circuit breakers.	

Optional Configurable User Inputs and Outputs	
User Configurable Inputs	2 Analog, 0-5 VDC 4 Dry Contact Digital
User Configurable Relay Outputs	14 NO/NC Relays 1 Common Fault Relay
Note: Programmable I/O is configurable by a Kohler-authorized technician	

Doosan Engine Data

The following Doosan engine data is displayed on the APM603 controller.

Parameter
Ambient Temperature
Coolant Temperature
ECU Runtime Hours
Engine Intake Air Pressure
Engine Intake Air Temperature
Engine Intercooler Temperature
Engine Speed
Fuel Pressure
Mechanical Engine Load
Oil Pressure
Oil Temperature

APM603 Available Options

- **Common Failure Relay** provides a relay output to signal a generator set fault.
- **Battery Charger** available with 6 amp, 10 amp, and 20 amp output for 12 and 24V DC voltage output. (Availability is generator model dependent.) The 10 amp and 20 amp models provide NFPA 110 charging and alarming capability.
- **Electrically Operated Circuit Breakers**
 - For paralleling systems
 - Available generator-mounted or remote-mounted
 - 24VDC
- **Ground Fault Relay** provides a relay output to signal a ground fault is detected.
- **Input/Output Module** for John Deere and Doosan powered models provides:
 - 2 analog inputs (0-5 VDC)
 - 4 digital input connections with connection to ground
 - 14 relay output connections (Form C, rated 10A, 120V)
 - 1 common fault relay output (NO, rated 2A, 24VDC)
- **Input/Output Module** for Kohler Diesel (KD) models provides:
 - 16 digital input connections with connection to ground
 - 8 relay output connections (Form C, rated 8A, 240 VAC or rated 0.5 A, 48 VDC)
- **Key Switch** to allow selection of RUN, OFF and AUTO modes. Lockable in the AUTO position by removing the key.
- **Remote Emergency Stop Switch** available as a wall mounted panel to remotely shut down the generator set.
- **Remote Monitoring Panel.** The Kohler® Remote Serial Annunciator (RSA) enables the operator to monitor the status of the generator set from a remote location, which may be required for NFPA 99 and NFPA 110 installations, and up to four Automatic transfer switches.
- **Shunt Trip Wiring** provides relay outputs to trip a shunt trip circuit breaker and to signal the common fault shutdowns. Contacts rated at 10 amps at 28 VDC or 120 VAC.

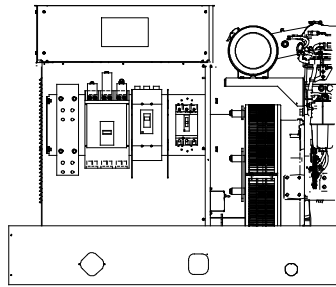
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Availability is subject to change without notice. Kohler Co. reserves the right to change the design or specifications without notice and without any obligation or liability whatsoever. Contact your local Kohler® generator set distributor for availability.

Standard Features

- The line circuit breaker interrupts the generator set output during a short circuit and protects the wiring when an overload occurs. Use the circuit breaker to manually disconnect the generator set from the load during generator set service.
- Circuit breaker kits are mounted to the generator set and are provided with load-side lugs and neutral bus bar.
- Kohler Co. offers a wide selection of molded-case line circuit breaker kits including single, dual, and multiple configurations for each generator set.
- Four types of line circuit breakers are available: (see page 2 for definitions and pages 3 and 4 for application details)
 - Magnetic trip
 - Thermal magnetic trip
 - Electronic trip
 - Electronic with ground fault (LSIG) trip
- In addition, line circuit breakers are offered with 80% and 100% ratings.
- Single line circuit breaker kits allow circuit protection of the entire electrical system load.
- Dual line circuit breaker kits allow circuit protection of selected priority loads from the remaining electrical system load.
- Multiple line circuit breaker kits with field connection barrier allow circuit protection for special applications (350- 2500 kW models and selected 80- 300 kW models).
- Up to four line circuit breakers can be used on 350- 2500 kW models.
- Line circuit breakers comply with the following codes and standards unless otherwise stated.
 - UL 489 Molded Case Circuit Breakers
 - UL 1077 Supplementary Protectors
 - UL 2200 Stationary Engine Generator Assemblies



**Multiple Circuit Breaker Kit with Neutral Bus Bar
180-300 kW Model Shown**

Line Circuit Breaker Types

Magnetic Trip

The magnetic trip features an electromagnet in series with the load contacts and a moveable armature to activate the trip mechanism. When a sudden and excessive current such as a short circuit occurs, the electromagnet attracts the armature resulting in an instantaneous trip.

Thermal Magnetic Trip

Thermal magnetic trip contains a thermal portion with a bimetallic strip that reacts to the heat produced from the load current. Excessive current causes it to bend sufficiently to trip the mechanism. The trip delay is dependent on the duration and excess of the overload current. Elements are factory-calibrated. A combination of both thermal and magnetic features allows a delayed trip on an overload and an instantaneous trip on a short circuit condition.

Electronic Trip

These line circuit breakers use electronic controls and miniature current transformers to monitor electrical currents and trip when preset limits are exceeded.

LI breakers are a combination of adjustable trip functions including long-time ampere rating, long-time delay, and instantaneous pickup. LSI breakers have all of the LI breaker features plus short-time pickup, short-time delay, and defeatable instantaneous pickup. LSIg breakers have all of the LSI breaker features plus ground-fault pickup and delay.

Electronic with Ground Fault Trip

The ground fault trip feature is referred to as LSIg in this document. Models with LSIg compare current flow in phase and neutral lines, and trip when current unbalance exists.

Ground fault trip units are an integral part of the circuit breaker and are not available as field-installable kits. The ground fault pickup switch sets the current level at which the circuit breaker will trip after the ground fault delay. Ground fault pickup values are based on circuit breaker sensor plug only and not on the rating plug multiplier. Changing the rating plug multiplier has no effect on the ground fault pickup values.

80% Rated Circuit Breaker

Most molded-case circuit breakers are 80% rated devices. An 80% rated circuit breaker can only be applied at 80% of its rating for continuous loads as defined by NFPA 70. Circuit conductors used with 80% rated circuit breakers are required to be rated for 100% of the circuit breaker's rating.

The 80% rated circuit breakers are typically at a lower cost than the 100% rated circuit breaker but load growth is limited.

100% Rated Circuit Breaker

Applications where all UL and NEC restrictions are met can use 100% rated circuit breakers where 100% rated circuits can carry 100% of the circuit breaker and conductor current rating.

The 100% rated circuit breakers are typically at a higher cost than the 80% rated circuit breaker but have load growth possibilities.

When applying 100% rated circuit breakers, comply with the various restrictions including UL Standard 489 and NEC Section 210. If any of the 100% rated circuit breaker restrictions are not met, the circuit breaker becomes an 80% rated circuit breaker.

Line Circuit Breaker Options

☐ Alarm Switch

The alarm switch indicates that the circuit breaker is in a tripped position caused by an overload, short circuit, ground fault, the operation of the shunt trip, an undervoltage trip, or the push-to-trip pushbutton. The alarm resets when the circuit breaker is reset.

☐ Auxiliary Contacts

These switches send a signal indicating whether the main circuit breaker contacts are in the open or closed position.

☐ Breaker Separators (350- 2500 kW)

Provides adequate clearance between breaker circuits.

☐ Bus Bars

Bus bar kits offer a convenient way to connect load leads to the generator set when a circuit breaker is not present.

15- 300 kW. Bus bar kits are available on alternators with leads for connection to the generator set when circuit breakers are not ordered.

350- 2500 kW. A bus bar kit is provided when no circuit breaker is ordered. Bus bars are also available in combination with circuit breakers or other bus bars on the opposite side of the junction box. On medium voltage (3.3 kV and above) units, a bus bar kit is standard (not applicable to KD models).

☐ Field Connection Barrier

Provides installer wiring isolation from factory connections.

☐ Ground Fault Annunciation

A relay contact for customer connection indicates a ground fault condition and is part of a ground fault alarm.

☐ Lockout Device (padlock attachment)

This field-installable handle padlock attachment is available for manually operated circuit breakers. The attachment can accommodate three padlocks and will lock the circuit breaker in the OFF position only.

☐ Lugs

Various lug sizes are available to accommodate multiple cable sizes for connection to the neutral or bus bar.

☐ Overcurrent Trip Switch

The overcurrent trip switch indicates that the circuit breaker has tripped due to overload, ground fault, or short circuit and returns to the deenergized state when the circuit breaker is reset.

☒ Shunt Trip, 12 VDC or 24 VDC

A shunt trip option provides a solenoid within the circuit breaker case that, when momentarily energized from a remote source, activates the trip mechanism. This feature allows the circuit breaker to be tripped by customer-selected faults such as alternator overload or overspeed. The circuit breaker must be reset locally after being tripped. Tripping has priority over manual or motor operator closing.

☒ Shunt Trip Wiring

Connects the shunt trip to the generator set controller. (standard on KD models with the APM802 controller)

☐ Undervoltage Trip, 12 VDC or 24 VDC

The undervoltage trips the circuit breaker when the control voltage drops below the preset threshold of 35%- 70% of the rated voltage.

15- 300 kW Line Circuit Breaker Specifications

100% Rating Circuit Breaker

Alt. Model	Ampere Range	Trip Type	C. B. Frame Size
4D/4E	15- 150	Thermal magnetic	HD
	60- 150	Electronic LI	
		Electronic LSI	
		Electronic LSIG	
	60- 150	Electronic LI	HG
		Electronic LSI	
		Electronic LSIG	
4P/4PX 4Q/4QX	15- 150	Thermal magnetic	HD
	60- 150	Electronic LI	
		Electronic LSI	
		Electronic LSIG	
	60- 150	Electronic LI	HG
		Electronic LSI	
		Electronic LSIG	
	175- 250	Thermal magnetic	JD
	250	Electronic LI	JD
		Electronic LSI	
		Electronic LSIG	
	250	Electronic LI	JG
		Electronic LSI	
		Electronic LSIG	
4RX 4S/4SX 4TX 4V 4UA 4M6226	15- 150	Thermal magnetic	HD
	60- 150	Electronic LI	
		Electronic LSI	
		Electronic LSIG	
	60- 150	Electronic LI	HG
		Electronic LSI	
		Electronic LSIG	
	175- 250	Thermal magnetic	JD
	250	Electronic LI	
		Electronic LSI	
		Electronic LSIG	
	250	Electronic LI	JG
		Electronic LSI	
		Electronic LSIG	
	400	Electronic LI	LG
		Electronic LSI	
		Electronic LSIG	
4UA 4M6226	1000- 1200	Electronic LSI	PG
	1200	Electronic LSIG	
		Electronic LSI	PJ
		Electronic LSIG	

100% Rating Electrically Operated Breakers

For use as paralleling breakers with the APM603 controller.

Generator-Mounted P-Frame, 24VDC Electrically Operated			
Alt. Model	Amps	Trip Unit	Frame
4RX 4S/4SX 4TX 4V	250	3.0 LI	PJ
	400	5.0 LSI	PJ
	600	3.0 LI	PL
	800	5.0 LSI	PL
4UA 4M6226	250	3.0 LI	PJ
	400	5.0 LSI	PJ
	600	3.0 LI	PL
	800	5.0 LSI	PL
	1000	3.0 LI	PL
	1200	5.0 LSI	PL

All circuit breakers listed in this table include line side bus and load side lugs, 24VDC motor operators, 2 type C auxiliary contacts, and 1 type C SDE overcurrent switch contact. No second breakers are allowed in combination with these breakers.

Interrupting Ratings

Circuit Breaker Frame Size	240 Volt, kA	480 Volt, kA	600 Volt, kA
HD	25	18	14
HG	65	35	18
HJ	100	65	25
JD	25	18	14
JG	65	35	18
JJ	100	65	25
LA	42	30	22
LG	65	35	18
MG			
PG			
PJ	100	65	25
PL	125	100	25

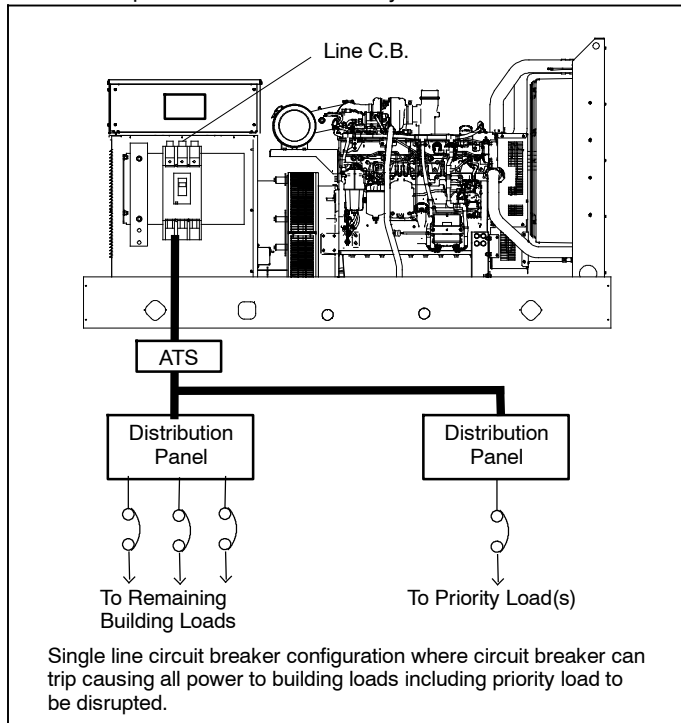
Circuit Breaker Lugs Per Phase (Al/Cu)

Frame Size	Ampere Range	Wire Range
E (480 V max.)	30- 100	Up to two wire terminals fitting 10-32 or 1/4-20 stud
H	15- 150	One #14 to 3/0
J	175	One 1/0 to 4/0
	200- 250	One 3/0 to 350 kcmil
LA	300- 400	One #1 to 600 kcmil or Two #1 to 250 kcmil
LG	400- 600	Two 2/0 to 500 kcmil AL/CU
M	700-800	Three 3/0 to 500 kcmil
P	600-800	Three 3/0 to 500 kcmil
	1000-1200	Four 3/0 to 500 kcmil
Mechanical Load Lugs Included with H, J, and LG LSIG Neutrals		
H	60- 150	One #14 to 3/0 AL/CU
J	250	One 3/0 to 350 kcmil AL/CU
LG	400- 600	Two 4/0 to 500 kcmil AL/CU

15- 300 kW Line Circuit Breaker Applications

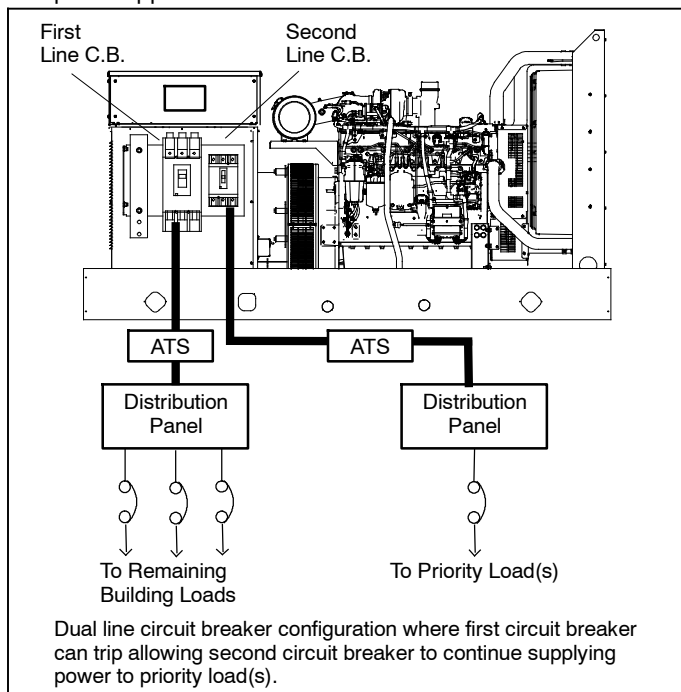
Single Circuit Breaker Installations

A generator set with a single circuit breaker installed typically feeds a single transfer switch and then a distribution panel. This allows protection of the entire system.



Multiple Circuit Breaker Installations

A generator set with dual circuit breakers installed is used to separate critical loads. Typically, one circuit breaker will feed a main transfer switch with noncritical loads and the other circuit breaker will feed a second transfer switch that feeds critical or priority loads. Multiple circuit breakers allow circuit protection for special applications.



Circuit Breaker Combinations

Alternator Model	First C. B. Frame	Second C. B. Frame	Third C. B. Frame	Trip Type
ALL except 4D/4E	H	—	—	All
	J	—	—	
	LA	—	—	
	LG	—	—	
4D/4E	H	—	—	Standard or LSIG
	H	H	—	No LSIG
4P/4PX 4Q/4QX	H	H or J	—	No LSIG
	J		—	
	LA		—	
	LG	H, J or LG	—	
4RX 4S/4SX 4TX 4V	M	—	—	All
	P	—	—	All
	H or J	H or J	—	No LSIG
	LA	H, J, or LA	—	
	LG	H, J, LA, or LG	—	
	M			
	P			
	H or J	H or J	H or J	
4UA 4M6226	M or P	—	—	All
	H or J	H or J	—	All
	LA	H, J, or LA	—	
	LG	H, J, LA, or LG	—	
	M or P	H, J, LA, or LG	—	
	P	P	—	
	H or J	H or J	H or J	
	LA	H or J	H or J	
		LA	H, J, or LA	
	LG	H or J	H or J	
		LA	H, J, or LA	
		LG	H, J, LA, or LG	
	M or P	H or J	H or J	
		LA	H, J, or LA	
		LG	H, J, or LG	

MULTIPLES OF RATED CURRENT

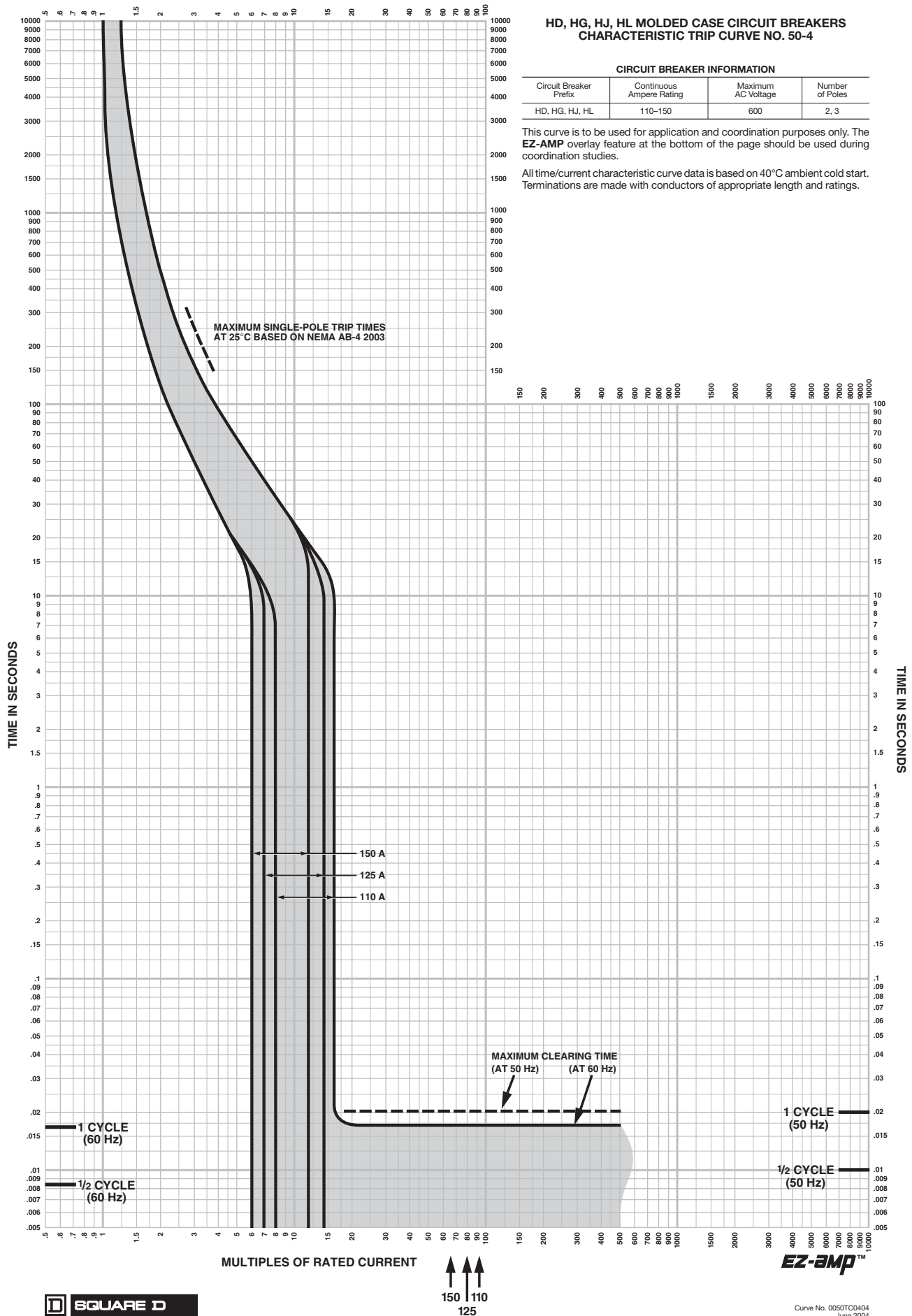
HD, HG, HJ, HL MOLDED CASE CIRCUIT BREAKERS CHARACTERISTIC TRIP CURVE NO. 50-4

CIRCUIT BREAKER INFORMATION

Circuit Breaker Prefix	Continuous Ampere Rating	Maximum AC Voltage	Number of Poles
HD, HG, HJ, HL	110-150	600	2, 3

This curve is to be used for application and coordination purposes only. The **EZ-AMP** overlay feature at the bottom of the page should be used during coordination studies.

All time/current characteristic curve data is based on 40°C ambient cold start. Terminations are made with conductors of appropriate length and ratings.



MULTIPLES OF RATED CURRENT

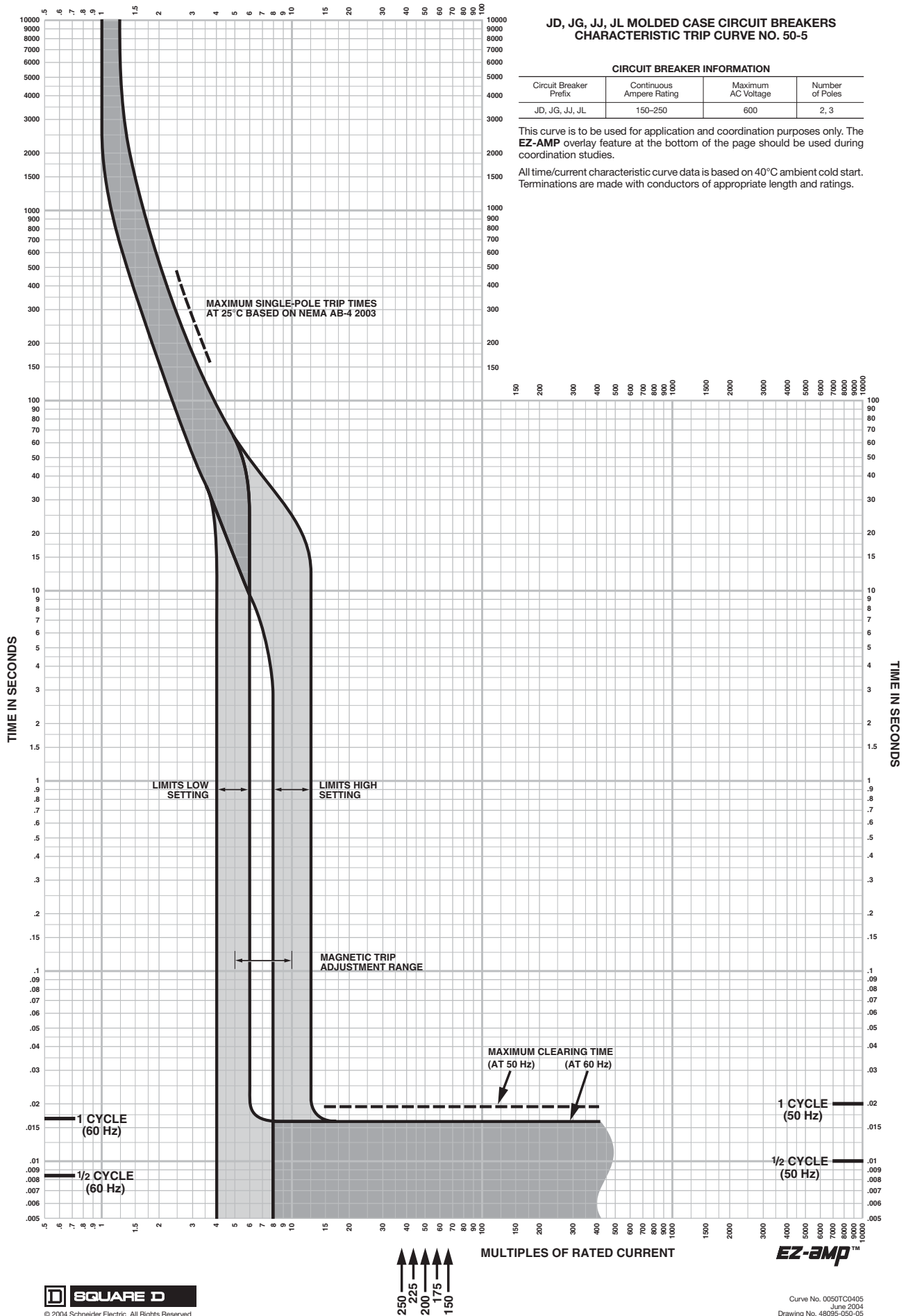
JD, JG, JJ, JL MOLDED CASE CIRCUIT BREAKERS CHARACTERISTIC TRIP CURVE NO. 50-5

CIRCUIT BREAKER INFORMATION

Circuit Breaker Prefix	Continuous Ampere Rating	Maximum AC Voltage	Number of Poles
JD, JG, JJ, JL	150-250	600	2, 3

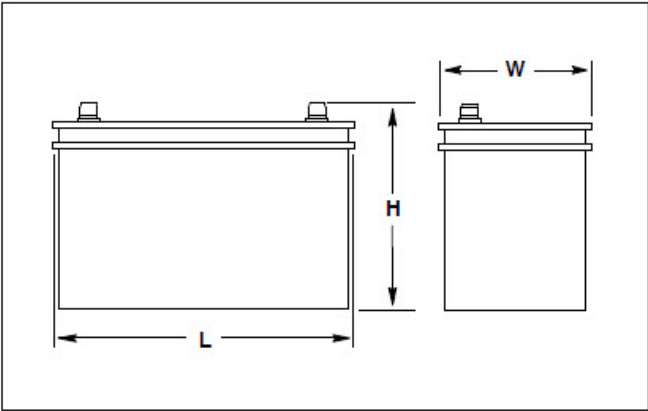
This curve is to be used for application and coordination purposes only. The **EZ-AMP** overlay feature at the bottom of the page should be used during coordination studies.

All time/current characteristic curve data is based on 40°C ambient cold start. Terminations are made with conductors of appropriate length and ratings.





Typical Overall Dimensions



Standard Features

- Kohler Co. selects batteries to meet the engine manufacturer's specifications and to comply with NFPA requirements for engine-cranking cycles.
- Heavy-duty starting batteries are the most cost-effective means of engine cranking and provide excellent reliability in generator set applications.
- Tough polypropylene cases protect against life-shortening vibration and impact damage.
- Batteries are rated according to SAE standard J-537.
- All batteries are 12-volts. Kits that contain two or four batteries are available for 24-volt systems and/or systems with redundant starters.
- Wet- and dry-charged batteries have lead-calcium or lead-antimony plates and use sulfuric acid electrolyte. Removable cell covers allow checking of electrolyte specific gravity.
- Absorbant glass mat (AGM) batteries are sealed and maintenance free.
- Batteries are for applications below and above 0°C (32°F).

Charge Type*	Battery Part Number	Battery Qty. per Size	BCI Group Size	Battery SAE Dimension, mm (in.)			Cold Cranking Amps at 18°C (0°F) Min.	Reserve Capacity Minutes at 27° (80°F) Min.	Battery Post Layout and Style
				L	W	H			
Wet	256984	1	24	273.0 (10.8)	173.0 (6.8)	228.6 (9.0)	650	130	D/1

Battery Specifications

Battery Post Layouts (A/C/D) and Styles (1/3)

A

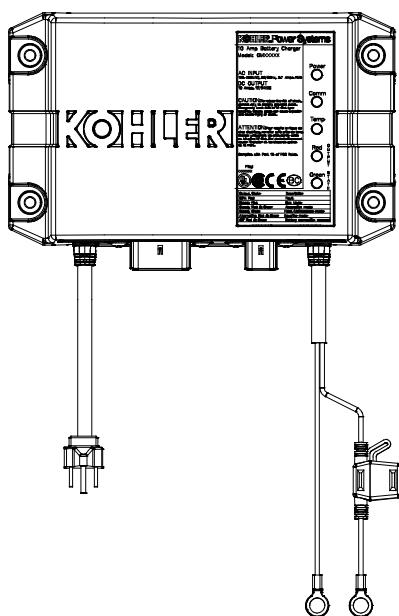
C

D

1

3

Notes: Dimensions are in mm; 25.4 mm equals 1 inch. BCI group numbers shown in *italics*.
Order stud kit 254427 to convert from Style 3 to Style 1.
Battery post layout letters and style numbers match drawing 244578 format.



The battery charger is a fully-automatic, high efficiency battery charger that charges batteries rapidly and safely. The battery charger is designed for an industrial environment.

The battery charger is designed for operation with an engine cranking battery.

The battery charger is universal voltage input capable, comes with a standard 120 V/60 Hz AC plug, and charges 12 VDC or 24 VDC battery systems.

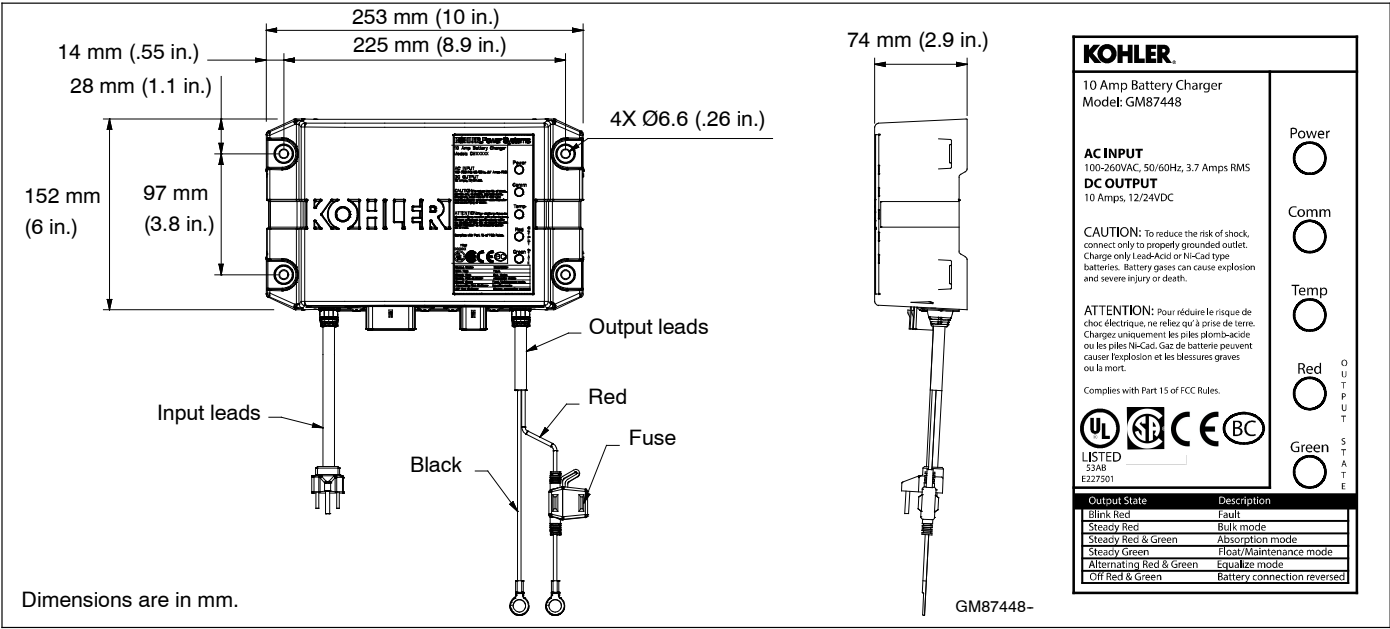
Five LED lights indicate power, communication status, temperature compensation status, charge curve, and charger status.

With the optional battery temperature sensor connected, the battery charger can adjust output voltages for optimal charging.

Standard Features

- 12 or 24 VDC output
 - Automatic voltage detection
- Automatic multi-stage charging modes
 - Recovery charge
 - Bulk charge
 - Absorption charge
 - Float charge
 - Equalize charge
- Charges the following type batteries:
 - Flooded lead acid (FLA)
 - AGM
 - Gel cell
 - High performance AGM
 - Nickel-cadmium (NiCad)
- 5 LED status indicators
- Durable potted assembly for waterproofing and vibration resistance
- Reverse-polarity protection
- Short-circuit protection
- Electronically limited output current
- Optional temperature compensation (FLA only)
- User adjustable parameters to support optimal manufacturer recommended charge curve.
- Code compliance:
 - UL 1236 Listed
 - NFPA 110, Level 1 compatible (when used with Kohler controller and connected to engine harness)
 - CSA - C22.2 No. 107.2-01
 - FCC - Title 47, Part 15 Class A
 - CE
 - IBC 2015
 - OSHPD

DC Output		AC Input		Overall Dimensions W x D x H	Shipping Weight	
Volts (Nominal)	Amps	Volts (Nominal)	Amps		kgs	lbs
12/24	10	100-260	3.7	253 mm x 152 mm x 74 mm (10.0 in x 6.0 in x 2.9 in)	3.6	7.9



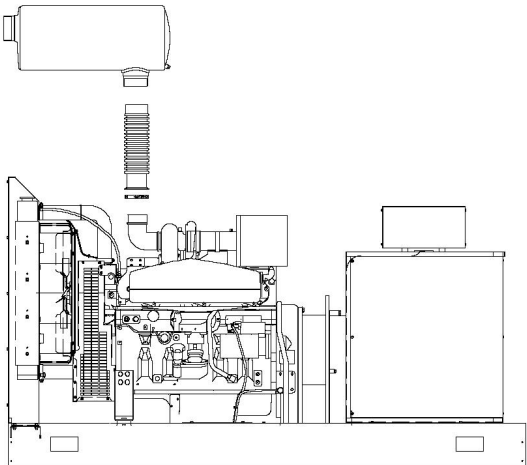
Specifications

AC Input	100–260 VAC
Frequency Input	50/60 Hz
DC Output	10 Amps @ 12 VDC or 10 Amps @ 24 VDC (On battery voltage regulation ±1%; current is electronically limited
Fuse Protection	15 amps ATC
Battery Types	Flooded Lead Acid (FLA) AGM Gel Cell High Performance AGM Nickel–Cadmium (NiCad)
Monitoring LED Indications	Power Communication Temperature compensation Output charger curve and charger status: - Red - Green
Environmental Operating Storage Relative Humidity Salt Spray Testing Corrosion Resistant	–20° to 70°C (–4° to 158° F) –40° to 85°C (–40° to 185° F) 5 to 95% (non-condensing) ASTM B117 From battery gases

Enclosure Environmental Resistant	From rain, snow, dust, and dripping water
Battery Connections Lead Length Battery Connections	1.8 m (6 ft.) red and black leads 9.5 mm (3/8 in.) ring terminals
AC Power Connections Lead Length Storage	1.8 m (6 ft.) Standard US style 3-prong AC plug
Available Options Temperature compensation	

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Kohler Co. offers silencers rated for hospital, critical, residential, and industrial applications and stainless steel flexible exhaust connectors for unhooded generator sets. The accessories described in this spec sheet apply to open units unless otherwise stated. These silencers withstand exhaust flow pressure and temperature without creating excessive engine backpressure.

Silencer Type	Noise Reduction dB(A)	Ambient Noise Level	Silencing Requirement
Critical	25-35	Low	High

Backpressure Considerations

Consult the generator set spec sheet to determine the maximum allowable exhaust backpressure. Add the backpressure produced by the silencer and exhaust piping together to determine the total exhaust system backpressure. Contact an authorized distributor/dealer to calculate backpressure in exhaust piping. Do not exceed the maximum allowable backpressure.

Legend

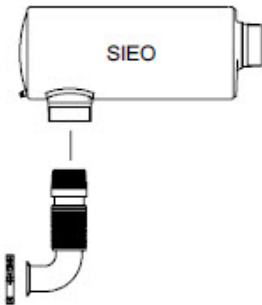
Use the information below for identifying the exhaust system components shown in the illustrations on the following pages.

Side Inlet End Outlet (SIEO)

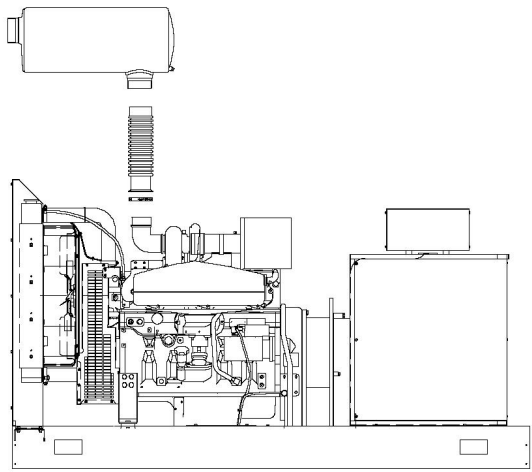

End Inlet End Outlet (EIEO)


Flex Exhaust


Flex Exhaust (wye)

Exhaust System Kits	Qty. in Kit	Size, In/Out	Note
Critical Silencer SIEO	-	4 NPT	



Use the illustrations on the following pages to select the available silencer type and exhaust system configuration best suited for your application. Refer to your authorized distributor/dealer for silencer and component kit numbers and for silencer dimensions.

Backpressure Considerations

Consult the generator set spec sheet to determine the maximum allowable exhaust backpressure. Add the backpressure produced by the silencer and exhaust piping together to determine the total exhaust system backpressure. Contact an authorized distributor/dealer to calculate backpressure in exhaust piping. Do not exceed the maximum allowable backpressure.

Legend

Use the information below for identifying the exhaust system components shown in the illustrations on the following pages.

Side Inlet End Outlet (SIEO)



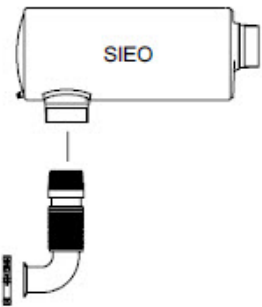
End Inlet End Outlet (EIEO)



Flex Exhaust



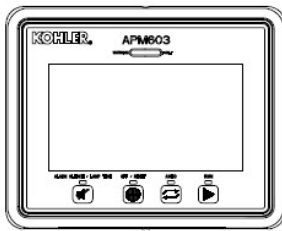
Flex Exhaust (wye)



Exhaust System Kits	Qty. in Kit	Size, In/Out	Note
Flex Exhaust 4NPT	-	-	Kit includes the exhaust clamp.



Integral Voltage Regulator with Kohler® APM603 Controllers and Menu-Driven Selections (80-3250 kW Generator Set Models)



**APM603 Controller
with Integral Voltage Regulator**

Voltage Regulators

The following information provides general features, specifications, and functions of available voltage regulators.

This information generally applies to a single generator set and multiple generator sets with paralleling applications. Refer to the respective generator set specification sheet and see your authorized distributor for information regarding specific voltage regulator applications and availability.

The voltage regulator is integral to the controller and uses patented high speed digital voltage regulator design providing $\pm 0.25\%$ no-load to full-load regulation using root-mean-square (RMS) voltage sensing.

Integral Voltage Regulators with APM603

Calibration	Range Settings	Default Selection
Voltage Adjustment	$\pm 10\%$ of System Voltage	System Voltage
Controller Gain	40 to 70 Hz	P: 1.3 I: 1.0 D: 0.25
Underfrequency Unload or Frequency Setpoint	40 to 70 Hz	0.5 Hz Below System Frequency (ECM)
Underfrequency Unload Scope	0-10% of Rated Voltage (Volts per Cycle)	15 Volts per Cycle at 480 Volts (3.1%)
Reactive Droop	0-10% of System Voltage	4% of System Voltage
VAR Control	-50% to 110%	0 kVAR
PF Adjust Control	-0.50 to 1.0 to 0.50	0.8 Lagging
VAR/PF Gain Adjustment	P: 0.3 to 3.00 I: 0.3 to 3.00 D: 0.3 to 3.00	P: 1.0 I: 1.0 D: 0.25



Specification/Feature	Integral with APM603
Generator Set Availability	80-500 kW
Type	Patented Hybrid Design
Status and Shutdown Indicators	LEDs and Text LCD Display
Operating Temperature	-40°C to 70°C (-40°F to 158°F)
Storage Temperature	-40°C to 85°C (-40°F to 185°F)
Humidity	5-95% Non-Condensing
Circuit Protection	Solid-State, Redundant Software and Fuses
Sensing, Nominal	100-600 Volts (L-L), 50-60 Hz
Sensing Mode	RMS, Single- or 3-Phase
Input Requirements	8-36 VDC
Continuous Output	5.0 ADC with GM88453 Activator Board
Maximum Output	7.8 ADC with GM88453 Activator Board
Transition Frequency	50-70 Hz
Exciter Field Resistance	4-30 Ohms with GM88453 Activator Board
No-Load to Full-Load Voltage Regulation	±0.25%
Thermal Drift	<0.5% (-40°C to 70°C) [-40°F to 158°F] Range
Response Time	3-phase: 1 mS 1-phase: 5 mS
System Voltage Adjust.	±10%
Voltage Adjustment	Controller Display
Remote Voltage Adjustment	Analog 0-5 VDC (±10%) Input Optional
Paralleling Capability	Full Load Share and Control plus Reactive Droop
VAR/PF Control Input	VAR Control Mode, PF Control Mode, System VAR Control, System PF Control

Integral Voltage Regulator with APM603 Controller

- A 7.5-inch color TFT touchscreen provides access to data.
- The controller provides an interface between the generator set and switchgear for paralleling applications incorporating multiple generator set and/or utility feeds.
- The controller can control Fast Response™ II, Fast Responset™ X, and PMG alternators using the GM88453 activator board.

Voltage Regulator Settings, APM603 Controller

- Voltage Regulator Configuration
 - Under Frequency Unload Settings
 - Single and Three Phase Sensing
 - Voltage Target
 - Voltage Regulator Gains

Paralleling Settings, APM603

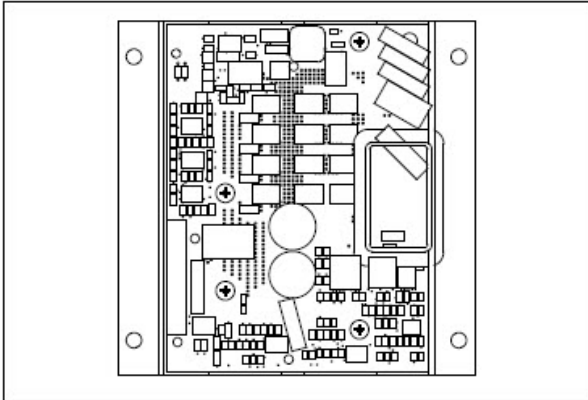
- Synchronizing parameters setup
 - Voltage matching
 - Frequency matching
 - Phase matching
 - Time delay
- Load sharing
 - kW sharing
 - kVAR sharing
 - Baseload settings
 - Droop

Paralleling Metering, APM603

- Paralleling State
 - Paralleling Mode
 - System Voltage
 - System Frequency
 - Connected Generators
 - Sync Status
 - Engine Speed



Activator Board GM88453



- Interfaces between the controller and alternator assembly using rotor field leads, auxiliary power windings, and optic board leads.
- Allows the Decision-Maker® controllers the ability to control a wound-field alternator using the same control signal as Fast Response™ alternator.
- Permits the generator set controller to control the current to the exciter field of a wound-field excited alternator.
- Contains two isolated relay driver outputs (RDO) rated at 250 mA. Provides RDO outputs indicating a field over-excitation condition and that the alternator is supplying voltage to the activator.

Modbus® is a registered trademark of Schneider Electric.

KOHLER®

ISO 9001
KOHLER
POWER SYSTEMS
NATIONALLY REGISTERED



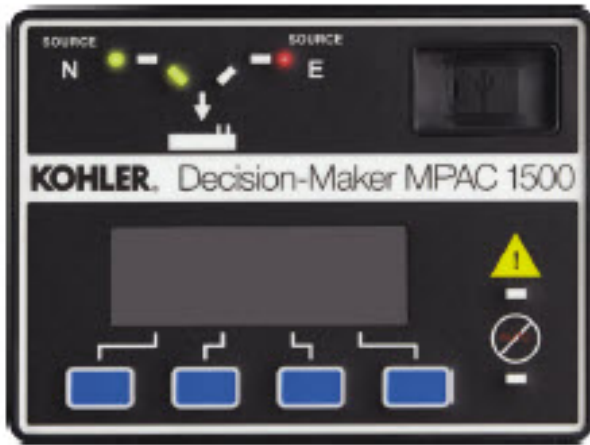
Transfer Switch Standard Features

- UL 1008 listed at 208-480 VAC, file #E108981
- CSA certification available
- IBC and OSHPD seismic certification available
- Bypass Isolation switches for uninterrupted power to the load during switch maintenance and testing
- Available in 2, 3, or 4 pole configurations
- Electrically operated, mechanically held mechanism
- High withstand and close-on ratings
- Fully rated for use as a manual 3-position transfer switch
- Heavy duty mechanical interlocks
- Bypass switch and contactor position indicator
- Drawout contactor for ease of maintenance
- Design suitable for emergency and standby applications on all classes of load, 100% tungsten rated through 400 amps
- Reliable, field-proven solenoid mechanism
- Switching mechanism lubricated for life
- Main shaft auxiliary position-indicating contacts
- Front- connected style available for some amperages
- Standard one-year limited warranty. Extended limited warranties are available.

Standard Transition Models (KBS)

- Standard-transition transfer time less than 100 milliseconds (6 cycles @ 60 Hz)
- Double-throw, mechanically interlocked design (break before make)
- Solid, switched, or overlapping neutral

Decision-Maker® MPAC 1500 Controller



- LCD display, 4 lines x 20 characters, backlit
 - Complete programming and viewing capability at the door using the keypad and LCD display
 - LED indicators: Source available, transfer switch position, service required (fault), and "not in auto"
 - Programmable voltage and frequency pickup and dropout settings
 - Programmable time delays
 - Programmable generator exerciser
 - Time-based load control
 - Current-based load control (current-sensing kit required)
 - Two programmable inputs and two programmable outputs
 - Up to four I/O extension modules available
 - Modbus communication standard
 - RS-485 communication standard
 - Ethernet communication standard
 - Three-source system
 - Prime power:
- For more information about Decision-Maker® MPAC 1500 features and functions, see specification sheet G11-128.

Codes and Standards

The ATS meets or exceeds the requirements of the following specifications:

- CSA C22.2 No. 178 certification at 600 VAC available, file # LR58301
- EN61000-4-4 Fast Transient Immunity Severity Level 4
- EN61000-4-5 Surge Immunity Class 4 (voltage sensing and programmable inputs only)
- EIC Specifications for EMI/EMC Immunity:
 - o CISPR 11, Radiated Emissions
 - o IEC 1000-4-2, Electrostatic Discharge
 - o IEC 1000-4-3, Radiated Electromagnetic Fields
 - o IEC 1000-4-4, Electrical Fast Transients (Bursts)
 - o IEC 1000-4-5, Surge Voltage
 - o IEC 1000-4-6, Conducted RF Disturbances
 - o IEC 1000-4-8, Magnetic Fields
 - o IEC 1000-4-11, Voltage Dips and Interruptions
- IEC 60947-6-1, Low Voltage Switchgear and Control Gear; Multifunction Equipment; Automatic Transfer Switching Equipment
- IEEE Standard 446, IEEE Recommended Practice for Emergency and Standby Power Systems for Commercial and Industrial Applications
- IEEE 472 (ANSI C37.90A) Ring Wave Test
- NEMA Standard ICS 10-2005, Electromechanical AC Transfer Switch Equipment
- NFPA 70, National Electric Code
- NFPA 99, Essential Electrical Systems for Health Care Facilities
- NFPA 110, Emergency and Standby Power Systems
- Underwriters Laboratories UL 508, Standard for Industrial Control Equipment
- Seismic certification in accordance with the International Building Code is available. (Accessory kit is required for seismic certification)
 - o IBC 2000, referencing ASCE 7-98 and ICC AC-156
 - o IBC 2003, referencing ASCE 7-02 and ICC AC-156
 - o IBC 2006, referencing ASCE 7-05 and ICC AC-156
 - o IBC 2009, referencing ASCE 7-05 and ICC AC-156
 - o IBC 2012, referencing ASCE 7-10 and ICC AC-156
- California OSHPD approval is available. (Accessory kit required.)
- Underwriters Laboratories UL 1008, Standard for Automatic Transfer Switches for Use in Emergency Standby Systems, file #E108981

Model KBS-DMWA-0260S , continued

Environmental Specifications	
Operating Temperature	-20°C to 70°C (-4°F to 158°F)
Storage Temperature	-40°C to 85°C (-40°F to 185°F)
Humidity	5% to 95% noncondensing

Input and Output Connection Specifications	
Component	Wire Size Range
Main board I/O terminals	#12-24 AWG
I/O module terminals	#14-24 AWG

Auxiliary Position Indication Contacts	
(rated 10 Amps @ 32 VDC/250 VAC)	
Switch Rating, amps	Number of Contacts Indicating Normal, Emergency
260	8, 8

UL-Listed Solderless Screw-Type Terminals for External Power Connections	
Normal, Emergency, and Load Terminals per Phase and Neutral	
Switch Rating, amps	Range of Wire Sizes, Copper or Aluminum*
260	(1) #4 AWG to 600 MCM
260	(2) 1/0 AWG to 250 MCM
* Use 75 degree C minimum Cu/AL wire for power connections	
* Use 75 degree C minimum Cu/AL wire for power connections	

Weights and Dimensions

See ADV drawings for weights and dimensions. Allow 15% additional weight for packing materials.

Withstand and Close-On Ratings (WCR)

Maximum current in RMS symmetrical amperes when coordinated with customer-supplied fuses or circuit breakers. All values are available symmetrical RMS amperes and tested in accordance with the withstand and close-on requirements of UL 1008. Application requirements may permit higher withstand ratings for certain size switches. Contact the factory for assistance.

Withstand and Closing Current Ratings in RMS Symmetrical Amperes								Short Time Ratings (sec.)**							
Current Limiting Fuses				Time-Based Rating*				480 V Max.				600 V Max.			
Switch Rating, Amps	Amps @ 480 V	Amps @ 600 V	Amps, Max.	Fuse Class	Amps @ 240 V	Amps @ 480 V	Amps @ 600 V	0.13	0.2	0.3	0.5	0.1	0.13	0.3	0.5
260	200,000	200,000	600	J	65,000	42000**	35,000	-	-	-	-	-	-	-	-
260	200,000	200,000	800	L	65,000	42000**	35,000	-	-	-	-	-	-	-	-
* Based on 0.050 seconds (approximately 3 cycles). Applicable to breakers with instantaneous trip elements.															
** Applicable to 2-pole, 3-pole, and conventional 4-pole switches only. Overlapping neutral switches have "any" breaker ratings of 35,000 A, 0.050 seconds at 480 V.															
*** Short time ratings are provided for applications involving breakers that utilize trip delay settings for system selective coordination.															

Ratings with Specific Manufacturer's Circuit Breaker

The following charts list power switching device withstand and close-on ratings (WCR) in RMS symmetrical amperes for specific manufacturers' circuit breakers. Circuit breakers are supplied by the customer.

Molded-Case Circuit Breakers					
Switch Rating, Amps	WCR, Amps, RMS	Voltage, Max.	Manufacturer	Type	Max. Size, Amps
260	50000	480	Eaton	HFDE, FDCE, HFD, FDC, LHH	225
260	50000	480	Eaton	JDC, JGH, JGC, JGU, JGX	250
260	50000	480	Eaton	HKD, HKDB, CHKD, CHKDB, KDC	400
260	50000	480	Eaton	HLD, CHLD, LDC, CLDC, LGH*, LGC*, LGU*, LGX*, NHH	600
260	50000	480	Eaton	MDL, CMDL, HMDL, CHMDL, NGS, NGH, NGC, MDLB, CMDLB, HMDLB, CHMDLB	800
260	50000	480	GE	SFL, SFP, FEN, FEH	250
260	50000	480	GE	TBC4	400
260	50000	480	GE	TBC6, TJL4V, TJL1S- 6S, SGL1, SGL4, SGL6, SGP1, SGP4, SGP6, FGN, FGH, FGL, FGP	600
260	50000	480	GE	TBC8, TKL4V, TKH8S- 12S, TKL8S- 12S, SKH8, SKL8, SKP8, TB8	800
260	50000	480	Siemens/ITE	HFD6, HFXD6, HHFD6, HHFXD6, CFD6, HFG, LFG	250

Model KBS-DMWA-0260S , continued

260	50000	480	Siemens/ITE	HJD6, HJXD6, SHJD6, HHJD6, HHJXD6, CJD6, SCJD6, HJG, LJG, LLG	400
260	50000	480	Siemens/ITE	HLD6, HLXD6, SHLD6, HHLD6, HHLXD6, CLD6, SCLD6, HLG	600
260	50000	480	Siemens/ITE	LMD, LMD6, LMXD, LMXD6, HLMD, HLMD6, HLMXD, HLMXD6, MD, MD6, MXD6, HMG, HMD6, HMXD6, SMD6, SHMD6, CMD6, SCMD6, LMG, MG	800
260	50000	480	Square D	KI, KC, CF250L, NSF250	250
260	50000	480	Square D	CK400N, CK400NN, CK400H, CK400HH, CJ400L, NSJ400	400
260	50000	480	Square D	LC,DJ,DL,LJ,LL,LR,LI,NSJ600	600
260	50000	480	Square D	CK800N, CK800NN, CK800H, CK800HH, MasterPact STR 28D, MJ, PK, PJ, PL	800
260	42000	600	Eaton	JGU, JGX	250
260	42000	600	Eaton	KDC	400
260	42000	600	Eaton	LDC, CLDC	600
260	42000	600	GE	TBC4	400
260	42000	600	GE	TBC6, SGL1, SGL4, SGL6, SGP1, SGP4, SGP6, FGP	600
260	42000	600	GE	TBC8, TKL4V, TKL8S- 12S, SKL8, SKP8	800
260	42000	600	Siemens/ITE	HJD, CFD6	250
260	42000	600	Siemens/ITE	HHJD6, HHJXD6, CJD6, SCJD6	400
260	42000	600	Siemens/ITE	HHLD6, HHLXD6, CLD6, SCLD6	600
260	42000	600	Siemens/ITE	HLMD6, HLMXD6, HMXD6, SHMD6, HMD6, CMD6, SCMD6, LMG, LNG, LPG, LGC*, LGU*, LGX*	800
260	50000	480	Cutler Hammer	HFDE, FDCE, HFD, FDC, LHH	225
260	50000	480	Cutler Hammer	JDC, JGH, JGC, JGU, JGX	250
260	50000	480	Cutler Hammer	HKD, HKDB, CHKD, CHKDB, KDC	400
260	50000	480	Cutler Hammer	HLD,CHLD, LDC, CLDC, LGH*, LGC*, LGU*, LGX*, NHH	600
260	50000	480	Cutler Hammer	MDL, CMDL, HMDL, CHMDL, NGS, NGH, NGC, MDLB, CMDLB, HMDLB, CHMDLB	800
260	50000	480	Square D	CK1000HL	1000
260	50000	480	Square D	CK1200NN, CK1200HH	1200
260	65000	480	Square D	JJ (Current Limiting)	250
260	100000	480	Square D	JL (Current Limiting)	250
260	200000	480	Square D	JR (Current Limiting)	250
260	42000	600	Cutler Hammer	JGU, JGX	250

Model KBS-DMWA-0260S , continued

260	42000	600	Cutler Hammer	KDC	400
260	42000	600	Cutler Hammer	LDC, CLDC	600
260	42000	600	Square D	KI, JL, JR, JJ, CF250L	250
260	42000	600	Square D	CK400H, CK400HH, CJ400L	400
260	42000	600	Square D	LI	600
260	42000	600	Square D	CK800H, CK800HH, MasterPact STR 28D, PK	800

Accessories

Accessories are available either factory-installed or as loose kits, unless otherwise noted.

Accessory Modules

The mounting kit holds up to five optional modules. The maximum total current draw is 300 mA. If an External Battery Module is installed, there is no current restriction.

- Alarm Module
- External Battery Supply Module
- Standard I/O Module
- High Power I/O Module

Current Sensing Kit

- Monitor current on all phases with 1% accuracy

Warranty

-

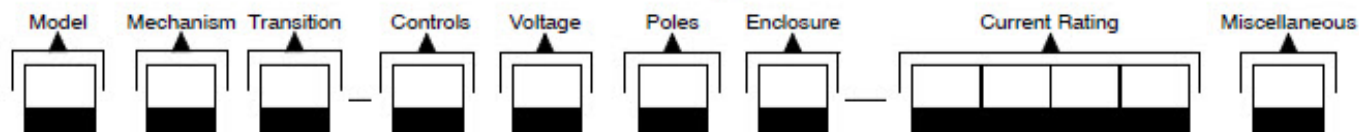
Standard Input/Output Module

Inputs	
Available Inputs	2
Input Definition	Contact closure
Current	5 mA Max
Connection Type	Terminal Strip
Wire Size	#14-24 AWG
Max Distance	700 feet
Outputs	
Outputs Available	6
Contact Type	Form C (SPDT)
Contact Voltage Rating	2 A @ 30 VDC 500 mA @ 125 VAC
Connection Type	Terminal Strip
Wire Size	#14-24 AWG

Warranty

Warranty-2-year

Model Designation



Record the transfer switch model designation in the boxes. The transfer switch model designation defines characteristics and ratings as explained below.

Model

K: Kohler

Mechanism

B: Bypass/Isolation Automatic Transfer Switch

Transition

S: Standard

P: Programmed-Transition

C: Closed-Transition

Controller

D: Decision-Maker® MPAC® 1500, Automatic

Voltage/Frequency

C: 208 Volts/60 Hz

D: 220 Volts/50 Hz

F: 240 Volts/60 Hz

G: 380 Volts/50 Hz

H: 400 Volts/50 Hz

J: 416 Volts/50 Hz

K: 440 Volts/60 Hz

M: 480 Volts/60 Hz

N: 600 Volts/60 Hz

P: 380 Volts/60 Hz

R: 220 Volts/60 Hz

Number of Poles/Wires

N: 2 Poles/3 Wires, Solid Neutral

T: 3 Poles/4 Wires, Solid Neutral

V: 4 Poles/4 Wires, Switched Neutral

W: 4 Poles/4 Wires, Overlapping Neutral

Enclosure

A: NEMA 1

B: NEMA 12

C: NEMA 3R

D: NEMA 4

F: NEMA 4X

G: Open Unit

Current, Amps *

0150 0800 2600

0225 1000 3000

0260 1200 4000

0400 1600

0600 2000

* Some selections are not available on all models. Contact the factory for 4000 amp models

Connections

S: Standard

F: Front (1600 and 2000 amp only)

Note: Some selections are not available on all models. Contact your Kohler distributor for availability.



Model KBS with Decision-Maker® MPAC 1500 Controller

Applicable Models

Model	Description
KCS	Standard-Transition Any Breaker ATS ‡
KCP	Programmed-Transition Any Breaker ATS ‡
KCC	Closed-Transition Any Breaker ATS §
KBS	Standard-Transition Bypass/Isolation ATS §
KBP	Programmed-Transition Bypass/Isolation ATS §
KBC	Closed-Transition Bypass/Isolation ATS §
KGS	Standard-Transition Bypass/Isolation ATS §
KGP	Programmed-Transition Bypass/Isolation ATS §
KEP	Service Entrance ATS §
‡ Available with automatic or non-automatic controller	
§ Available with automatic controller only	

Decision-Maker® MPAC 1500 Controller Standard Features

- Microprocessor-based controller
- Environmentally sealed user interface
- LCD display, 4 lines x 20 characters, backlit
- Dynamic function keypad with tactile feedback pushbuttons allows complete programming and viewing capability at the door
- LED indicators: Source available, transfer switch position, service required (fault), and not in auto
- Broadrange voltage sensing (208–600 VAC) on all phases
- Phase-to-phase sensing and monitoring with 0.5% accuracy on both sources
- Frequency sensing with 0.5% accuracy on both sources
- Anti-single phasing protection
- Phase rotation sensing for three-phase systems
- Real-time clock with automatic adjust for daylight saving time and leap year
- Run time clock and operation counter
- Time-stamped event log
- Fail-safe transfer for loaded test and exercise functions
- DIP switches: password disable and maintenance
- Isolated RS-485 ports for Modbus connections (9.6, 19.2, and 57.6 kbps)
- Standard Ethernet communications with RJ45 connector for 10/100 ethernet connection
- Modbus® RTU and Modbus® TCP/IP protocols (Modbus® register map available)
- USB port. Connect a personal computer and use Kohler® SiteTech™ software to view events and adjust settings *
- Available in automatic and non-automatic versions; see supervised transfer control switch on page 5

Programmable Features

- Programming and monitoring methods:
 - Monitoring and password-protected programming at the door using the keypad and display
 - Program using a PC with Kohler SiteTech software *
- Over/undervoltage and over/underfrequency for all phases of the normal and emergency sources
- Adjustable time delays
- Load/no load/auto-load test and load/no-load exercise functions
- Programmable inputs and outputs
- Load bank control for exercise or test
- Time-based and current-based† load control, nine individual time delays for selected loads
- In-phase monitor (3-phase only)
- Password protection, three security levels

* SiteTech software is available to Kohler-authorized distributors and dealers.

† Requires current sensing kit.

Modbus is a registered trademark of Schneider Electric.

Decision-Maker® MPAC 1500 Controller Features

User Interface LED Indicators

- Contactor position: source N and source E
- Source available: source N and source E
- Service required (fault indication)
- Not in automatic mode

LCD Display

- System status
- Line-to-line voltage
- Line-to-neutral voltage
- Active time delays
- Source frequency
- Preferred source selection
- System settings
- Common alarms
- Load current, each phase (current sensing kit required)
- Inputs and outputs
- Faults
- Time/date
- Address
- Event history
- Maintenance records
- Exerciser schedule
- Exerciser mode
- Time remaining on active exercise

Dynamic Function Tactile Keypad Operations

- Scroll up/down/forward/back
- Increase/decrease/save settings
- End time delay
- Start/end test or exercise
- Reset fault
- Lamp test

DIP Switches

- Maintenance mode
- Password disable

Event History

- View time and date-stamped events on the display or on a personal computer equipped with Kohler® SiteTech™ software. *
- Download complete event history files using Kohler SiteTech software and a PC connected to the USB port. *

Main Logic Board Inputs and Outputs

- Two (2) programmable inputs
- Two (2) programmable outputs

Communications

- Ethernet communications with RJ-45 connector for 10/100 ethernet connection
- Isolated RS-485 ports for Modbus communications
- Modbus® RTU and Modbus® TCP/IP protocols (Modbus® register map available)
- USB Port. Use SiteTech software to upload or download files and adjust transfer switch settings *
 - Application software
 - Event history files
 - Language files
 - Parameter settings
 - Usage reports
 - Feature configuration

Programmable Features

- System voltage, 208–600 VAC †
- System frequency, 50/60 Hz †
- Single/three-phase operation †
- Standard/programmed/closed-transition operation †
- Bypass/isolation enable/disable †
- Service entrance enable/disable †
- Preferred source selection allows the normal or emergency source to be used when both sources are available (alarm module required)
- Phase rotation: ABC/BAC/none selection with error detection
- Voltage and frequency pickup and dropout settings
- Voltage unbalance, enable/disable
- In-phase monitor: enable/disable and phase angle
- Transfer commit/no commit
- Source/source mode: utility/gen, gen/gen, utility/utility, or utility/gen/gen for 3-source systems
- Passwords, system and test
- Three-source system setup allows the use of one utility source and two generator sets
- Time, date, automatic daylight saving time enable/disable
- Time delays (see table)
- Exerciser: calendar mode, loaded/unloaded up to 21 events
- Test: loaded/unloaded/auto load (1–60 minutes)
- Remote test: loaded/unloaded
- Automatic override on generator failure (loaded test and exercise)
- Peak shave delay enable/disable
- Current monitoring (current sensing kit required)
- Load control pre/post-transfer delays, 9 individual time delays for selected loads
- Current-based load control settings: high/low current levels and load add/remove priority for 9 separate loads (current sensing kit required)
- Prime power sequence alternates between two generator sets with adjustable generator set runtimes
- Resettable historical data

* SiteTech software is available to Kohler-authorized distributors and dealers.

† System parameters are factory-set per order.

Modbus is a registered trademark of Schneider Electric.

Decision-Maker® MPAC 1500 Controller Features, Continued

Programmable Inputs

- Bypass contactor disable (for bypass/isolation switches)
- Forced transfer to OFF (programmed-transition models only; requires load shed accessory)
- Inhibit transfer
- Low battery voltage (external battery supply module required)
- Peak shave/area protection input
- Remote common fault
- Remote test
- Remote end time delay
- Remotely monitored inputs, four (4) available
- Service disconnect (for service entrance models)
- Three-source system disable

Programmable Outputs

- Alarm silenced
- Audible alarm
- Chicago alarm control
- Common alarm events
- Contactor position
- Exercise active
- Fail to open, source 1/source 2 (service entrance models)
- Fail to close, source 1/source 2 (service entrance models)
- Failure to acquire preferred source
- Failure to acquire standby source
- Failure to transfer
- Generator engine start, source N and E
- I/O module faults
- In-phase monitor synch
- Load bank control
- Load control active (pre/post transfer delay, up to 9 outputs)
- Loss of phase fault, source N and E
- Low battery fault (external battery supply module required)
- Maintenance mode
- Non-emergency transfer
- Not in automatic mode
- Over/underfrequency faults, source N and E (generator)
- Over/undervoltage faults, source N and E
- Peak shave/area protection active
- Phase rotation error, source N and E
- Preferred source supplying load
- Software-controlled relay outputs (four maximum)
- Source available, preferred and standby
- Standby source supplying load
- Test active
- Three-source system disable
- Transfer switch auxiliary contact fault
- Transfer switch auxiliary contact open
- Voltage unbalance, source N and E

Voltage and Frequency Sensing		
Parameter	Default	Adjustment Range
Undervoltage dropout	90% of pickup	75%-98%
Undervoltage pickup	90% of nominal	85%-100%
Overvoltage dropout *	115% of nominal*	106%-135%
Overvoltage pickup	95% of dropout	95%-100%
Unbalance enable	Disable	Enable/Disable
Unbalance dropout	20%	5%-20%
Unbalance pickup	10%	3%-18%
Voltage dropout time	0.5 sec.	0.1-9.9 sec.
Underfrequency dropout	99% of pickup	95%-99%
Underfrequency pickup	90% of nominal	80%-95%
Overfrequency dropout	101% of pickup	101%-115%
Overfrequency pickup	110% of nominal	105%-120%
Frequency dropout time	3 sec.	0.1-15 sec.
* 690 volts, maximum. Default = 110% for 600 volt applications.		

Adjustable Time Delays		
Time Delay	Default	Adjustment Range
Engine start, Source S2	3 sec.	0-6 sec. †
Engine start, Source S1 (gen/gen)	3 sec.	
Engine cooldown, Source S2	5 min.	0-60 min.
Engine cooldown, S1 (gen/gen)	5 min.	
Fail to acquire standby source	1 min.	
Fail to acquire preferred source	1 min.	
Transfer, preferred to standby	3 sec.	
Transfer, standby to preferred	15 min.	
Transfer, off to standby	1 sec.	1 sec. - 60 min.
Transfer, off to preferred	1 sec.	
Fail to synchronize	60 sec.	10 sec - 15 min.
Auto load test termination after transfer	1 sec.	1 sec.-60 min.
Prime power run duration	6 min.	6 min. - 100 days (6 min. increments)
Load Control Time Delays:		
Pretransfer to preferred	0 sec.	0-60 min.
Post-transfer to preferred	0 sec.	
Pretransfer to standby	0 sec.	
Post-transfer to standby	0 sec.	
Load add Source1/Source2	0 sec.	
Load remove Source1/Source2	0 sec.	
Note: Time delays are adjustable in 1 second increments, except as noted.		
† Engine start time delay can be extended to 60 minutes with an External Battery Supply Module Kit.		

Accessory Modules

The mounting kit holds up to five optional modules.

Module Current Draw Specifications, mA	
Alarm Module	75
Standard I/O Module	75
High Power I/O Module	100
Maximum Total Current *	300
* If an External Battery Module is installed, there is no current restriction.	

Standard Input/Output Module

Inputs	
Available Inputs	2
Input Definition	Contact closure
Current	5 mA Max
Connection Type	Terminal Strip
Wire Size	#14-24 AWG
Max Distance	700 feet
Outputs	
Outputs Available	6
Contact Type	Form C (SPDT)
Contact Voltage Rating	2 A @ 30 VDC 500 mA @ 125 VAC
Connection Type	Terminal Strip
Wire Size	#14-24 AWG

High-Power Input/Output Module

Inputs	
Available Inputs	2
Input Definition	Contact closure
Current	5 mA Max
Connection Type	Terminal Strip
Wire Size	#14-24 AWG
Max Distance	700 feet
Outputs	
Outputs Available	3
Contact Type	Form C (SPDT)
Contact Voltage Rating	12 A @ 24 VDC 12 A @ 250 VAC 10 A @ 277 VAC 2 A @ 480 VAC
Connection Type	Terminal Strip
Wire Size	#14-24 AWG
Environmental Specifications	
Temperature	-40°C to 85°C (-40°F to 185°F)
Humidity	35% to 85% noncondensing

Alarm Module

- 90 dB Audible alarm
- Any alarm function can be programmed to trigger the audible alarm
- Chicago alarm function
- Preferred source selection
- Supervised transfer control (supervised transfer control switch required)
- Connection for external alarm

External Alarm Connection Specifications	
Wire Size	#12-22 AWG Cu
Contact Voltage Rating	500 mA @ 120 VAC 250 mA @ 240 VAC

External Battery Supply Module

- Energizes the ATS controls using an external battery when no source power is available
- Allows extended engine start time delays
- Allows the use of any combination of accessory modules (no current draw restriction, maximum of five modules total)
- Connects to one or two batteries, 12 VDC or 24 VDC system
- Current draw, 140 mA @ 12 VDC, 86 mA @ 24 VDC
- Provides low external battery voltage indication to the transfer switch controller
- Reverse-polarity protected

Other Controller Accessories

Accessories are available either factory-installed or as loose kits, unless otherwise noted.

☐ Controller Disconnect Switch

- Disconnects power to controller without disconnecting load
- Mounts inside the enclosure

☒ Current Sensing Kit

- Monitor current on all phases with 1% accuracy

☐ Digital Meter

- Measure and display voltage, current, frequency, and power for both sources
- Programmable visual alarms for high voltage, low voltage, and high current
- Three digital outputs
- Serial port for optional network connections
- Password-protected programming menus
- Joystick operation
- Factory-installed
- Joystick operation
- Available factory-installed
- For more information, see TT-1506, Digital Power Meters.

☐ Line-to-Neutral Voltage Monitoring

- Monitors all line-to-neutral voltages

☐ Load Shed Kit

- Forced transfer from Emergency to OFF for programmed-transition models
- Customer-supplied signal (contact closure) is required for the forced transfer to OFF function
- Factory-installed only

☐ Padlockable User Interface Cover

- Provides additional protection against unauthorized access
- Standard on NEMA 3R enclosures

☐ RSA III Remote Serial Annunciator

- Monitors the generator set
- Monitors Normal and Emergency source status and connection
- Monitors ATS common alarm
- Allows remote testing of the ATS
- For more information about RSA III features and functions, see specification sheet G6-139.

☐ Supervised Transfer Control Switch

- Standard on models with non-automatic controls
- Optional for models with automatic controls
- Auto, manual, and transfer positions
- Automatic and non-automatic modes
- Alarm module required

Supervised Transfer Control Switch Operation for Automatic and Non-Automatic Transfer Switches		
Switch Position	Automatic Switches	Non-Automatic Switches
AUTO	• Automatically transfers to the standby source, when available, if the preferred source is lost. • Transfers back to the preferred source when it becomes available.	
MANUAL	• Automatically transfers to an available source if the connected source is lost. • Test, peak shave, and loaded exercise commands will transfer to the standby source. • Does not automatically transfer back to preferred when both sources are available.	• Does not automatically transfer to an available source when the connected source is lost. • Test, peak shave, and loaded exercise commands are ignored. • Does not automatically transfer back to preferred when both sources are available. • Transfers only when the switch is manually moved to the TRANSFER position as described below.
TRANSFER (momentary switch position)	• Does not initiate an engine start sequence. Generator set engine must be signalled to start by an event such as a loss of utility, loaded test, loaded exercise, etc. • Allows transfer to the other source, if available. An event such as a loss of utility, loaded exercise, or loaded test must first initiate the transfer sequence. • Time delays will operate. Wait for time delays to expire, or press the End Time Delay button. • Operates pre- and post-transfer load control time delays if both sources are available. • MANUAL TRANSFER is displayed when the ATS is ready to transfer.	



KOHLER CO., Kohler, Wisconsin 53044 USA
Phone 920-457-4441, Fax 920-459-1646
For the nearest sales and service outlet in the
US and Canada, phone 1-800-544-2444
KOHLERPower.com

Environmental Specifications	
Operating Temperature	-20°C to 70°C (-4°F to 158°F)
Storage Temperature	-40°C to 85°C (-40°F to 185°F)
Humidity	5% to 95% noncondensing

Main Board I/O Specifications	
Output contact type	Isolated form C (SPDT)
Output contact rating	1 amp @ 30 VDC, 500 mA @120 VAC
I/O terminals wire size	#12-24 AWG

Availability is subject to change without notice. Kohler Co. reserves the right to change the design or specifications without notice and without any obligation or liability whatsoever. Contact your local Kohler® Power Systems distributor for availability.

DISTRIBUTED BY:

KINSLEY POWER SYSTEMS
14 CONNECTICUT SOUTH DRIVE
EAST GRANBY, CT 06026
860-844-6100
www.KinsleyPower.com

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Alternator Data

TECHNICAL INFORMATION BULLETIN
Alternator Data Sheet
Alternator Model: 4R12X
Frequency: 60 Hz
Speed: 1800 RPM
Leads: 12 (6 Lead, 600 Volt)

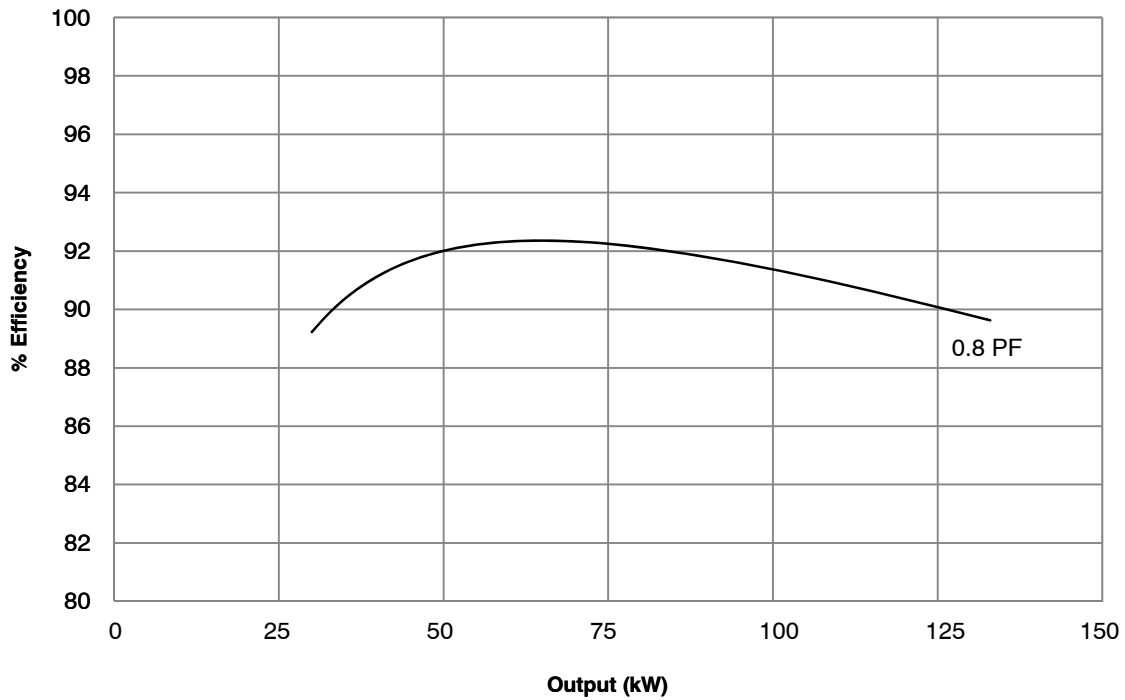
Voltage L-N/L-L	Phase	Power Factor	Connection	kW* (kVA)						
				Class B	Class F				Class H	
				80°C Continuous	90°C Lloyds	95°C ABS	105°C Continuous	130°C Standby	125°C Continuous	150°C Standby
139/240 277/480	3	0.8	Wye	111.0 (138.5)	116.0 (145.0)	119.0 (148.5)	124.0 (155.0)	134.5 (168.0)	132.5 (165.5)	142.5 (178.0)
127/220 254/440	3	0.8	Wye	107.5 (134.0)	113.5 (141.5)	115.5 (144.0)	121.5 (151.5)	130.5 (163.0)	128.5 (160.5)	138.0 (172.5)
120/208 240/416	3	0.8	Wye	104.0 (130.0)	108.5 (135.5)	111.0 (138.5)	115.0 (143.5)	124.0 (155.0)	123.0 (153.5)	131.0 (163.5)
110/190 220/380	3	0.8	Wye	94.0 (117.5)	98.0 (122.5)	100.0 (125.0)	104.0 (130.0)	112.0 (140.0)	111.0 (138.5)	118.5 (148.0)
120/240	3	0.8	Delta	104.0 (130.0)	108.5 (135.5)	111.0 (138.5)	115.0 (143.5)	124.0 (155.0)	123.0 (153.5)	131.0 (163.5)
120/240	1	1.0	Dogleg	76.0 (76.0)	79.5 (79.5)	81.0 (81.0)	84.0 (84.0)	91.0 (91.0)	90.0 (90.0)	96.0 (96.0)
347/600	3	0.8	Wye	102.0 (127.0)	108.0 (135.0)	111.0 (138.0)	116.0 (145.0)	130.0 (162.0)	127.0 (158.0)	139.0 (173.0)

* All data tested in accordance with IEEE Standard 115. Kohler Co. reserves the right to change the design or specifications without notice and without any obligation or liability whatsoever.

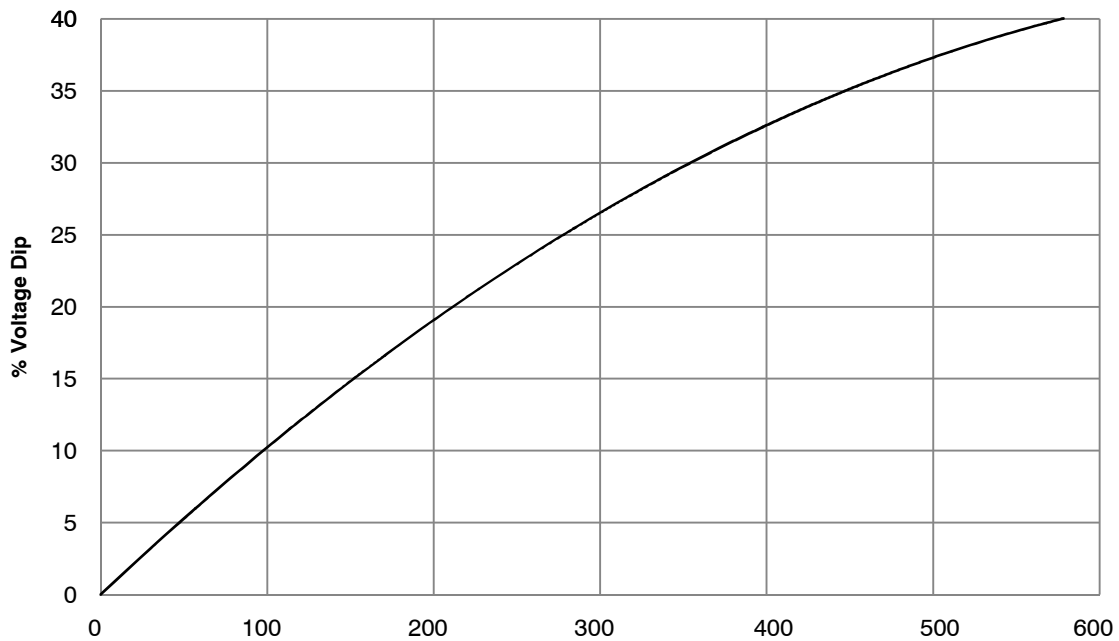
Submittal Data: 139/240 Volts, 0.8 PF, 1800 RPM, 60 Hz, 3 Phase, 130°C Rise

	Symbol	PerUnit	Ohms		Symbol	Value
Typical Cold Resistances				Typical Time Constants		
Phase Resistance		0.029	0.010	Armature Short Circuit	T _a	0.008 sec.
Rotor Resistance		17.83	6.112	Transient Short Circuit	T' _d	0.074 sec.
Typical Reactances				Transient Open Circuit	T' _{do}	0.860 sec.
Synchronous				Typical Field Current		
Direct	X _d	2.973	1.019	Full Load	I _{fFL}	18.9 amps
Quadrature	X _q	1.526	0.523	No Load	I _{fNL}	5.2 amps
Transient				Typical Short Circuit Ratio		0.336
Unsaturated	X' _{du}	0.290	0.099	Harmonic Distortion		
Saturated	X' _d	0.255	0.087	RMS Total Harmonic Distortion		3.25%
Subtransient				Max. Single Harmonic		5th
Direct	X'' _d	0.122	0.042	Deviation Factor (No Load, L-L)		<5%
Quadrature	X'' _q	0.116	0.040	Telephone Influence Factor		<50
Negative Sequence	X ₂	0.119	0.041	Insulation Class		
Zero Sequence	X ₀	0.010	0.003	per NEMA MG1 -1.66		H
				Phase Rotation		ABC

**4R12X, 60 Hz, 139/240, 277/480 Volts, Wye
TYPICAL ALTERNATOR EFFICIENCY***

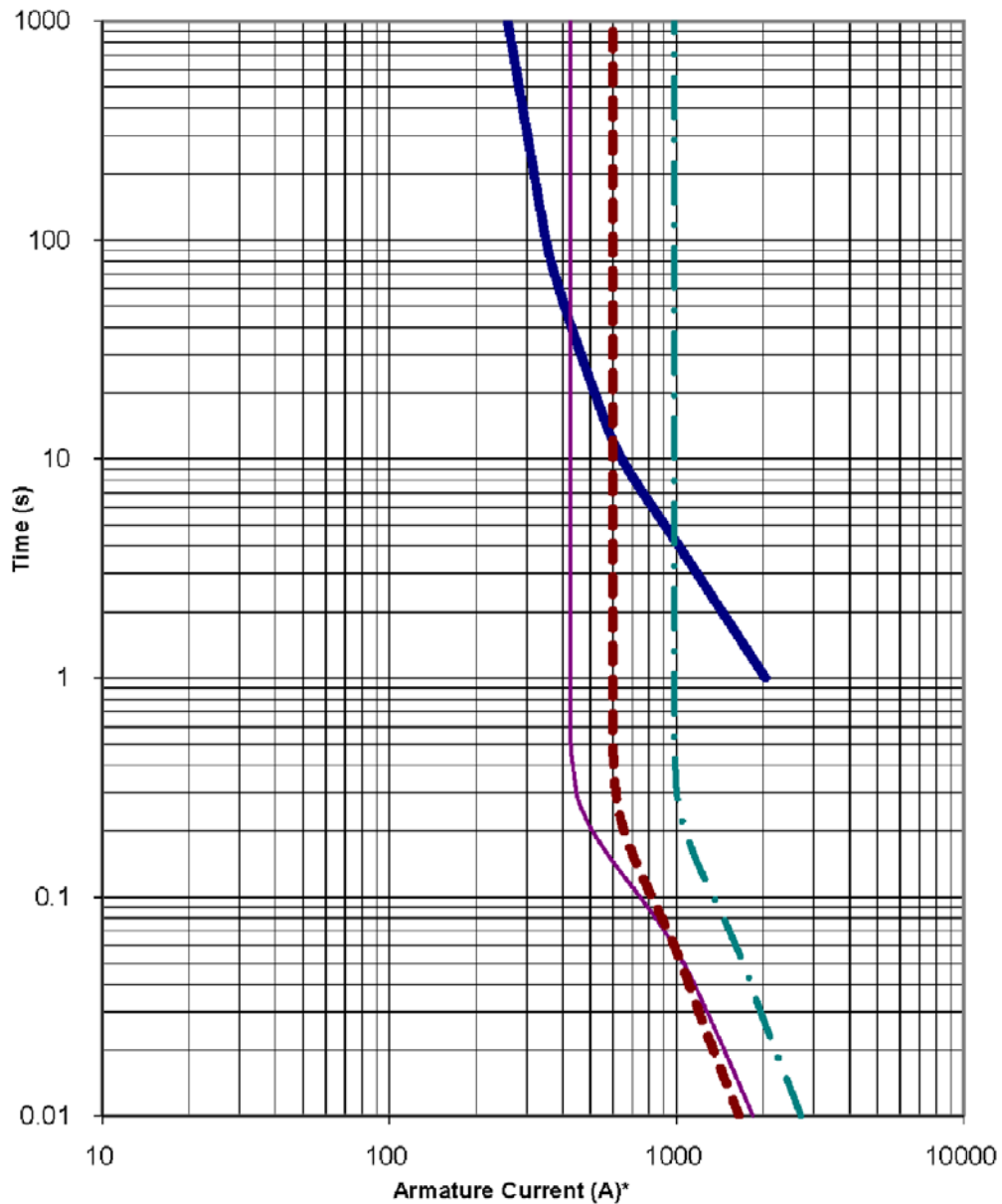


**4R12X, 60 Hz, 139/240, 277/480 Volts, Wye
TYPICAL MOTOR STARTING CHARACTERISTICS***



* All data tested in accordance with IEEE Standard 115. Kohler Co. reserves the right to change the design or specifications without notice and without any obligation or liability whatsoever.

4R12X, 60 Hz, High Wye Connection SHORT CIRCUIT DECREMENT CURVE



*Instantaneous current (t=0) is asymmetric. Divide by 1.73 for symmetric.



Sound Data

TECHNICAL INFORMATION BULLETIN
Generator Set Sound Data Sheet

			Sound Pressure Data in dB(A)		
Generator Set Model	Hz	Load		Open Unit, Isolated Exhaust	
100REOZJF	60	100% Load		84.5	
		No Load		80.9	

Note: Sound pressure data is the logarithmic average of eight perimeter measurement points at a distance of 7 m (23 ft.), except Raw Exhaust data which is a single measurement point at 1 m (3.3 ft.) from the mouth of a straight pipe exhaust.

				Sound Pressure Levels, dB(A)								
Load	Distance, m (ft)		Measurement Clock Position	Octave Band Center Frequency (Hz)								Overall Level
				63	125	250	500	1000	2000	4000	8000	
100% Load	7 (23)	Open Unit, Isolated Exhaust	3:00	58.3	62.6	72.9	74.1	76.8	78.8	74.3	69.6	83.5
			1:30	58.2	63.5	71.9	72.5	77.2	79.2	77.1	72.6	85.6
			12:00-Engine	53.3	62.0	72.4	75.3	75.1	76.7	71.5	66.0	81.9
			10:30	54.3	61.8	73.1	75.9	78.2	79.3	75.5	70.8	84.4
			9:00	60.2	65.5	73.8	77.7	78.0	78.7	75.2	72.9	84.5
			7:30	54.7	65.6	74.1	76.6	75.8	76.9	76.9	72.7	84.9
			6:00-Alternator	52.6	64.0	72.8	74.8	74.6	75.6	78.3	75.6	85.7
			4:30	47.8	62.1	71.5	75.9	76.8	77.6	75.9	71.0	84.1
			8-pos. log avg.	56.3	63.6	72.9	75.6	76.7	78.0	76.0	72.1	84.5

The generator set manufacturer reserves the right to change the design or specifications without notice and without any obligation or liability whatsoever. © 2015 by Kohler Co. All rights reserved.



Emissions Data



100REOZJF

60 HZ. DIESEL INDUSTRIAL GENERATOR SET
EMISSION DATA SHEET

ENGINE INFORMATION			
Model:	John Deere, 4045HF285I	Bore:	106mm (4.19 in.)
Nameplate BHP @ 1800 RPM:	158	Stroke:	127mm (5.00 in.)
Type:	4-Cycle, 4 Cylinder, Inline	Displacement:	4.5 L (276 cu. in.)
Aspiration:	Turbocharged, Charge Air-Cooled	EPA Family:	LJDXL04.5119
Compression Ratio	19:1	EPA Certificate:	LJDXL04.5119-005

PERFORMANCE DATA: Engine bkW @ Stated Load Fuel Consumption (g/kWh) Exhaust Gas Flow (m³/min) Exhaust Temperature (°C)	Table 1			
	1/4 Standby	1/2 Standby	3/4 Standby	Full Standby
	29.50	59.00	88.50	118.00
	272.10	254.70	237.40	222.40
				22.80
				580.00
EXHAUST EMISSION DATA: HC (Total Unburned Hydrocarbons) NOx (Oxides of Nitrogen as NO2) CO (Carbon Monoxide) PM (Particular Matter)	Table 2 EPA CERTIFICATE DATA			
	0.15			
	3.4			
	1.3			
	0.17			
Values are in g/kWh unless otherwise noted				

TEST METHODS AND CONDITIONS
The EPA Certificate Data in Table 2 is a weighted average value per ISO 8528 D2.

Data and specifications subject to change without notice



UNITED STATES ENVIRONMENTAL PROTECTION AGENCY
2020 MODEL YEAR
CERTIFICATE OF CONFORMITY
WITH THE CLEAN AIR ACT

OFFICE OF TRANSPORTATION
AND AIR QUALITY
ANN ARBOR, MICHIGAN 48105

Certificate Issued To: Deere & Company
(U.S. Manufacturer or Importer)
Certificate Number: LJDXL04.5119-005

Effective Date:

06/24/2019

Expiration Date:

12/31/2020

Byron J. Bunker, Division Director
Compliance Division

Issue Date:

06/24/2019

Revision Date:

N/A

Model Year: 2020

Manufacturer Type: Original Engine Manufacturer

Engine Family: LJDXL04.5119

Mobile/Stationary Indicator: Stationary

Emissions Power Category: 75<=kW<130

Fuel Type: Diesel

After Treatment Devices: No After Treatment Devices Installed

Non-after Treatment Devices: Electronic Control, Smoke Puff Limiter, Engine Design
Modification, Non-standard Non-After Treatment Device Installed

Pursuant to Section 111 and Section 213 of the Clean Air Act (42 U.S.C. sections 7411 and 7547) and 40 CFR Part 60, and subject to the terms and conditions prescribed in those provisions, this certificate of conformity is hereby issued with respect to the test engines which have been found to conform to applicable requirements and which represent the following engines, by engine family, more fully described in the documentation required by 40 CFR Part 60 and produced in the stated model year.

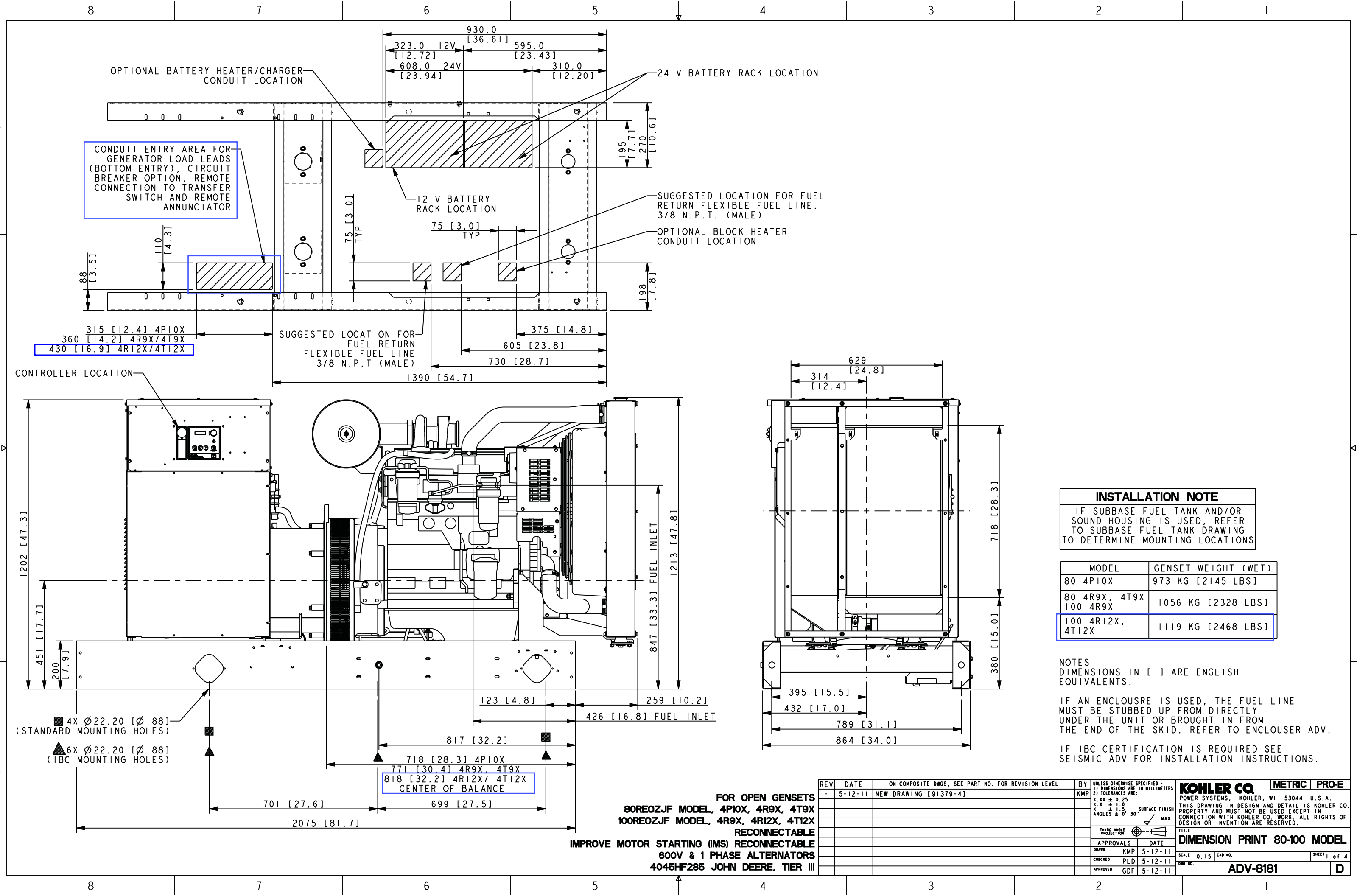
This certificate of conformity covers only those new compression-ignition engines which conform in all material respects to the design specifications that applied to those engines described in the documentation required by 40 CFR Part 60 and which are produced during the model year stated on this certificate of the said manufacturer, as defined in 40 CFR Part 60.

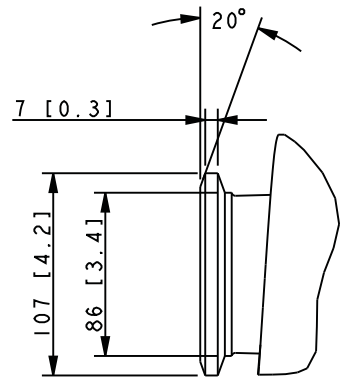
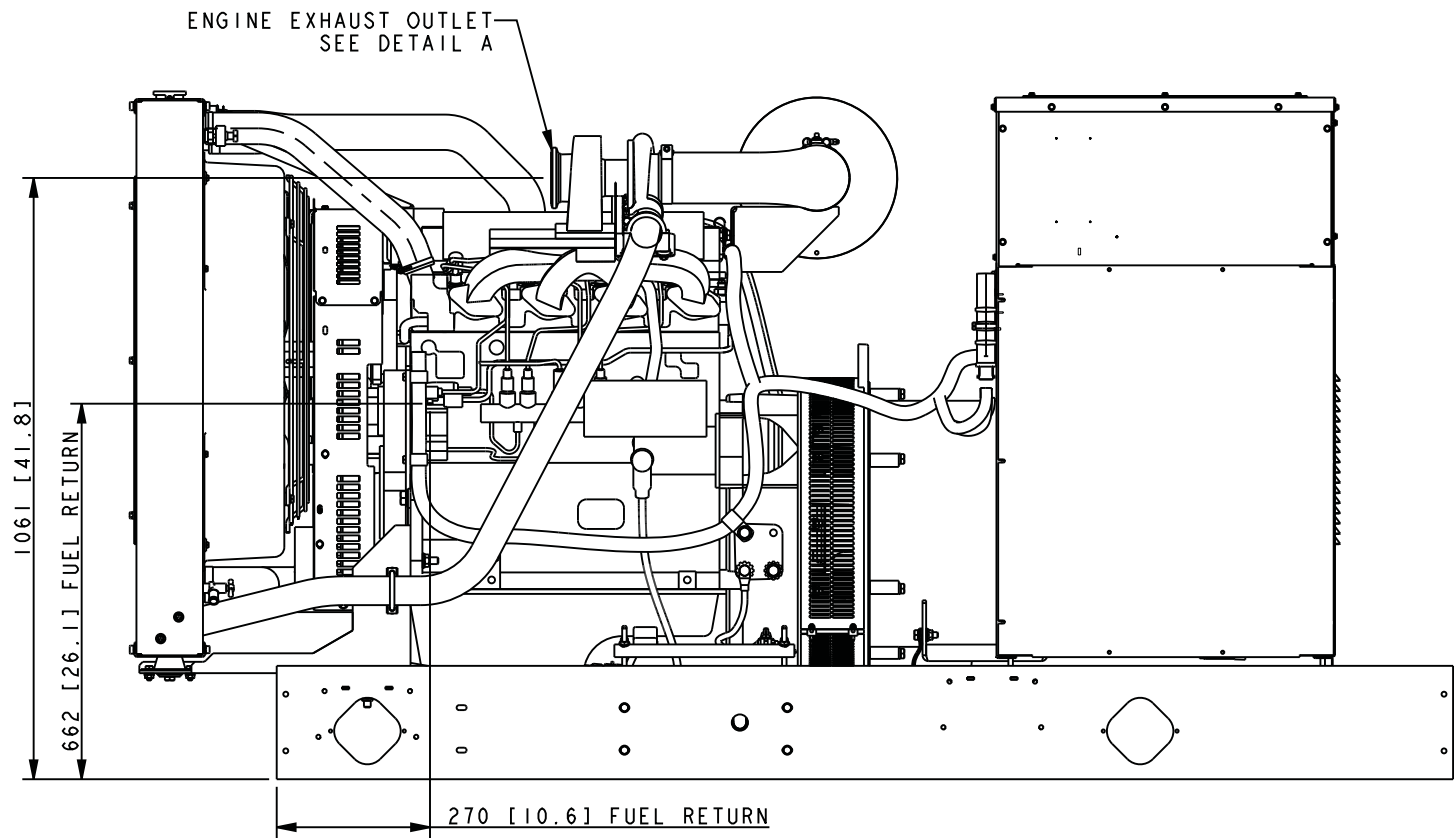
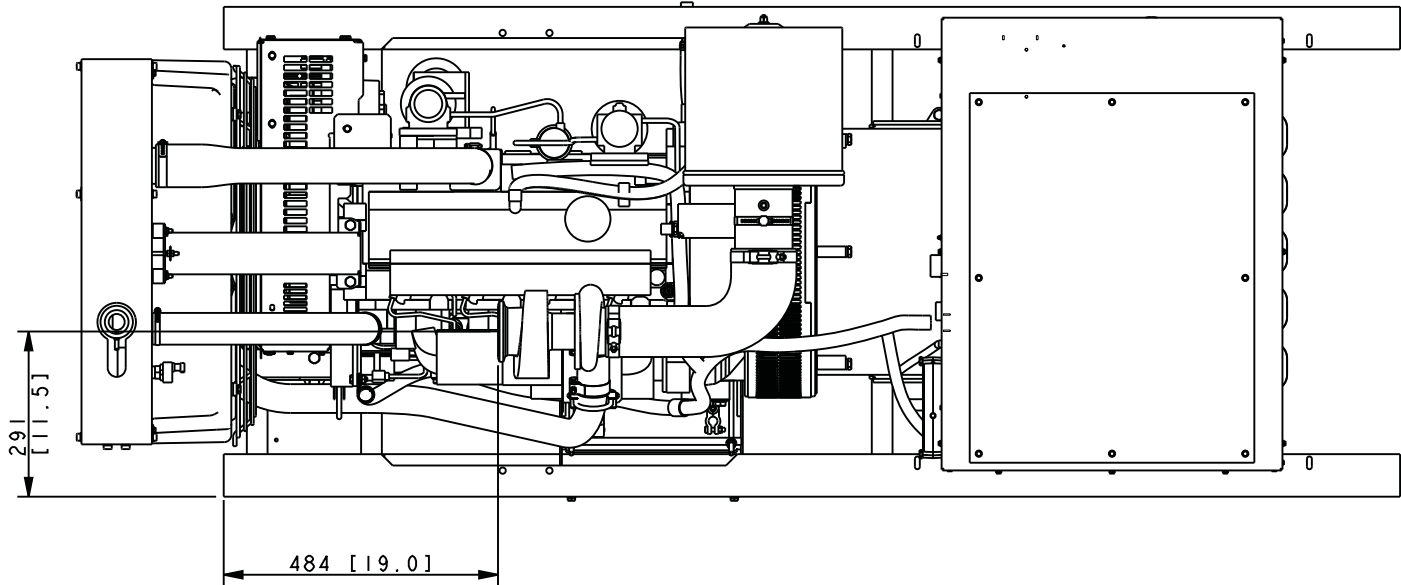
It is a term of this certificate that the manufacturer shall consent to all inspections described in 40 CFR 1068 and authorized in a warrant or court order. Failure to comply with the requirements of such a warrant or court order may lead to revocation or suspension of this certificate for reasons specified in 40 CFR Part 60. It is also a term of this certificate that this certificate may be revoked or suspended or rendered void *ab initio* for other reasons specified in 40 CFR Part 60.

This certificate does not cover engines sold, offered for sale, or introduced, or delivered for introduction, into commerce in the U.S. prior to the effective date of the certificate.



Dimensional Drawings

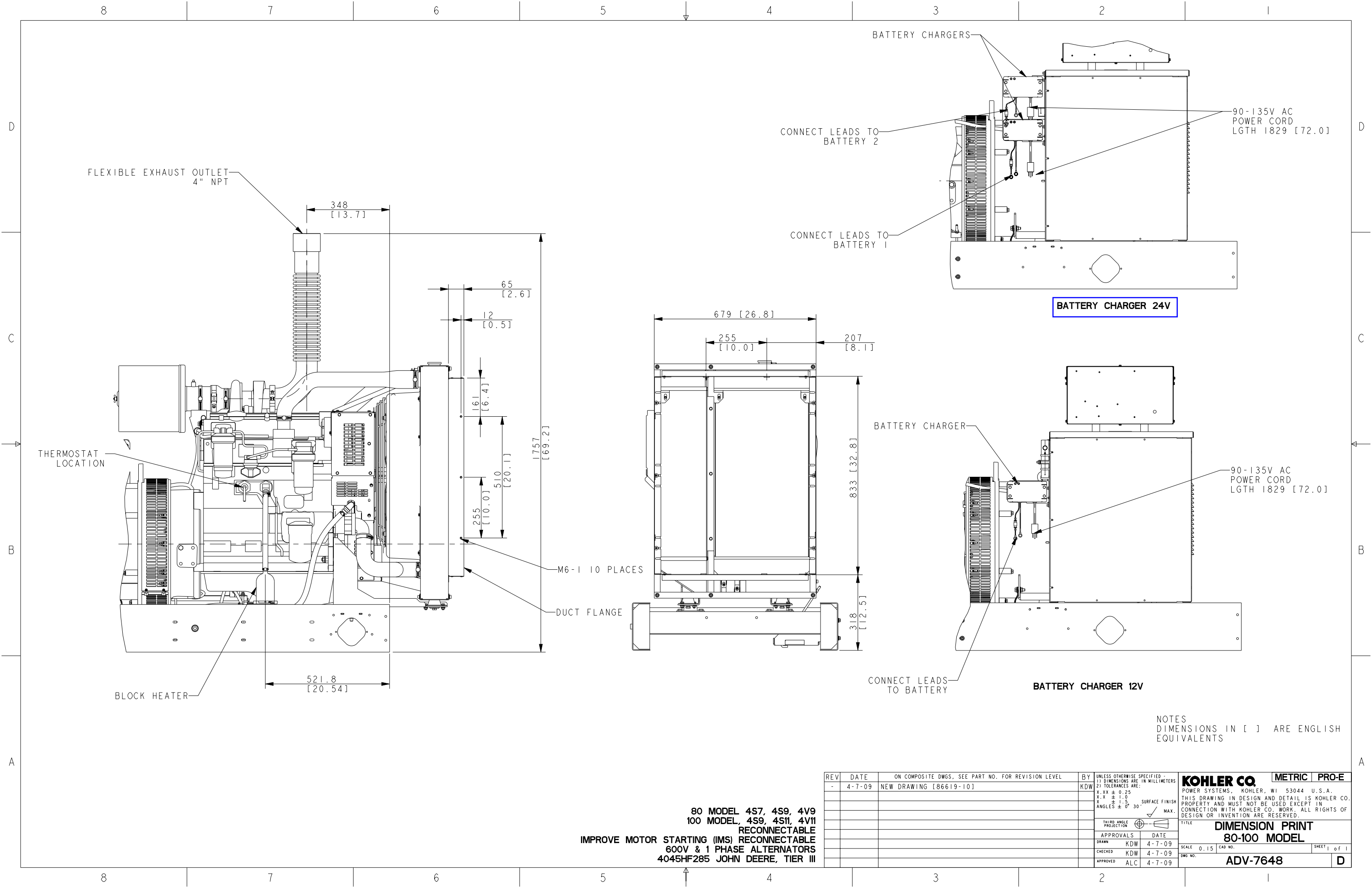




DETAIL A
EXHAUST OUTLET
SCALE 0.50

FOR OPEN GENSETS
80REOZJF MODEL, 4P10X, 4R9X, 4T9X
100REOZJF MODEL, 4R9X, 4R12X, 4T12X
FUEL RETURN AND EXHAUST
LOCATION WITH 34 IN. SKID

REV	DATE	ON COMPOSITE DWGS, SEE PART NO. FOR REVISION LEVEL	BY	UNLESS OTHERWISE SPECIFIED - 1) DIMENSIONS ARE IN MILLIMETERS 2) TOLERANCES ARE:	SURFACE FINISH MAX. THIRD ANGLE PROJECTION	TITLE DIMENSION PRINT 80-100 MODEL
-	5-19-11	NEW DRAWING [91379-4]	KMP	X.XX ± 0.25 X.X ± 1.0 X ± 1.5 ANGLES ± 0° 30'		
					APPROVALS	DATE
					DRAWN KMP	5-19-11
					CHECKED PLD	5-19-11
					APPROVED GDF	5-19-11
POWER SYSTEMS, KOHLER, WI 53044 U.S.A. THIS DRAWING IN DESIGN AND DETAIL IS KOHLER CO. PROPERTY AND MUST NOT BE USED EXCEPT IN CONNECTION WITH KOHLER CO. WORK. ALL RIGHTS OF DESIGN OR INVENTION ARE RESERVED.						SCALE 0.15 CAD NO. SHEET 3 of 4
ADV-8181						D



80 MODEL 4S7, 4S9, 4V9
100 MODEL, 4S9, 4S11, 4V11
RECONNECTABLE
IMPROVE MOTOR STARTING (IMS) RECONNECTABLE
600V & 1 PHASE ALTERNATORS
4045HF285 JOHN DEERE, TIER III

NOTES
DIMENSIONS IN [] ARE ENGLISH
EQUIVALENTS

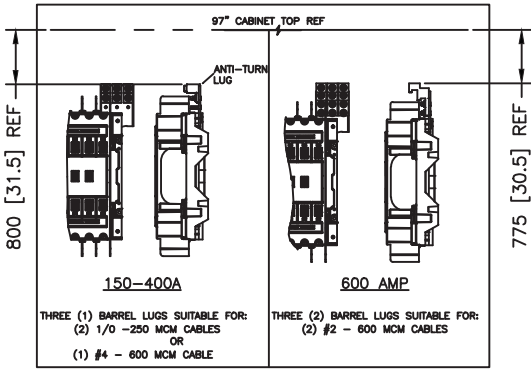
REV	DATE	ON COMPOSITE DWGS. SEE PART NO. FOR REVISION LEVEL	BY	UNLESS OTHERWISE SPECIFIED - 1) DIMENSIONS ARE IN MILLIMETERS 2) TOLERANCES ARE:	KOHLER CO. METRIC PRO-E	
-	4-7-09	NEW DRAWING (86619-10)	KDW	X.XX ± 0.25 X.X ± 1.0 X ± 1.5 ANGLES ± 0° 30'	POWER SYSTEMS, KOHLER, WI 53044 U.S.A. THIS DRAWING IN DESIGN AND DETAIL IS KOHLER CO. PROPERTY AND MUST NOT BE USED EXCEPT IN CONNECTION WITH KOHLER CO. WORK. ALL RIGHTS OF DESIGN OR INVENTION ARE RESERVED.	
				SURFACE FINISH MAX.	TITLE DIMENSION PRINT 80-100 MODEL	
				THIRD ANGLE PROJECTION	SCALE 0.15 CAD NO. SHEET 1 of 1	
				APPROVALS DATE	DWG NO. ADV-7648	
				DRAWN KDW 4-7-09	D	
				CHECKED KDW 4-7-09		
				APPROVED ALC 4-7-09		

REV	DATE	REVISION	BY
-	8-26-13	NEW DRAWING [CTS4441]	BTW

- GENERAL NOTES
- TYPE 12/4/4X ENCLOSURE. FREE STANDING. FLOOR MOUNTED.
 - MATERIAL: 12 GAUGE STEEL, TYPE 12/4 - CARBON STEEL; TYPE 4X - 304 STAINLESS STEEL.
 - FINISH: ANSI #1 GRAY, POLYESTER POWDER. UL RECOGNIZED.
 - CONSTRUCTION IS IN ACCORDANCE WITH UL 1008.
 - PADLOCKING PROVISIONS ARE INCLUDED ON TYPE 4/4X; KEY LOCKING LATCHING HANDLES ON TYPE 12; ISOLATION HANDLE. THE TRANSFER SWITCH ISOLATION HANDLE MAY BE PADLOCKED WITH THE TRANSFER SWITCH IN THE FULLY ISOLATED (DISCONNECTED POSITION).
 - RECOMMENDED FRONT CLEARANCE: 27 INCHES MINIMUM.
 - A 20% RATED GROUND BUS IS PROVIDED.
 - A FULL RATED NEUTRAL CONNECTION FOR EACH SOURCE AND THE LOAD IS OPTIONAL. WHEN PROVIDED IT IS IN ONE OF THE FOLLOWING FORMATS AS SPECIFIED BY THE CATALOG NO. NEUTRAL TYPE:
TYPE A: SOLID (COPPER BUS) NEUTRAL
TYPE B: SWITCHED NEUTRAL POLE
TYPE C: OVERLAPPING NEUTRAL POLE (NOT AVAILABLE ON 7A0TB & 7ACTB UNITS)
 - APPROXIMATE WEIGHT: 1601 LBS.
 - STANDARD OUTLINE FOR A FOUR POLE 600 AMP TRANSFER SWITCH WITH BYPASS/ISOLATION SWITCH SHOWN. SEE DETAIL "A" FOR LUG CONFIGURATION OF SOLID NEUTRAL.
 - BOTH BYPASS SWITCH MANUAL OPERATION HANDLE & TRANSFER SWITCH CARRIAGE MANUAL CRANK HANDLE CAN BE REMOVED. ALSO NOTE THAT THE TRANSFER SWITCH CARRIAGE MANUAL CRANK HANDLE CAN BE LEFT IN PLACE AND FOLDED DOWN.

- CABLING NOTES
- ALL SIZES SUPPLIED STANDARD WITH MECHANICAL (SCREW TYPE) LUGS. (SEE AMP SIZE BELOW)
A. LUG MATERIAL: ALUMINUM ALLOY 6061-T6 WITH ELECTRO TIN PLATED FINISH.
B. SCREW MATERIAL: ALUMINUM ALLOY 6262-T6 WITH ELECTRO TIN PLATED FINISH.
C. UL LISTED, CSA CERTIFIED.
D. LUG MAX WIRE TIGHTENING TORQUE PER UL 486B; SEE TABLE BELOW.
 - OPTIONAL COPPER CRIMP LUGS MAY BE SUPPLIED. CONSULT FACTORY.
A. LUG MATERIAL: HIGH CONDUCTIVITY WROUGHT COPPER FINISH, ELECTRO TIN PLATED.
B. UL LISTED, CSA CERTIFIED.
C. LUG MOUNTING HARDWARE TIGHTENING TORQUE: (REFER TO WITHSTAND CURRENT RATING LABEL PROVIDED ON EACH TRANSFER SWITCH).
 - SUITABLE WIRE BENDING SPACE IS PROVIDED.
 - GROUND LUGS ARE PROVIDED STANDARD AS FOLLOWS:
(6) 600MCM AL/CU CABLES FOR 600 AMP
(3) 600MCM AL/CU CABLES FOR 150-400 AMP.
 - CONSULT FACTORY FOR OTHER TERMINATION REQUIREMENTS.

SIZE	CABLE ACCOMMODATIONS (PER PHASE & NEUTRAL)	LUG TORQUE
150-400	SCREW TYPE (STANDARD) - (2) 1/0 - 250 MCM AL/CU	500 IN.-LBS.
	OR (1) #4 - 600 MCM AL/CU	
600	SCREW TYPE (STANDARD) - (2) #2 - 600 MCM AL/CU	375 IN.-LBS.



DETAIL "A"
SOLID NEUTRAL

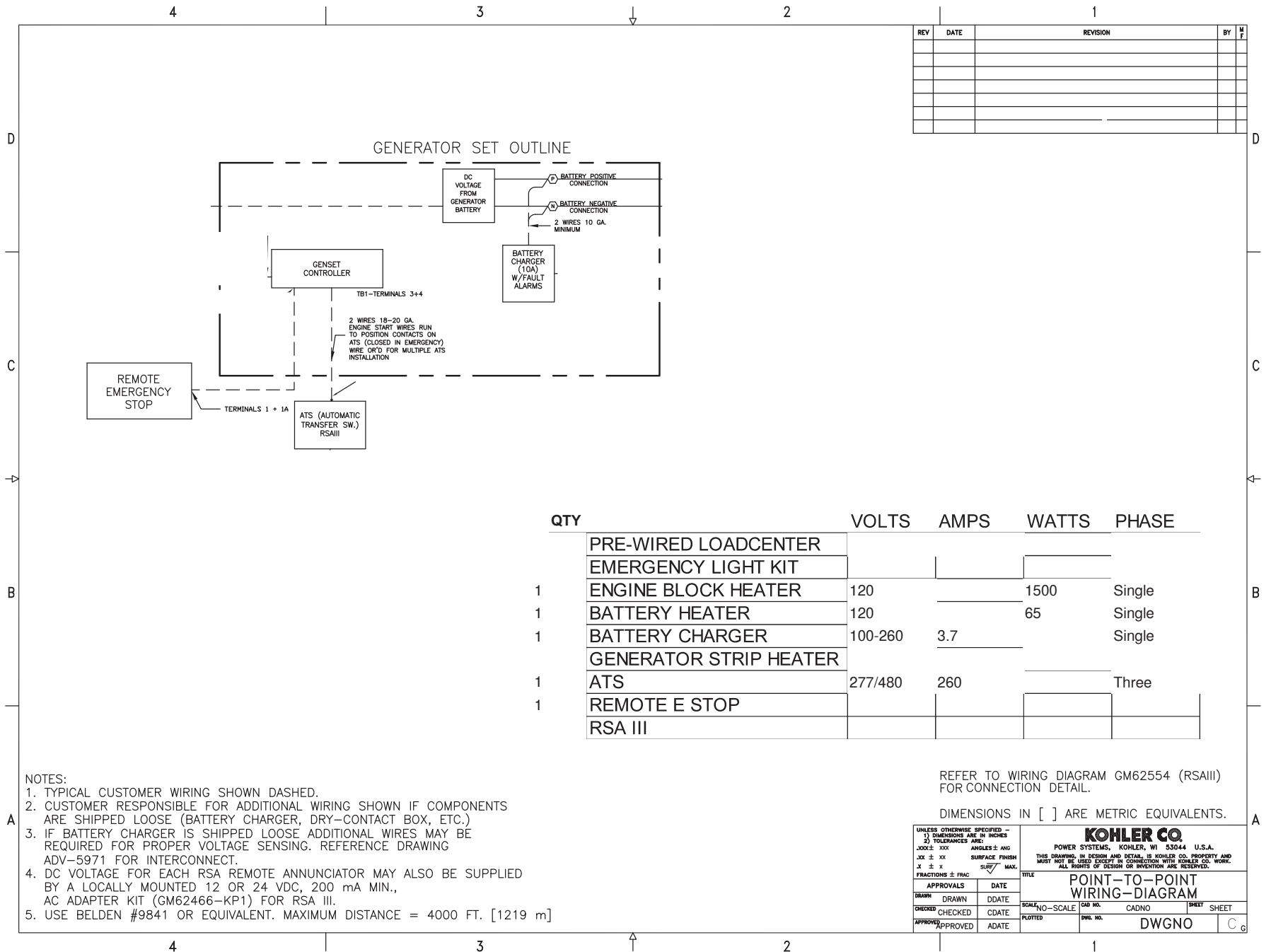
METRIC CAD FILE

DIMENSIONS IN [] ARE INCHES.

UNLESS OTHERWISE SPECIFIED - 1) DIMENSIONS ARE IN MILLIMETERS 2) TOLERANCES ARE: X.XX ± X.X ± X ± ANGLES ± SURFACE FINISH ✓ MAX.				KOHLER CO. POWER SYSTEMS, KOHLER, WI 53044 U.S.A. THIS DRAWING, IN DESIGN AND DETAIL, IS KOHLER CO. PROPERTY AND MUST NOT BE USED EXCEPT IN CONNECTION WITH KOHLER CO. WORK. ALL RIGHTS OF DESIGN OR INVENTION ARE RESERVED.			
APPROVALS				TITLE			
DRAWN BTW				DATE 8-26-13			
CHECKED BTW				DATE 8-26-13			
APPROVED MTL				DATE 8-26-13			
SCALE NONE				SHEET 3-3			
PLOT DATE				ADV-8600			



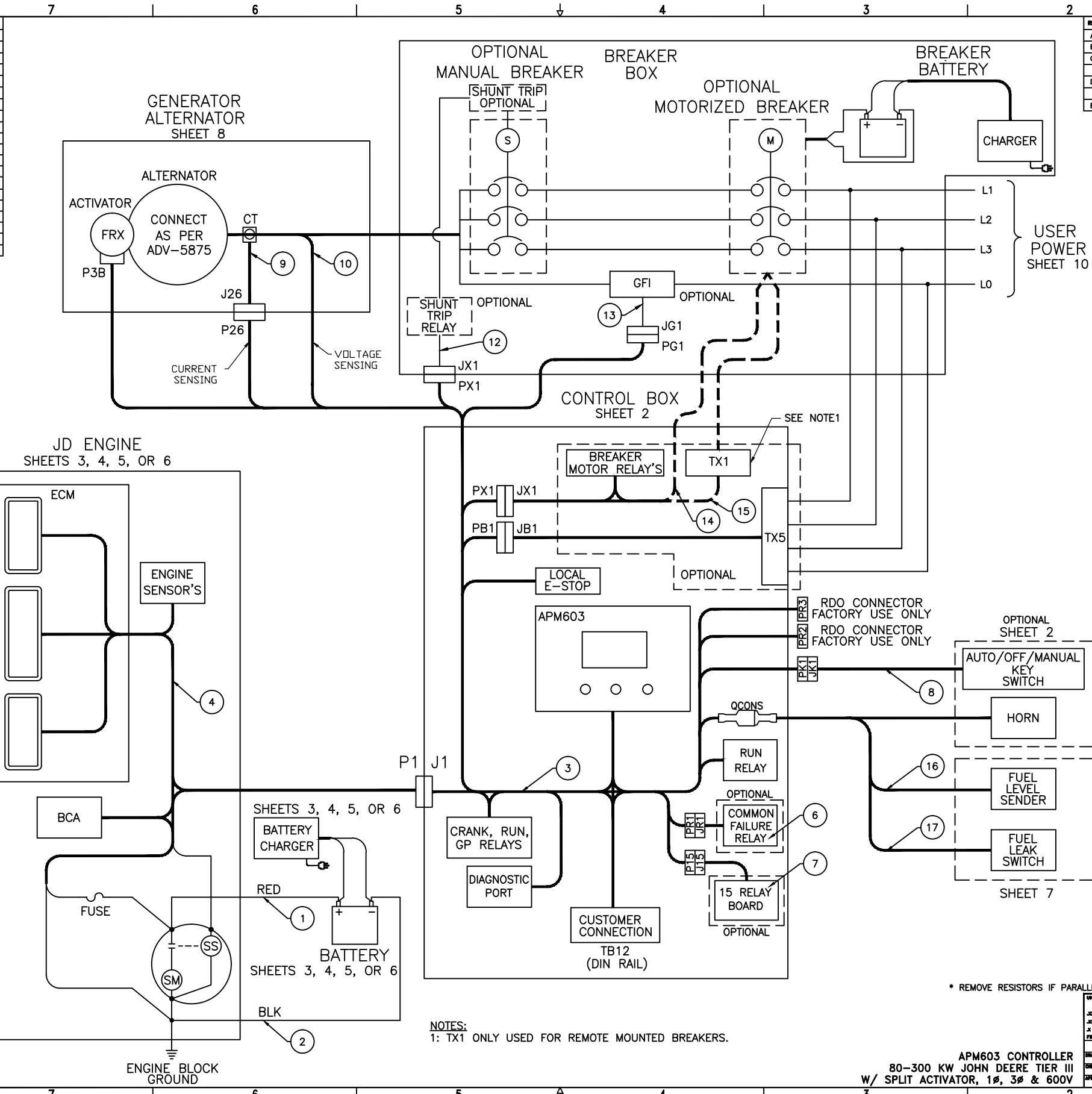
Wiring Schematics



REV	DATE	REVISION	BY	APP
A	10-11-18	SEE SHEET 2 [CT191090]	TLK	
B	2-05-19	SEE SHEETS 10 & 11 [CT193517]	SMH	
C	3-18-19	(A-1,-2)CHART UPDATED FOR SPEED AND VOLTAGE BIAS; SEE SHEET 2 [CT194009]	SMH	
D	5-13-19	(A,B,C-1) TB12-42 THRU 44 ADDED, MOVED VOLTAGE AND SPEED BIAS ON TB12, SEE SHEET 2 [CT195741]	TEV	
E	9-10-19	SEE SHEET 2 [CT198508]	TEV	

REV	DATE	REVISION	BY	APP
A	10-11-18	SEE SHEET 2 [CT191090]	TLK	
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E	9-10-19	SEE SHEET 2 [CT198508]	TEV	

REV	DATE	REVISION	BY	APP
A	10-11-18	SEE SHEET 2 [CT191090]	TLK	
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D	5-13-19	(A,B,C-1) TB12-42 THRU 44 ADDED, MOVED VOLTAGE AND SPEED BIAS ON TB12, SEE SHEET 2 [CT195741]	TEV	
E	9-10-19	SEE SHEET 2 [CT198508]	TEV	



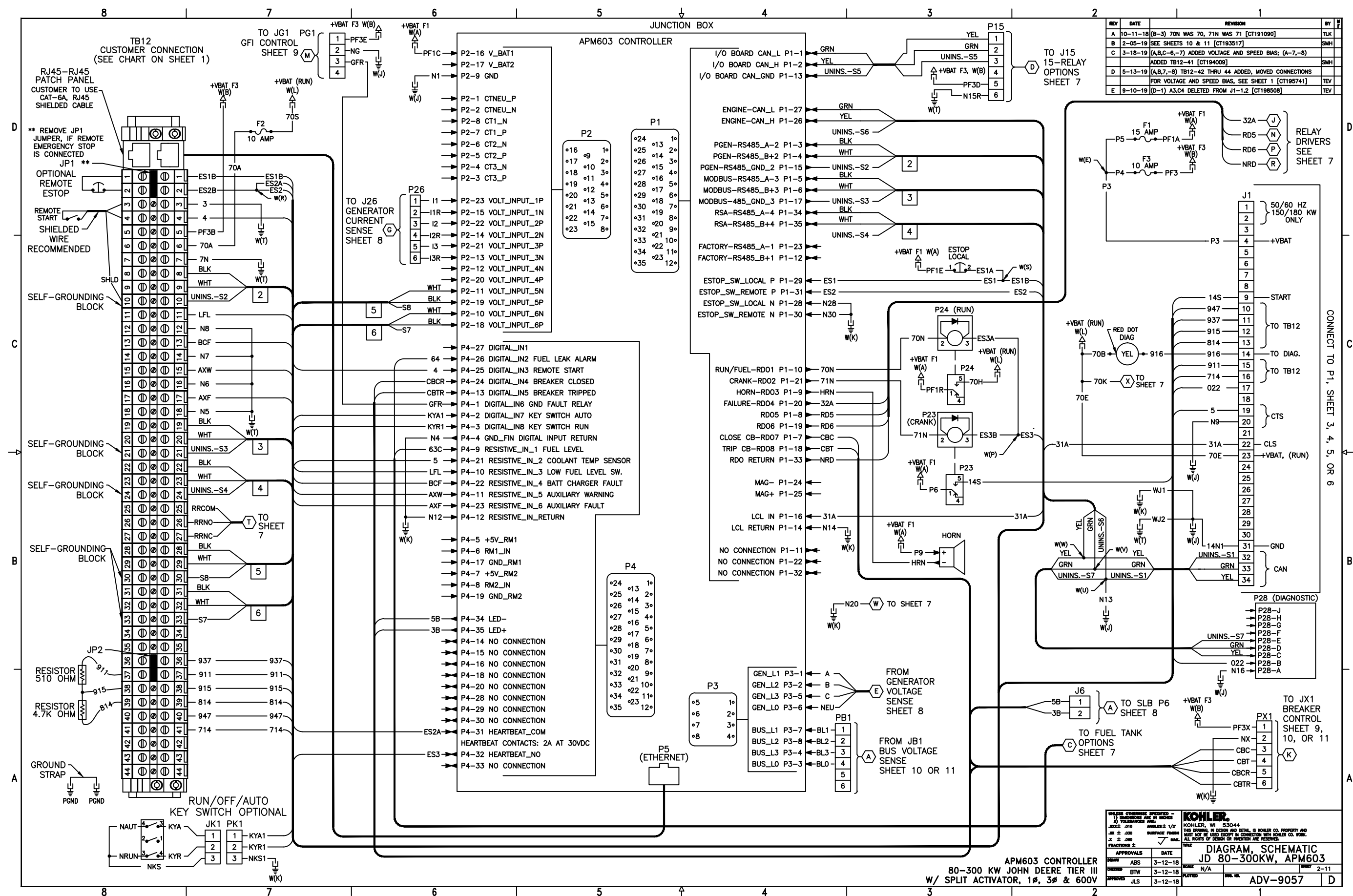
FUNCTION	POS	SIGNAL DESCRIPTION
REMOTE E-STOP	1	REMOTE EMERGENCY STOP
REMOTE START	2	REMOTE START SIGNAL
CUSTOMER INTERFACE	3	FUSED BATTERY POWER
CUSTOMER INTERFACE	4	BATT VOLTS WHEN RUNNING
CUSTOMER INTERFACE	5	BATTERY NEGATIVE
CUSTOMER INTERFACE	6	ISOLATED RS-485 #2 (PGEN)
CUSTOMER INTERFACE	7	ISOLATED RS-485 #3 (MODBUS/PGEN)
CUSTOMER INTERFACE	8	NON-ISOLATED RS-485 #4 (MODBUS RSA)
CUSTOMER INTERFACE	9	COMMON CONTACT
CUSTOMER INTERFACE	10	NORMALLY OPEN CONTACT
CUSTOMER INTERFACE	11	NORMALLY CLOSED CONTACT
CUSTOMER INTERFACE	12	SPEED BIAS (+)
CUSTOMER INTERFACE	13	SPEED BIAS (-)
CUSTOMER INTERFACE	14	VOLTAGE BIAS (+)
CUSTOMER INTERFACE	15	VOLTAGE BIAS (-)
CUSTOMER INTERFACE	16	SHIELD
CUSTOMER INTERFACE	17	SPARE
CUSTOMER INTERFACE	18	SPARE
CUSTOMER INTERFACE	19	ISOCHRONOUS JP2 IN
CUSTOMER INTERFACE	20	DROOP JP2 OUT
CUSTOMER INTERFACE	21	+5 VDC REF
CUSTOMER INTERFACE	22	SIGNAL (+2.5 VDC NORM)
CUSTOMER INTERFACE	23	SENSOR RETURN
CUSTOMER INTERFACE	24	CONSULT FACTORY
CUSTOMER INTERFACE	25	SPARE
CUSTOMER INTERFACE	26	SPARE
CUSTOMER INTERFACE	27	SPARE
CUSTOMER INTERFACE	28	SPARE
CUSTOMER INTERFACE	29	SPARE
CUSTOMER INTERFACE	30	SPARE
CUSTOMER INTERFACE	31	SPARE
CUSTOMER INTERFACE	32	SPARE
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CUSTOMER INTERFACE	40	SPARE
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CUSTOMER INTERFACE	42	SPARE
CUSTOMER INTERFACE	43	SPARE
CUSTOMER INTERFACE	44	SPARE

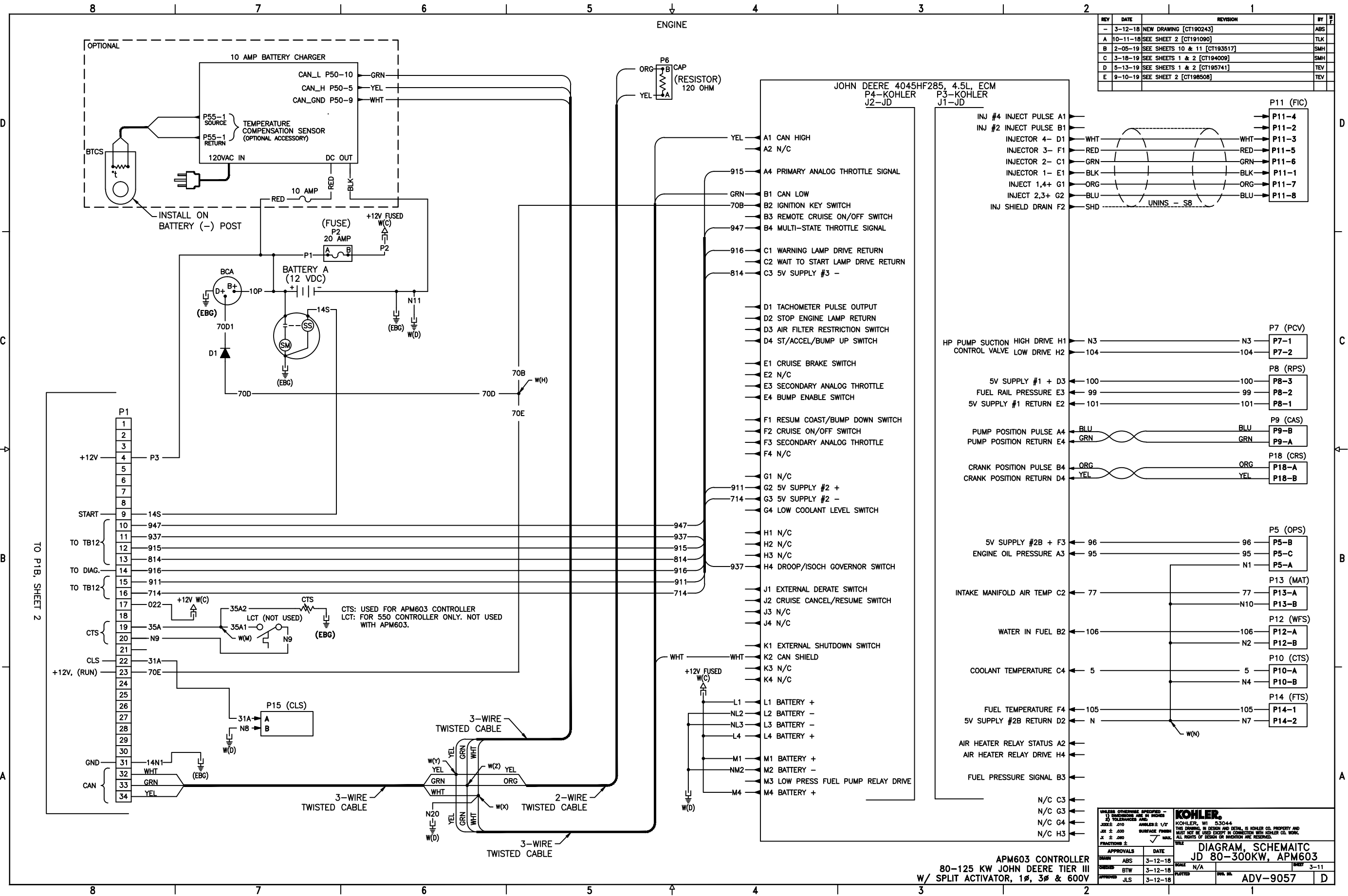
APPROVALS	DATE	SCALE	BY
DESIGNED ABS	3-12-18	N/A	1-11
CHECKED BTW	3-12-18	PLOTTED	
APPROVED JLS	3-12-18		

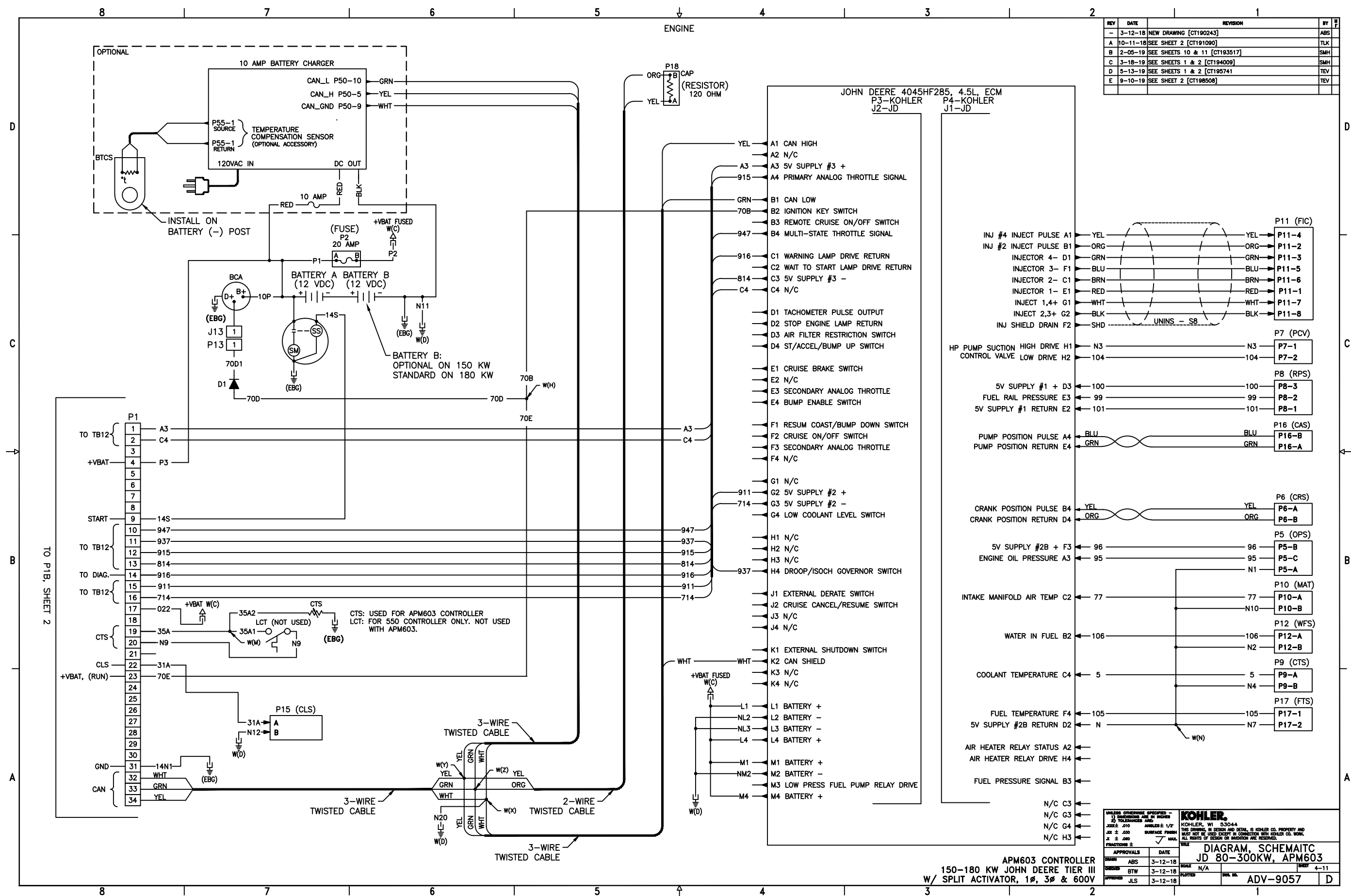
APM603 CONTROLLER
80-300 KW JOHN DEERE TIER III
W/ SPLIT ACTIVATOR, 1Ø, 3Ø & 600V

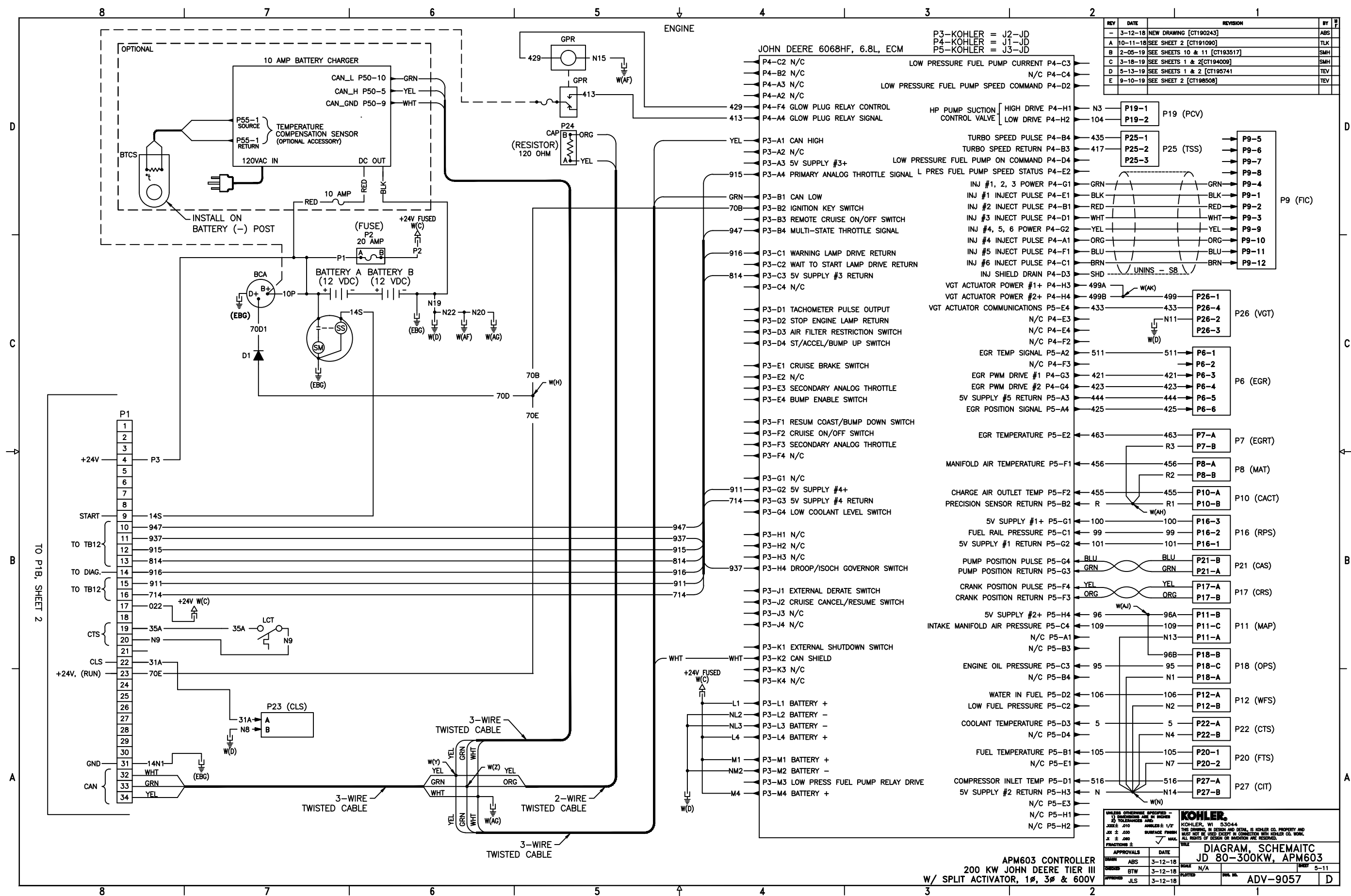
APPROVALS	DATE	SCALE	BY
DESIGNED ABS	3-12-18	N/A	1-11
CHECKED BTW	3-12-18	PLOTTED	
APPROVED JLS	3-12-18		

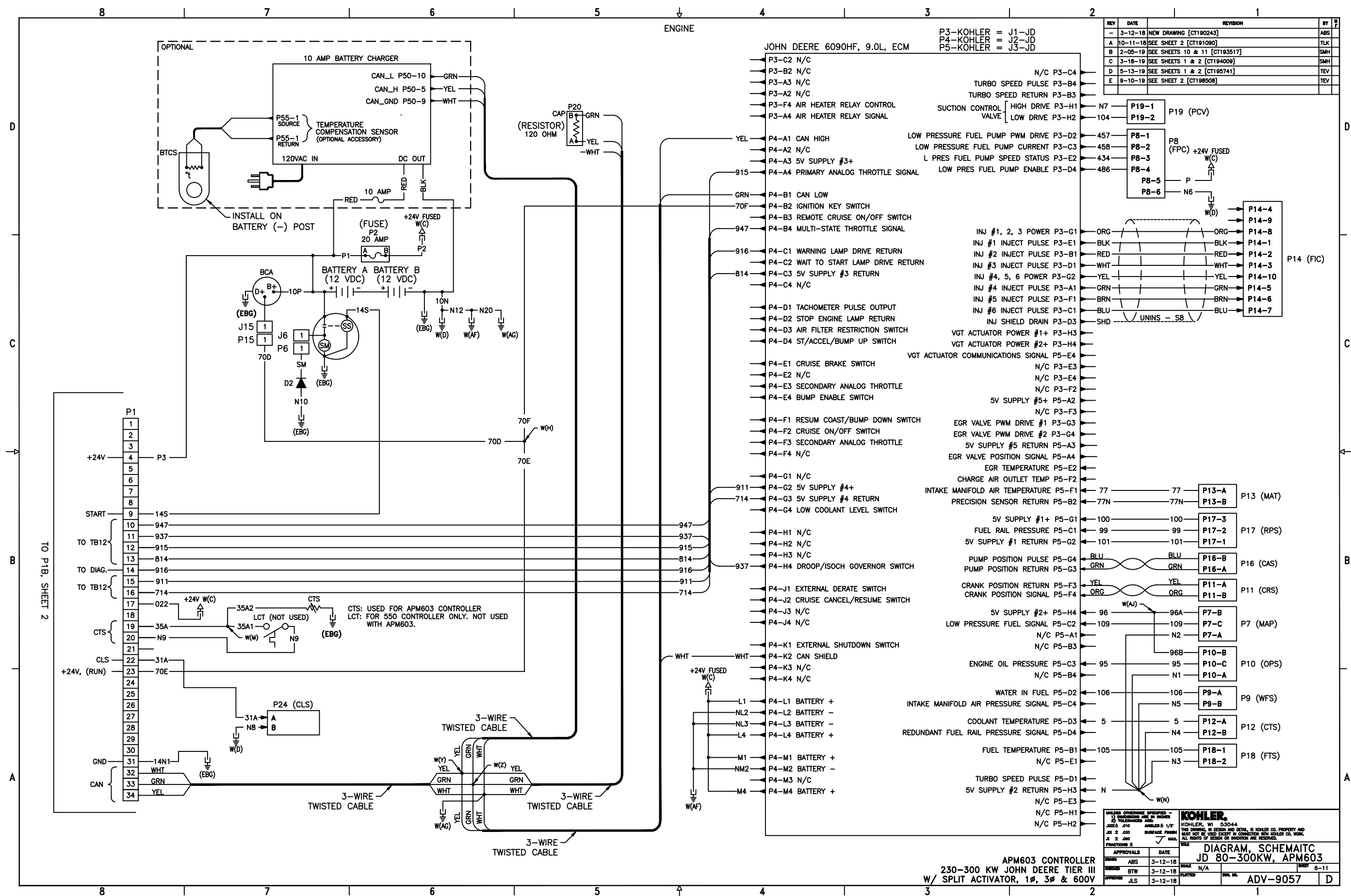
APPROVALS	DATE	SCALE	BY
DESIGNED ABS	3-12-18	N/A	1-11
CHECKED BTW	3-12-18	PLOTTED	
APPROVED JLS	3-12-18		

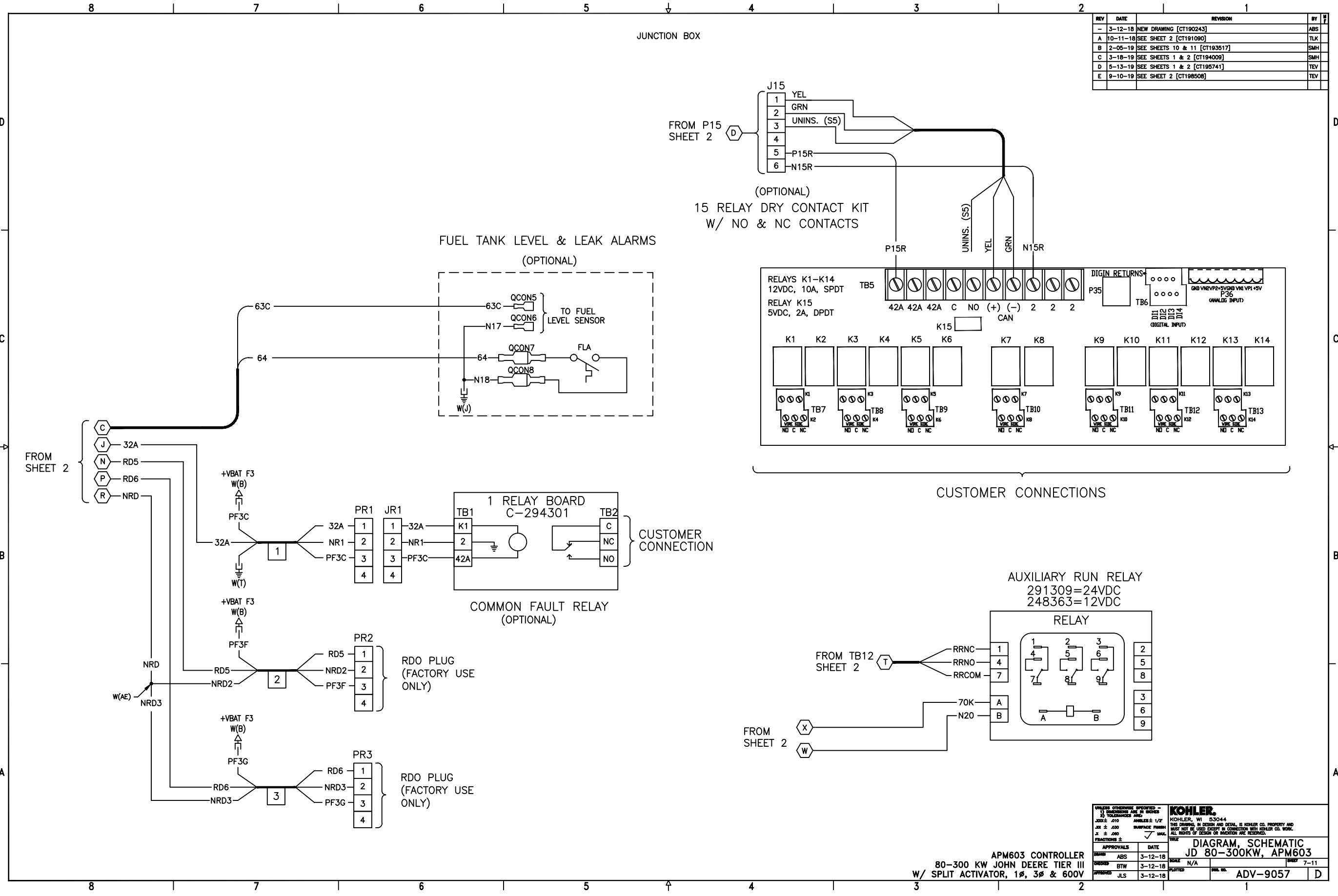




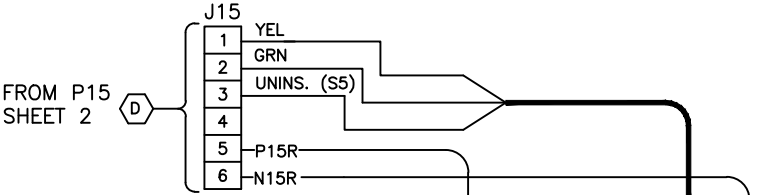




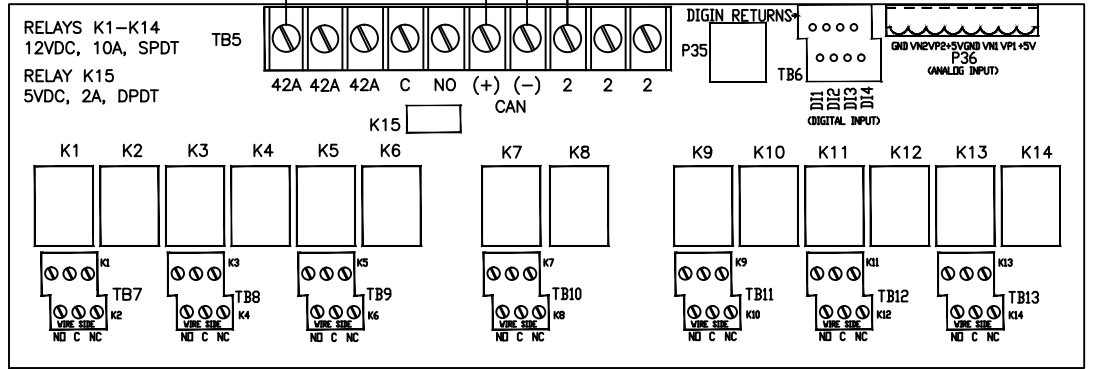




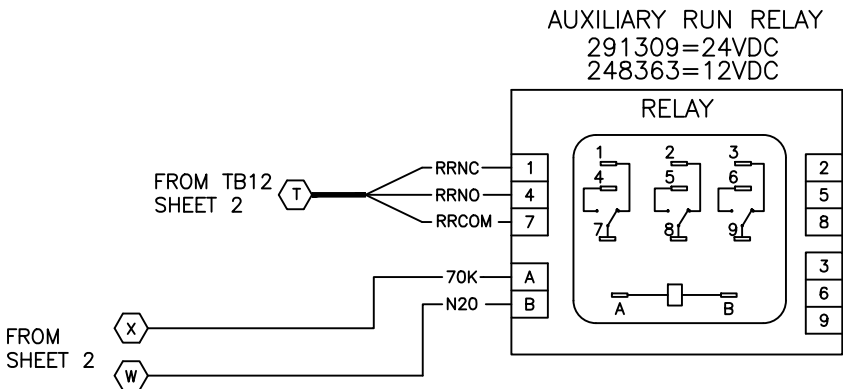
JUNCTION BOX



(OPTIONAL)
15 RELAY DRY CONTACT KIT
W/ NO & NC CONTACTS



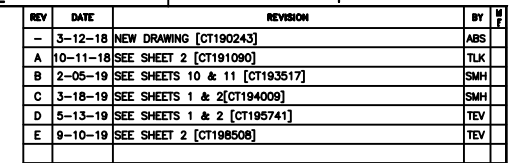
CUSTOMER CONNECTIONS



REV	DATE	REVISION	BY	APP
-	3-12-18	NEW DRAWING [CT190243]	ABS	
A	10-11-18	SEE SHEET 2 [CT191090]	TLK	
B	2-06-19	SEE SHEETS 10 & 11 [CT193517]	SMH	
C	3-18-19	SEE SHEETS 1 & 2 [CT194009]	SMH	
D	5-13-19	SEE SHEETS 1 & 2 [CT195741]	TEV	
E	9-10-19	SEE SHEET 2 [CT198508]	TEV	

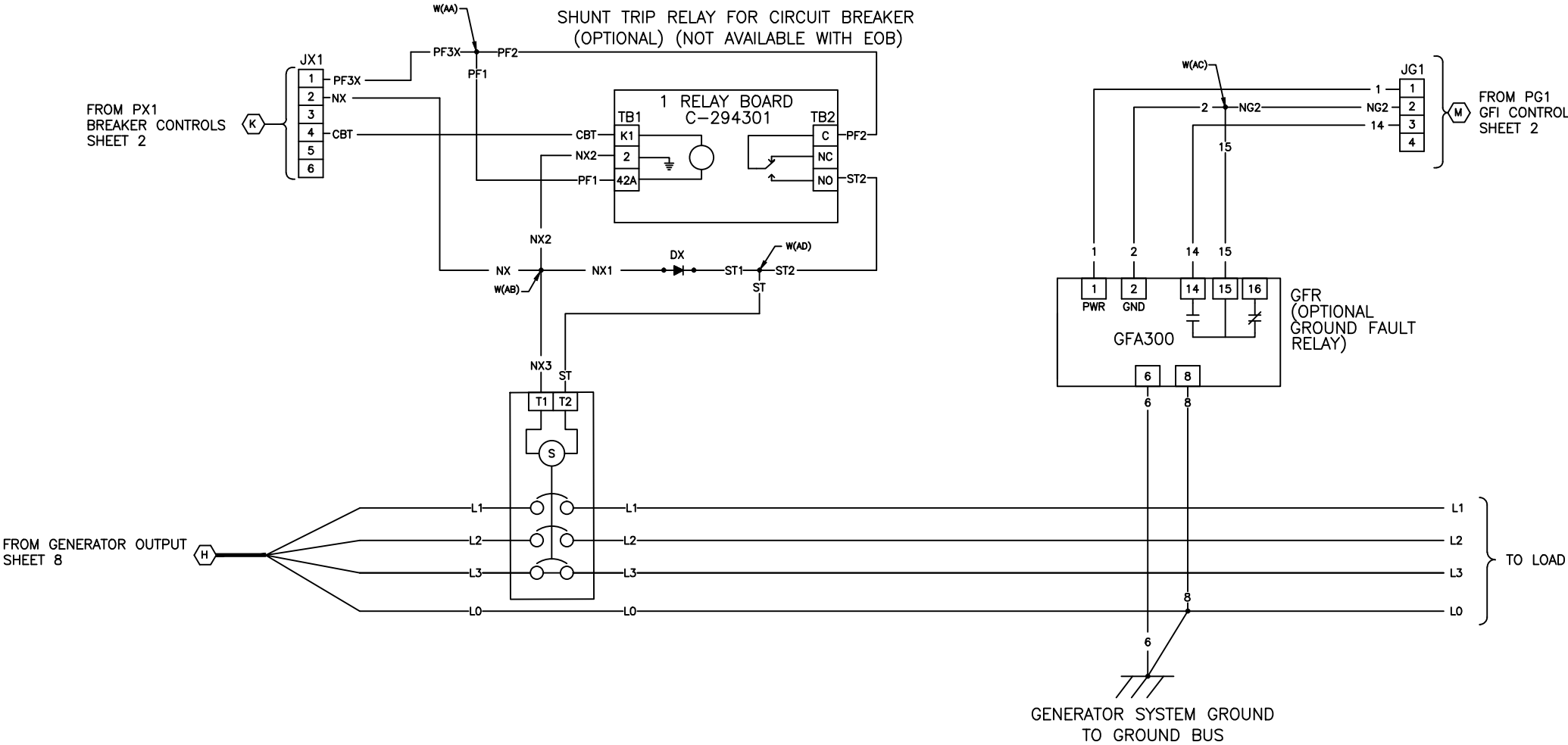
APM603 CONTROLLER
80-300 KW JOHN DEERE TIER III
W/ SPLIT ACTIVATOR, 1Ø, 3Ø & 600V

UNLESS OTHERWISE SPECIFIED - 1) DIMENSIONS ARE IN INCHES 2) TOLERANCES ARE: DIMENSIONS ± .010 HOLE ± .030 SHAFT ± .005 FRACTIONS ± .005		KOHLER. KOHLER, WI 53044 THIS DRAWING, IN WHOLE OR IN PART, IS KOHLER CO. PROPERTY AND MAY NOT BE USED EXCEPT IN CONNECTION WITH KOHLER CO. WORK. ALL RIGHTS OF DESIGN OR INVENTION ARE RESERVED.	
APPROVALS		DATE	
DESIGNED	ABS	3-12-18	
CHECKED	BTW	3-12-18	
APPROVED	JLS	3-12-18	
TITLE		7-11	
DIAGRAM, SCHEMATIC JD 80-300KW, APM603		SCALE N/A	
DRAWN BY		ADV-9057	
D			



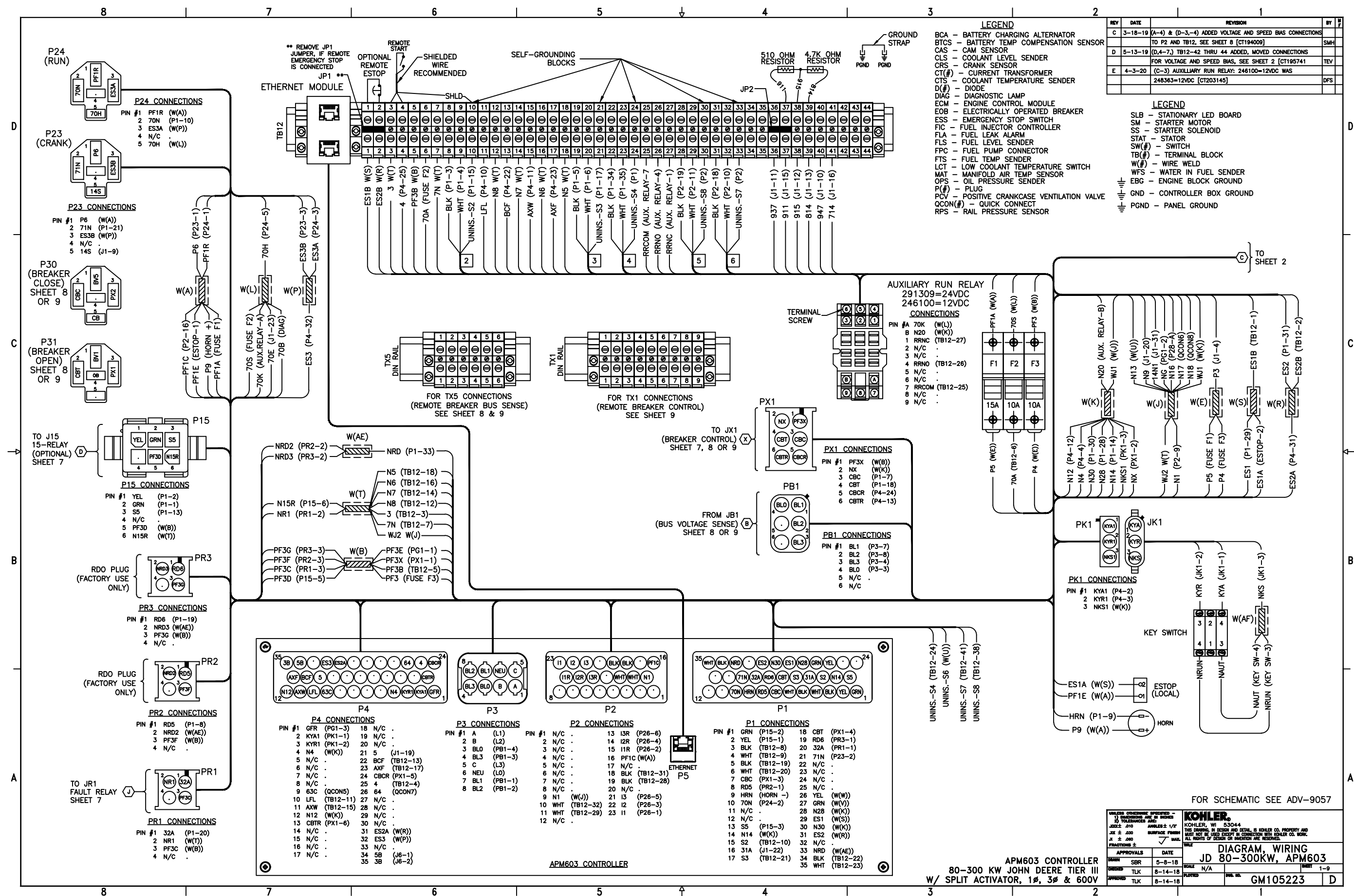
UNLESS OTHERWISE SPECIFIED - 1) DIMENSIONS ARE IN INCHES 2) TOLERANCES ARE: FRACTIONS DECIMALS .0005 .010 .001 .015 .002 .020 .003 .025 .004 .030 .005 .035 .006 .040 .007 .045 .008 .050 .009 .055 .010 .060 .011 .065 .012 .070 .013 .075 .014 .080 .015 .085 .016 .090 .017 .095 .018 .100 .019 .105 .020 .110 .021 .115 .022 .120 .023 .125 .024 .130 .025 .135 .026 .140 .027 .145 .028 .150 .029 .155 .030 .160 .031 .165 .032 .170 .033 .175 .034 .180 .035 .185 .036 .190 .037 .195 .038 .200 .039 .205 .040 .210 .041 .215 .042 .220 .043 .225 .044 .230 .045 .235 .046 .240 .047 .245 .048 .250 .049 .255 .050 .260 .051 .265 .052 .270 .053 .275 .054 .280 .055 .285 .056 .290 .057 .295 .058 .300 .059 .305 .060 .310 .061 .315 .062 .320 .063 .325 .064 .330 .065 .335 .066 .340 .067 .345 .068 .350 .069 .355 .070 .360 .071 .365 .072 .370 .073 .375 .074 .380 .075 .385 .076 .390 .077 .395 .078 .400 .079 .405 .080 .410 .081 .415 .082 .420 .083 .425 .084 .430 .085 .435 .086 .440 .087 .445 .088 .450 .089 .455 .090 .460 .091 .465 .092 .470 .093 .475 .094 .480 .095 .485 .096 .490 .097 .495 .098 .500 .099 .505 .100 .510 .101 .515 .102 .520 .103 .525 .104 .530 .105 .535 .106 .540 .107 .545 .108 .550 .109 .555 .110 .560 .111 .565 .112 .570 .113 .575 .114 .580 .115 .585 .116 .590 .117 .595 .118 .600 .119 .605 .120 .610 .121 .615 .122 .620 .123 .625 .124 .630 .125 .635 .126 .640 .127 .645 .128 .650 .129 .655 .130 .660 .131 .665 .132 .670 .133 .675 .134 .680 .135 .685 .136 .690 .137 .695 .138 .700 .139 .705 .140 .710 .141 .715 .142 .720 .143 .725 .144 .730 .145 .735 .146 .740 .147 .745 .148 .750 .149 .755 .150 .760 .151 .765 .152 .770 .153 .775 .154 .780 .155 .785 .156 .790 .157 .795 .158 .800 .159 .805 .160 .810 .161 .815 .162 .820 .163 .825 .164 .830 .165 .835 .166 .840 .167 .845 .168 .850 .169 .855 .170 .860 .171 .865 .172 .870 .173 .875 .174 .880 .175 .885 .176 .890 .177 .895 .178 .900 .179 .905 .180 .910 .181 .915 .182 .920 .183 .925 .184 .930 .185 .935 .186 .940 .187 .945 .188 .950 .189 .955 .190 .960 .191 .965 .192 .970 .193 .975 .194 .980 .195 .985 .196 .990 .197 .995 .198 1.000 .199 1.005 .200 1.010 .201 1.015 .202 1.020 .203 1.025 .204 1.030 .205 1.035 .206 1.040 .207 1.045 .208 1.050 .209 1.055 .210 1.060 .211 1.065 .212 1.070 .213 1.075 .214 1.080 .215 1.085 .216 1.090 .217 1.095 .218 1.100 .219 1.105 .220 1.110 .221 1.115 .222 1.120 .223 1.125 .224 1.130 .225 1.135 .226 1.140 .227 1.145 .228 1.150 .229 1.155 .230 1.160 .231 1.165 .232 1.170 .233 1.175 .234 1.180 .235 1.185 .236 1.190 .237 1.195 .238 1.200 .239 1.205 .240 1.210 .241 1.215 .242 1.220 .243 1.225 .244 1.230 .245 1.235 .246 1.240 .247 1.245 .248 1.250 .249 1.255 .250 1.260 .251 1.265 .252 1.270 .253 1.275 .254 1.280 .255 1.285 .256 1.290 .257 1.295 .258 1.300 .259 1.305 .260 1.310 .261 1.315 .262 1.320 .263 1.325 .264 1.330 .265 1.335 .266 1.340 .267 1.345 .268 1.350 .269 1.355 .270 1.360 .271 1.365 .272 1.370 .273 1.375 .274 1.380 .275 1.385 .276 1.390 .277 1.395 .278 1.400 .279 1.405 .280 1.410 .281 1.415 .282 1.420 .283 1.425 .284 1.430 .285 1.435 .286 1.440 .287 1.445 .288 1.450 .289 1.455 .290 1.460 .291 1.465 .292 1.470 .293 1.475 .294 1.480 .295 1.485 .296 1.490 .297 1.495 .298 1.500 .299 1.505 .300 1.510 .301 1.515 .302 1.520 .303 1.525 .304 1.530 .305 1.535 .306 1.540 .307 1.545 .308 1.550 .309 1.555 .310 1.560 .311 1.565 .312 1.570 .313 1.575 .314 1.580 .315 1.585 .316	
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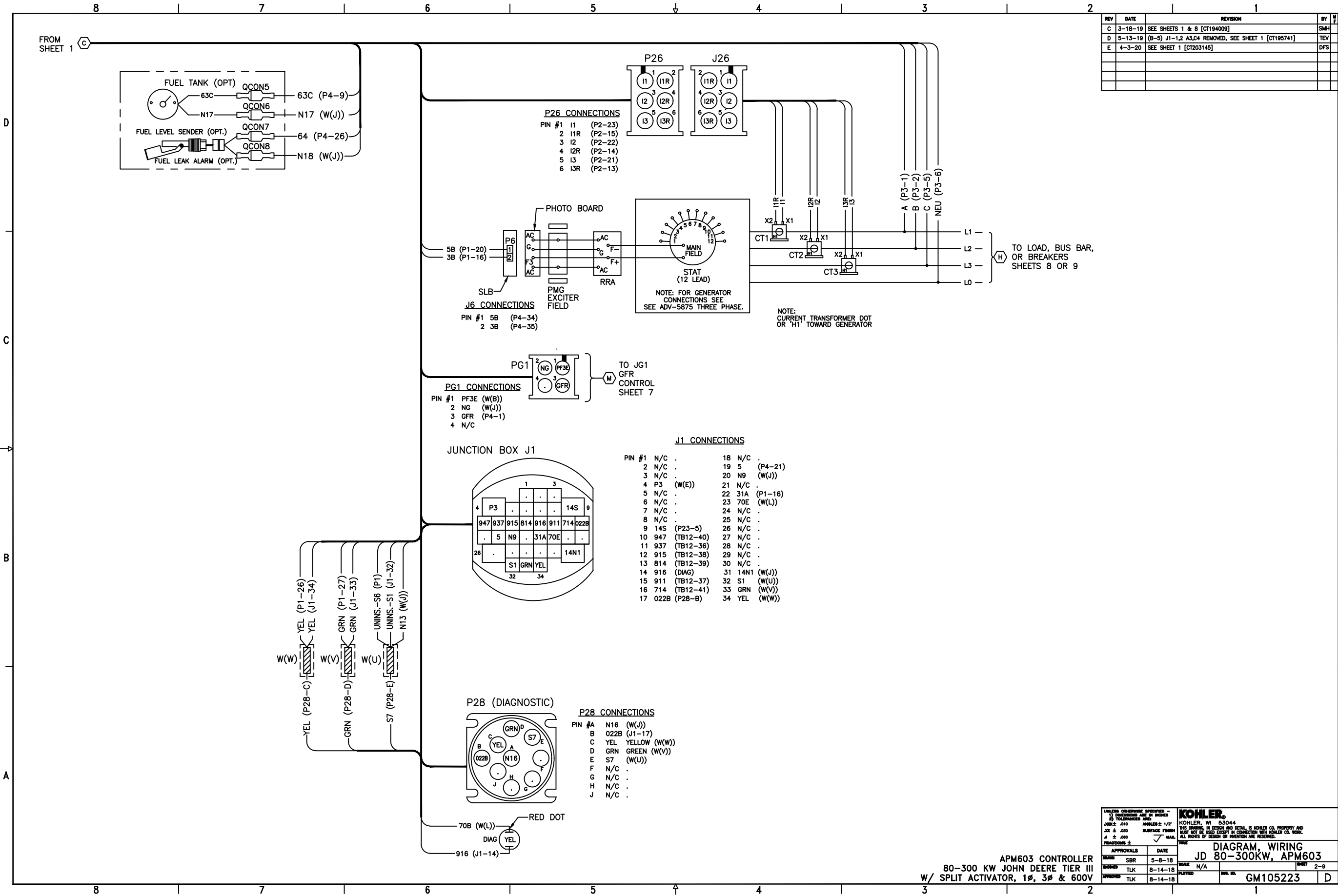
REV	DATE	REVISION	BY	APP
-	3-12-18	NEW DRAWING [CT190243]	ABS	
A	10-11-18	SEE SHEET 2 [CT191090]	TLK	
B	2-05-19	SEE SHEETS 10 & 11 [CT193517]	SMH	
C	3-18-19	SEE SHEETS 1 & 2 [CT194009]	SMH	
D	5-13-19	SEE SHEETS 1 & 2 [CT195741]	TEV	
E	9-10-19	SEE SHEET 2 [CT198508]	TEV	



SHUNT TRIP OPTION
APM603 CONTROLLER
80-300 KW JOHN DEERE TIER III
W/ SPLIT ACTIVATOR, 1Ø, 3Ø & 600V

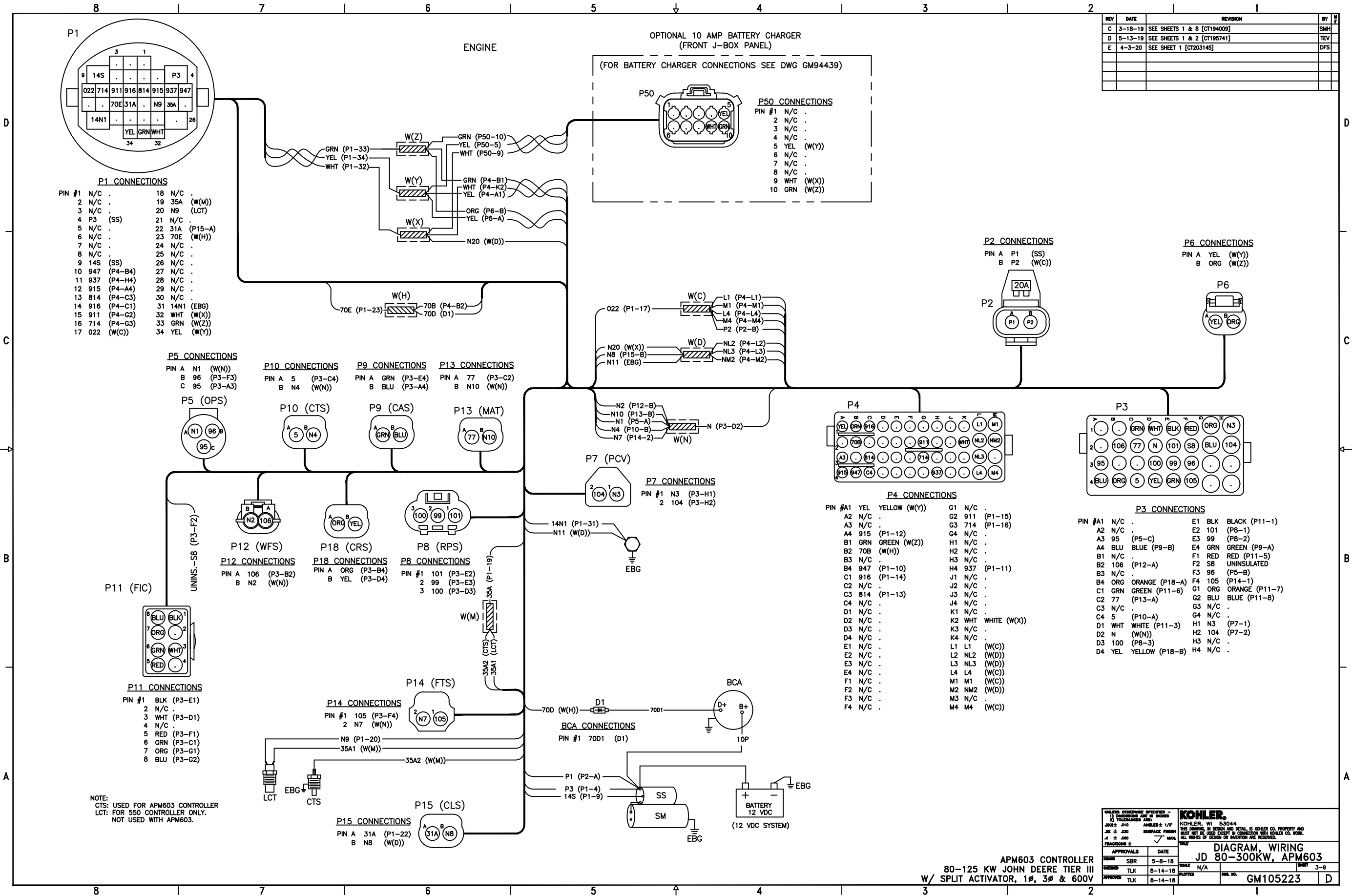
UNLESS OTHERWISE SPECIFIED - 1) DIMENSIONS ARE IN INCHES 2) TOLERANCES ARE: JXS ± .010 JXS ± .030 FRACTIONS ±				KOHLER. KOHLER, WI 53044 THIS DRAWING, IN DESIGN AND DETAIL, IS KOHLER CO. PROPERTY AND MAY NOT BE USED EXCEPT IN CONNECTION WITH KOHLER CO. WORK. ALL RIGHTS OF DESIGN OR INVENTION ARE RESERVED.			
APPROVALS				DATE			
DESIGN	ABS	3-12-18		SCALE	N/A	SHEET 9-11	
DESIGNED	BTW	3-12-18		PLOTTED		REV. NO. ADV-9057	
APPROVED	JLS	3-12-18				D	

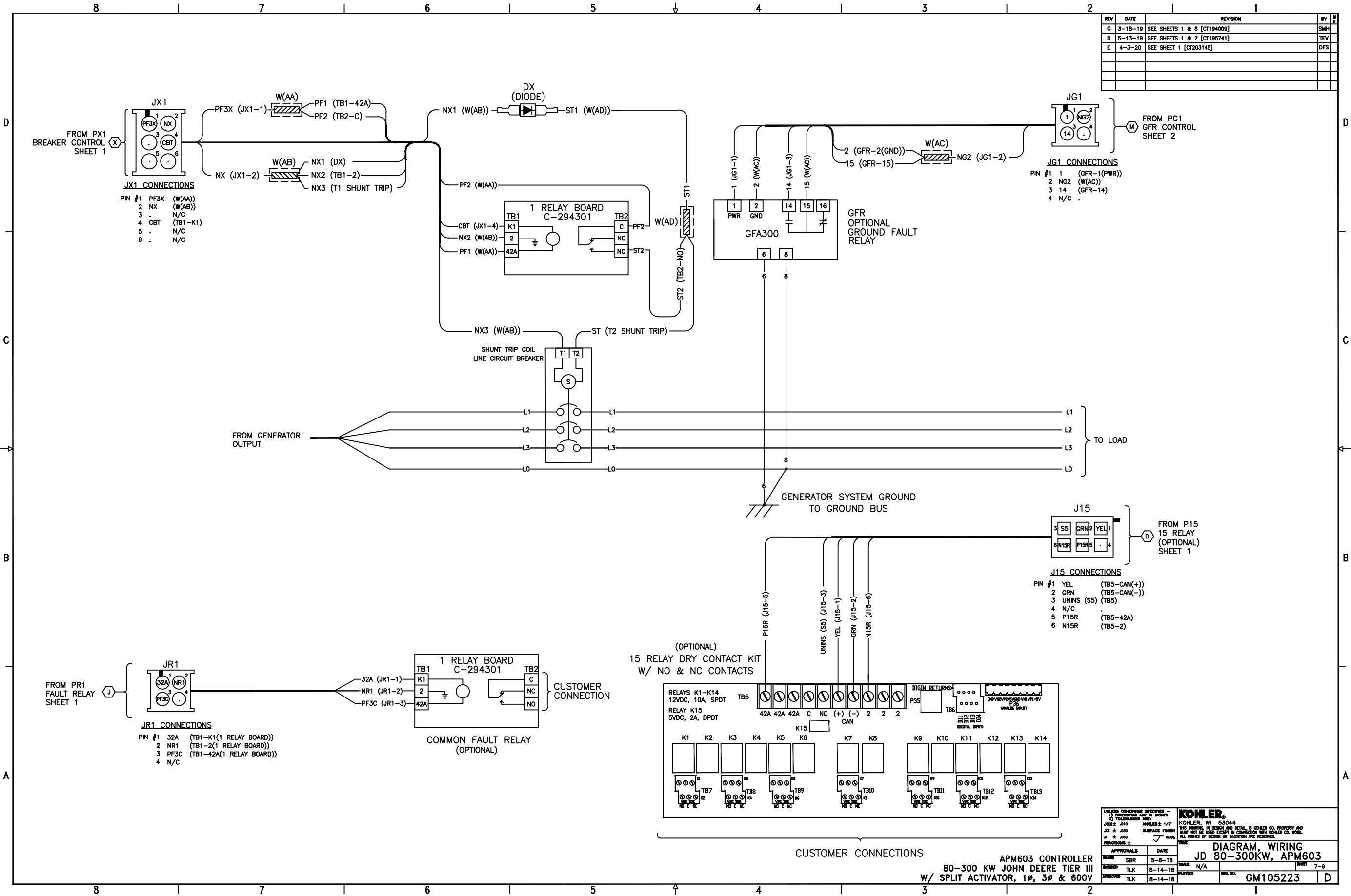


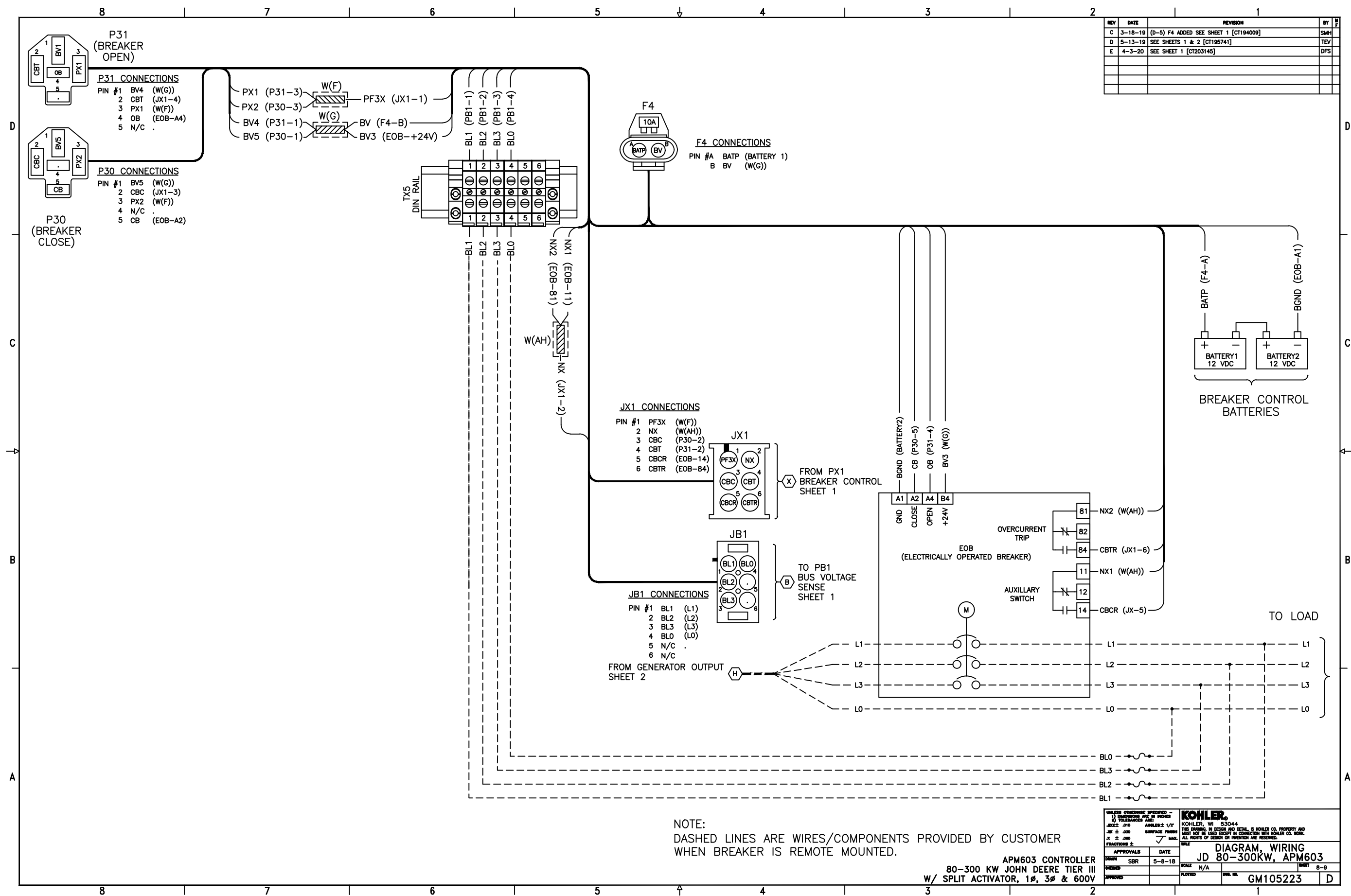


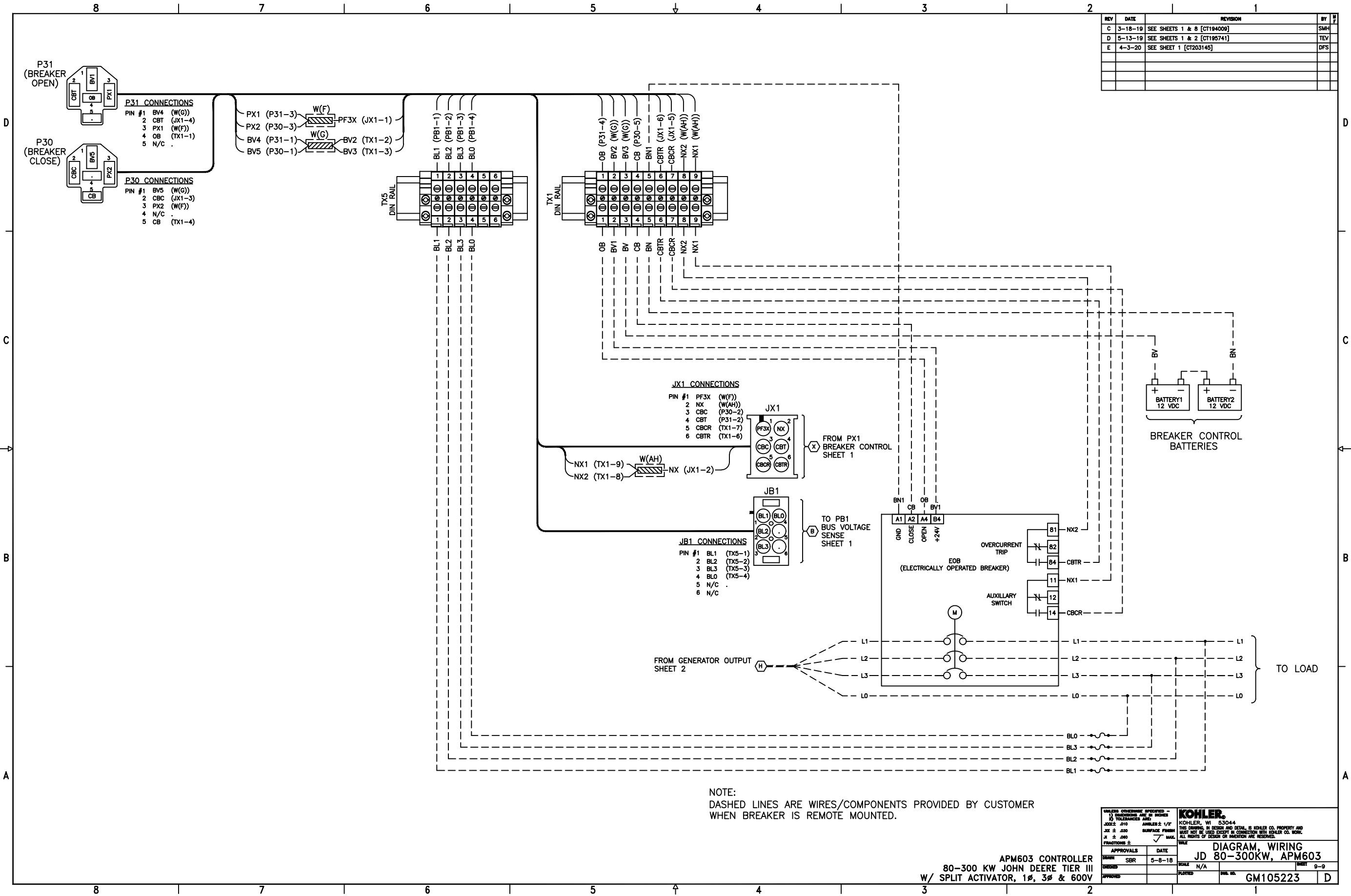
REV	DATE	REVISION	BY
C	3-18-19	SEE SHEETS 1 & 8 [CT194009]	SMH
D	5-13-19	(B-5) J1-1,2 A3,C4 REMOVED, SEE SHEET 1 [CT195741]	TEV
E	4-3-20	SEE SHEET 1 [CT203145]	DFS

UNLESS OTHERWISE SPECIFIED - 1) DIMENSIONS ARE IN INCHES TOLERANCES ARE: .005 ± .010 ANGLES ± 1/2° .0625 ± .005 SURFACE FINISH .125 ± .005		KOHLER. KOHLER, WI 53044 THIS DRAWING, IN DESIGN AND DETAIL, IS KOHLER CO. PROPERTY AND WILL NOT BE USED EXCEPT IN CONNECTION WITH KOHLER CO. WORK. ALL RIGHTS OF DESIGN OR INVENTION ARE RESERVED.	
APPROVALS		DATE	
DESIGN	SR	5-8-18	
CHECKED	TLK	8-14-18	
APPROVED	TLK	8-14-18	
TITLE		SCALE	
DIAGRAM, WIRING JD 80-300KW, APM603		SHEET 2-9	
PWA NO.		GM105223	



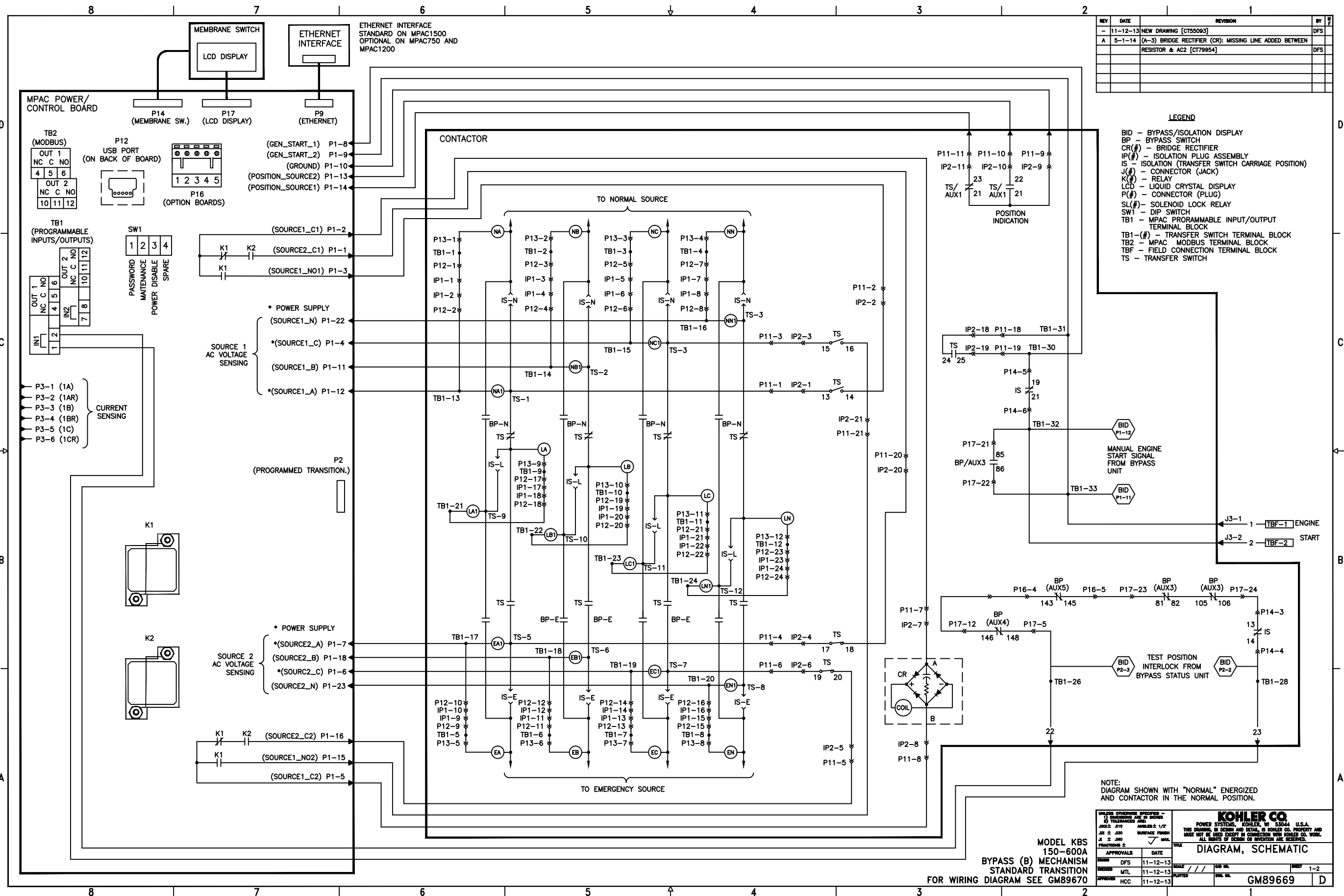






REV	DATE	REVISION	BY	APP
C	3-18-19	SEE SHEETS 1 & 8 [CT194009]	SMH	
D	5-13-19	SEE SHEETS 1 & 2 [CT195741]	TEV	
E	4-3-20	SEE SHEET 1 [CT203145]	DPS	

UNLESS OTHERWISE SPECIFIED - 1) DIMENSIONS ARE IN INCHES 2) TOLERANCES ARE: FRACTIONS ± DECIMALS ± ANGLES ± SURFACE FINISH MATERIAL		KOHLER. KOHLER, WI 53044 THIS DRAWING, IN DESIGN AND DETAIL, IS KOHLER CO. PROPERTY AND WILL NOT BE USED EXCEPT IN CONNECTION WITH KOHLER CO. WORK. ALL RIGHTS OF DESIGN OR INVENTION ARE RESERVED.	
APPROVALS DESIGN SBR CHECKED APPROVED		DATE 5-8-18	SHEET 9-9
SCALE N/A		PWA NO. GM105223	



REV	DATE	REVISION	BY	CHK
-	11-12-13	NEW DRAWING [CTSS0093]	DFS	
A	5-1-14	(A-3) BRIDGE RECTIFIER (CR): MISSING LINE ADDED BETWEEN RESISTOR & AC2 [CT79954]	DFS	

- LEGEND**
- BID - BYPASS/ISOLATION DISPLAY
 - BP - BYPASS SWITCH
 - CR(##) - BRIDGE RECTIFIER
 - IP(##) - ISOLATION PLUG ASSEMBLY
 - IS - ISOLATION (TRANSFER SWITCH CARRIAGE POSITION)
 - J(##) - CONNECTOR (JACK)
 - K(##) - RELAY
 - LCD - LIQUID CRYSTAL DISPLAY
 - P(##) - CONNECTOR (PLUG)
 - SL(##) - SOLENOID LOCK RELAY
 - SW1 - DIP SWITCH
 - TB1 - MPAC PROGRAMMABLE INPUT/OUTPUT TERMINAL BLOCK
 - TB1-(##) - TRANSFER SWITCH TERMINAL BLOCK
 - TB2 - MPAC MODBUS TERMINAL BLOCK
 - TBF - FIELD CONNECTION TERMINAL BLOCK
 - TS - TRANSFER SWITCH

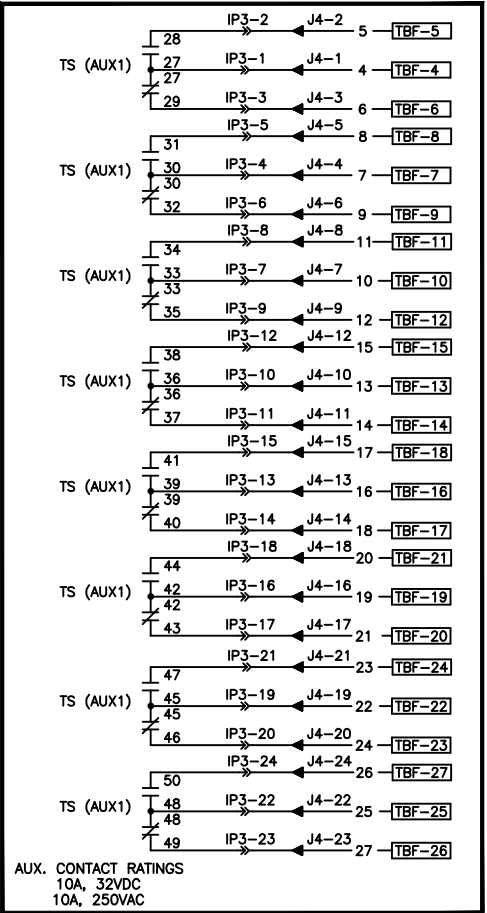
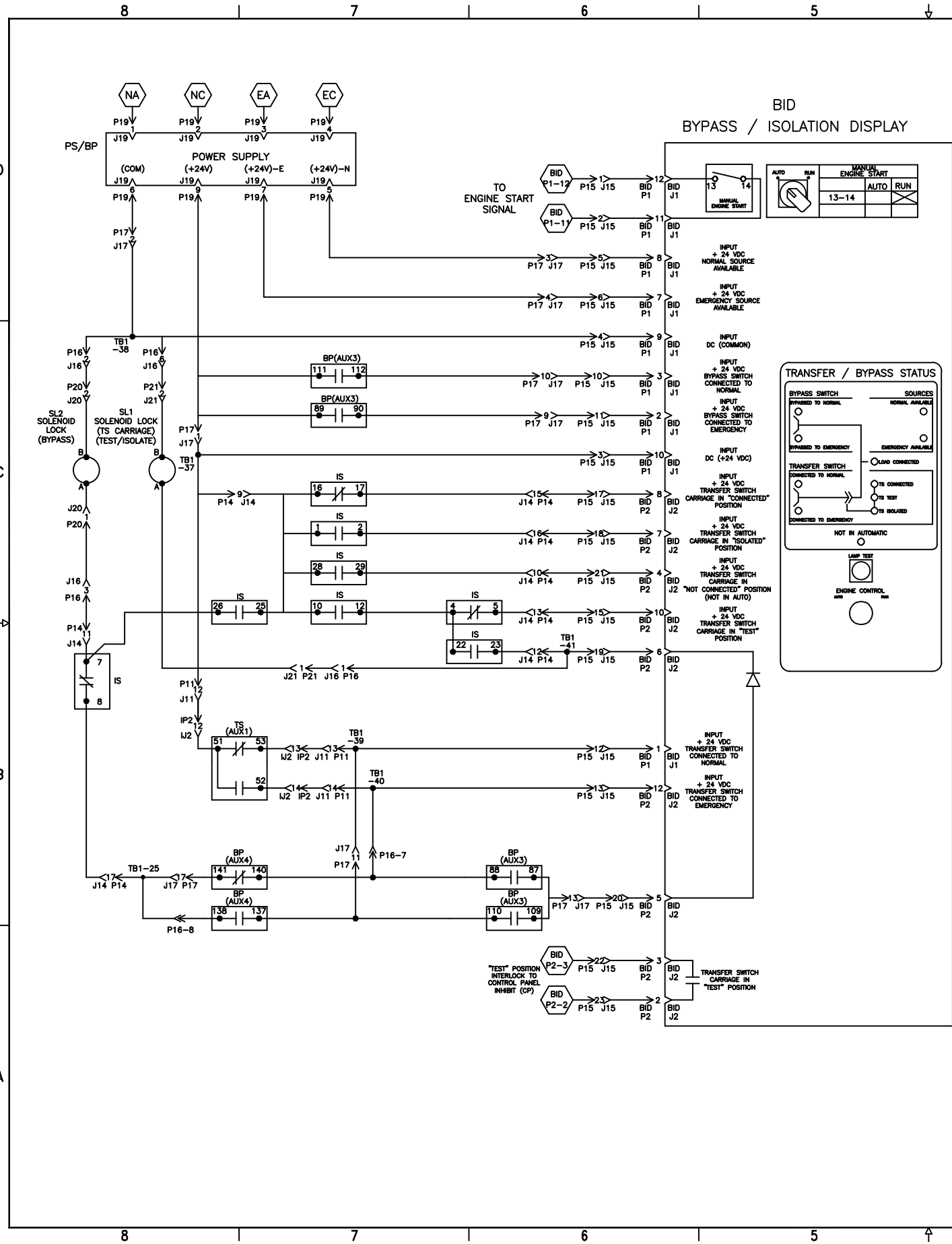
NOTE: DIAGRAM SHOWN WITH "NORMAL" ENERGIZED AND CONTACTOR IN THE NORMAL POSITION.

UNLESS OTHERWISE SPECIFIED -		KOHLER CO.	
1) DIMENSIONS ARE IN INCHES		POWER SYSTEMS, KOHLER, WI 53044 U.S.A.	
2) TOLERANCES ARE:		THIS DRAWING, IN DESIGN AND DETAIL, IS KOHLER CO. PROPERTY AND	
3) SURFACE FINISH		MAY NOT BE USED EXCEPT IN CONNECTION WITH KOHLER CO. WORK.	
4) ALL RIGHTS OF DESIGN OR INVENTION ARE RESERVED.			
APPROVALS		TITLE	
DESIGN	DATE	DIAGRAM, SCHEMATIC	
DFS	11-12-13		
DESIGNED	DATE		
MTL	11-12-13		
APPROVED	DATE		
HCC	11-12-13		

MODEL KBS 150-600A
BYPASS (B) MECHANISM
STANDARD TRANSITION
FOR WIRING DIAGRAM SEE GM89670

1-2
GM89669
D

REV	DATE	REVISION	BY	CHK
-	11-12-13	NEW DRAWING [CT55093]	DFS	
A	5-1-14	SEE SHEET 1 [CT79954]	DFS	



BYPASS SWITCH AUXILIARY CONTACTS			
BP AUXILIARY CONTACT	STATUS (*)	BP SWITCH POSITION (AUX3)	
		EMERG	NORMAL
81-82	●		
83-84	●		
85-86	●		
87-88	●		
89-90	●		
91-92	●		
93-94	●		
101-102	●		
103-104	●		
105-106	●		
107-108	●		
109-110	●		
111-112	●		
113-114	●		
115-116	●		
117-118	●		
119-120	●		
121-122	●		
123-124	●		
125-126	●		
127-128	●		

BYPASS SWITCH OPERATOR AUXILIARY CONTACTS					
BP AUXILIARY CONTACT	STATUS (*)	BP SWITCH HANDLE POSITION			
		NORMAL (CW)	OFF	EMERGENCY (CCW)	BYPASS (±90°)
140-141	●				
142-143	●				
144-145	●				
146-147	●				
148-149	●				
150-151	●				
152-153	●				
154-155	●				
156-157	●				
158-159	●				
160-161	●				
162-163	●				
164-165	●				
166-167	●				
168-169	●				
170-171	●				
172-173	●				
174-175	●				
176-177	●				
178-179	●				
180-181	●				
182-183	●				
184-185	●				
186-187	●				
188-189	●				
190-191	●				
192-193	●				
194-195	●				
196-197	●				
198-199	●				
200-201	●				

ISOLATION (TRANSFER SWITCH CARRIAGE POSITION) AUXILIARY CONTACTS					
IS AUXILIARY CONTACT	STATUS (*)	TRANSFER SWITCH CARRIAGE POSITION (AUX3)			
		CONNECT	TEST	ISOLATE	
1-2	●				
3-4	●				
5-6	●				
7-8	●				
9-10	●				
11-12	●				
13-14	●				
15-16	●				
17-18	●				
19-20	●				
21-22	●				
23-24	●				
25-26	●				
27-28	●				
29-30	●				
31-32	●				
33-34	●				
35-36	●				
37-38	●				
39-40	●				
41-42	●				
43-44	●				
45-46	●				
47-48	●				
49-50	●				
51-52	●				
53-54	●				
55-56	●				
57-58	●				
59-60	●				
61-62	●				
63-64	●				
65-66	●				
67-68	●				
69-70	●				
71-72	●				
73-74	●				
75-76	●				
77-78	●				
79-80	●				
81-82	●				
83-84	●				
85-86	●				
87-88	●				
89-90	●				
91-92	●				
93-94	●				
95-96	●				
97-98	●				
99-100	●				

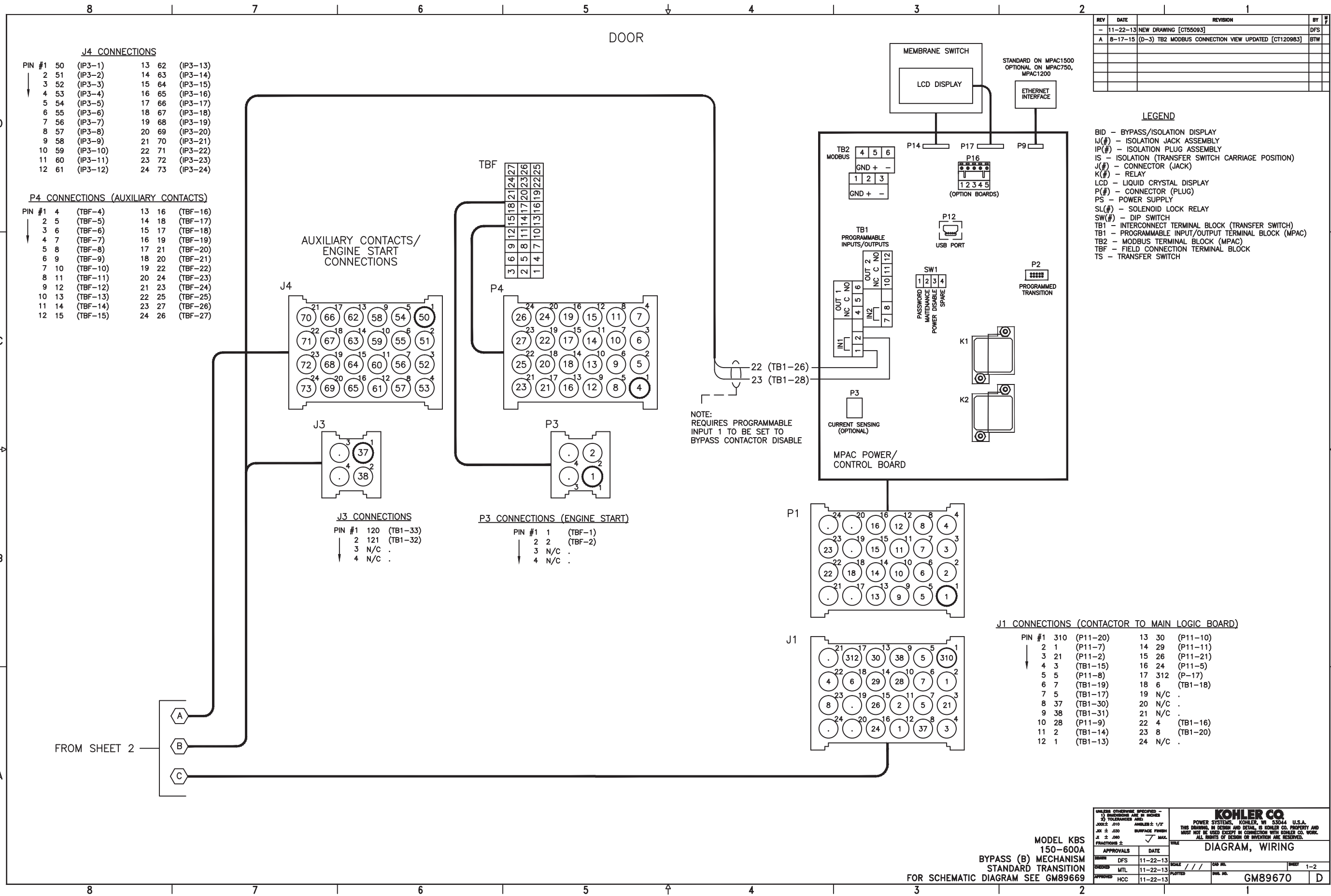
(*) CONTACT AVAILABILITY STATUS:
 ● CONTACT PROVIDED & USED IN CIRCUITRY
 "BLANK" CONTACT NOT USED, IF PHYSICALLY AVAILABLE,
 CONTACT IS FOR FACTORY USE ONLY

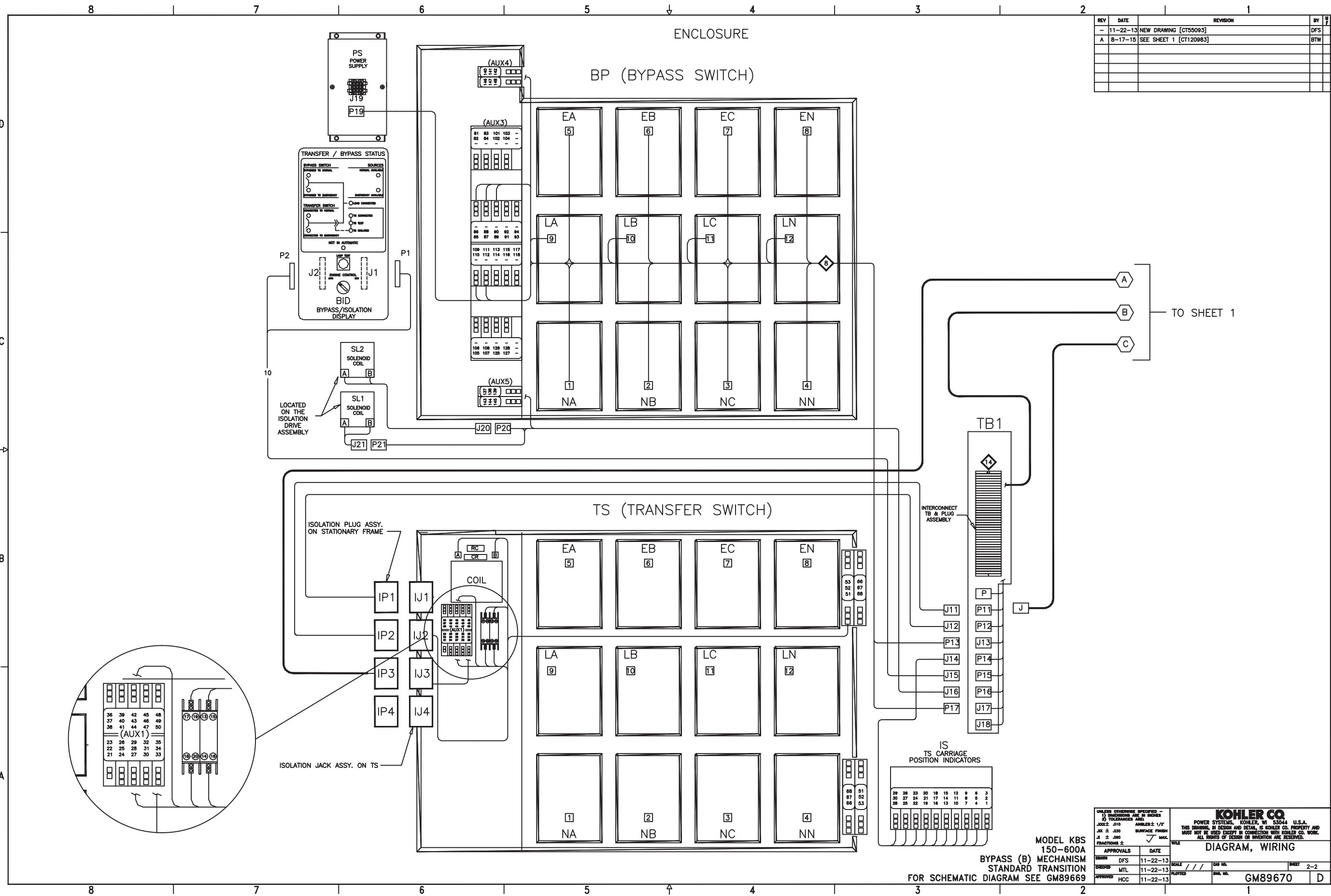
TS CONTROL CONTACTS					
TS	SOLENOID POSITION				
	CLOSED NORMAL	BEFORE TDC*	BEFORE ←TDC	CLOSED EMERG	
13-14					
15-16					
17-18					
19-20					

TDC (TOP DEAD CENTER)
 TRANSFER SWITCH TEST & ADJUSTMENT PROCEDURE
 SPECIFIES CONTROL CUT-OFF (CONTACT OPENING)
 SETTING.

MODEL KBS
 150-600A
 BYPASS (B) MECHANISM
 STANDARD TRANSITION
 FOR WIRING DIAGRAM SEE GM89670

UNLESS OTHERWISE SPECIFIED -			
1) DIMENSIONS ARE IN INCHES	2) TOLERANCES ARE:	3) FINISHES ARE:	4) SURFACE FINISH:
± .010	± .010	± .010	± .010
± .020	± .020	± .020	± .020
± .030	± .030	± .030	± .030
± .040	± .040	± .040	± .040
± .050	± .050	± .050	± .050
± .060	± .060	± .060	± .060
± .070	± .070	± .070	± .070
± .080	± .080	± .080	± .080
± .090	± .090	± .090	± .090
± .100	± .100	± .100	± .100
± .120	± .120	± .120	± .120
± .150	± .150	± .150	± .150
± .200	± .200	± .200	± .200
± .250	± .250	± .250	± .250
± .300	± .300	± .300	± .300
± .350	± .350	± .350	± .350
± .400	± .400	± .400	± .400
± .450	± .450	± .450	± .450
± .500	± .500	± .500	± .500
± .550	± .550	± .550	± .550
± .600	± .600	± .600	± .600
± .650	± .650	± .650	± .650
± .700	± .700	± .700	± .700
± .750	± .750	± .750	± .750
± .800	± .800	± .800	± .800
± .850	± .850	± .850	± .850
± .900	± .900	± .900	± .900
± .950	± .950	± .950	± .950
± 1.000	± 1.000	± 1.000	± 1.000
± 1.050	± 1.050	± 1.050	± 1.050
± 1.100	± 1.100	± 1.100	± 1.100
± 1.150	± 1.150	± 1.150	± 1.150
± 1.200	± 1.200	± 1.200	± 1.200
± 1.250	± 1.250	± 1.250	± 1.250
± 1.300	± 1.300	± 1.300	± 1.300
± 1.350	± 1.350	± 1.350	± 1.350
± 1.400	± 1.400	± 1.400	± 1.400
± 1.450	± 1.450	± 1.450	± 1.450
± 1.500	± 1.500	± 1.500	± 1.500
± 1.550	± 1.550	± 1.550	± 1.550
± 1.600	± 1.600	± 1.600	± 1.600
± 1.650	± 1.650	± 1.650	± 1.650
± 1.700	± 1.700	± 1.700	± 1.700
± 1.750	± 1.750	± 1.750	± 1.750
± 1.800	± 1.800	± 1.800	± 1.800
± 1.850	± 1.850	± 1.850	± 1.850
± 1.900	± 1.900	± 1.900	± 1.900
± 1.950	± 1.950	± 1.950	± 1.950
± 2.000	± 2.000	± 2.000	± 2.000
± 2.050	± 2.050	± 2.050	± 2.050
± 2.100	± 2.100	± 2.100	± 2.100
± 2.150	± 2.150	± 2.150	± 2.150
± 2.200	± 2.200	± 2.200	± 2.200
± 2.250	± 2.250	± 2.250	± 2.250
± 2.300	± 2.300	± 2.300	± 2.300
± 2.350	± 2.350	± 2.350	± 2.350
± 2.400	± 2.400	± 2.400	± 2.400
± 2.450	± 2.450	± 2.450	± 2.450
± 2.500	± 2.500	± 2.500	± 2.500
± 2.550	± 2.550	± 2.550	± 2.550
± 2.600	± 2.600	± 2.600	± 2.600
± 2.650	± 2.650	± 2.650	± 2.650
± 2.700	± 2.700	± 2.700	± 2.700
± 2.750	± 2.750	± 2.750	± 2.750
± 2.800	± 2.800	± 2.800	± 2.800
± 2.850	± 2.850	± 2.850	± 2.850
± 2.900	± 2.900	± 2.900	± 2.900
± 2.950	± 2.950	± 2.950	± 2.950
± 3.000	± 3.000	± 3.000	± 3.000
± 3.050	± 3.050	± 3.050	± 3.050
± 3.100	± 3.100	± 3.100	± 3.100
± 3.150	± 3.150	± 3.150	± 3.150
± 3.200	± 3.200	± 3.200	± 3.200
± 3.250	± 3.250	± 3.250	± 3.250
± 3.300	± 3.300	± 3.300	± 3.300
± 3.350	± 3.350	± 3.350	± 3.350
± 3.400	± 3.400	± 3.400	± 3.400
± 3.450	± 3.450	± 3.450	± 3.450
± 3.500	± 3.500	± 3.500	± 3.500
± 3.550	± 3.550	± 3.550	± 3.550
± 3.600	± 3.600	± 3.600	± 3.600
± 3.650	± 3.650	± 3.650	± 3.650
± 3.700	± 3.700	± 3.700	± 3.700
± 3.750	± 3.750	± 3.750	± 3.750
± 3.800	± 3.800	± 3.800	± 3.800
± 3.850	± 3.850	± 3.850	± 3.850
± 3.900	± 3.900	± 3.900	± 3.900
± 3.950	± 3.950	± 3.950	± 3.950
± 4.000	± 4.000	± 4.000	± 4.000
± 4.050	± 4.050	± 4.050	± 4.050
± 4.100	± 4.100	± 4.100	± 4.100
± 4.150	± 4.150	± 4.150	± 4.150
± 4.200	± 4.200	± 4.200	± 4.200
± 4.250	± 4.250	± 4.250	± 4.250
± 4.300	± 4.300	± 4.300	± 4.300
± 4.350	± 4.350	± 4.350	± 4.350
± 4.400	± 4.400	± 4.400	± 4.400
± 4.450	± 4.450	± 4.450	± 4.450
± 4.500	± 4.500	± 4.500	± 4.500
± 4.550	± 4.550	± 4.550	± 4.550
± 4.600	± 4.600	± 4.600	± 4.600
± 4.650	± 4.650	± 4.650	± 4.650
± 4.700	± 4.700	± 4.700	± 4.700
± 4.750	± 4.750	± 4.750	± 4.750
± 4.800	± 4.800	± 4.800	± 4.800
± 4.850	± 4.850	± 4.850	± 4.850
± 4.900	± 4.900	± 4.900	± 4.900
± 4.950	± 4.950	± 4.950	± 4.950
± 5.000	± 5.000	± 5.000	± 5.000





REV	DATE	REVISION	BY
-	11-22-13	NEW DRAWING [CT55093]	DFS
A	8-17-15	SEE SHEET 1 [CT120983]	BTW

UNLESS OTHERWISE SPECIFIED - 1) DIMENSIONS ARE IN INCHES 2) TOLERANCES ARE: FRACTIONS ± DECIMALS ± ANGLES ± SURFACE FINISH HCC				KOHLER CO. POWER SYSTEMS, KOHLER, WI 53044 U.S.A. THIS DRAWING, IN DESIGN AND DETAIL, IS KOHLER CO. PROPERTY AND MAY NOT BE USED EXCEPT IN CONNECTION WITH KOHLER CO. WORK. ALL RIGHTS OF DESIGN OR INVENTION ARE RESERVED.			
APPROVALS		DATE		TITLE		SHEET	
DFS		11-22-13		DIAGRAM, WIRING		2-2	
MTL		11-22-13		SCALE		GM89670	
HCC		11-22-13		PLOTED		D	